

NP-Completeness of Deterministic Communication Complexity via Relaxed Interlacing

Serge Gaspers*

Tao Zixu He[†]

Simon Mackenzie*

Abstract

We prove that computing the deterministic communication complexity of a Boolean function, given its truth table, is NP-complete in the standard protocol-tree-depth model, addressing a meta-complexity question raised by Yao in 1979. The reduction is from $\{0,1\}$ -Vector Bin Packing and produces, in polynomial time, a communication matrix whose optimal protocol depth exhibits a one-bit gap between satisfiable and unsatisfiable instances.

The main technical contribution is the *relaxed-interlacing* framework that makes this reduction possible. It replaces exponential-size Cartesian products with polynomial-size almost t -wise independent column sets, a pseudorandom substitute for full products, while preserving the lower-bound and protocol-control statements needed for the reduction. We develop these statements in two stages: first for classical interlacing, where projection arguments give clean lower bounds and separation statements, and then for relaxed interlacing, where a bridge lemma recovers the classical lower-bound and separation statements with controlled density loss. This leads to an extension theorem that lifts the classical lower bound to the relaxed setting and a near-exact separation theorem that lifts the corresponding protocol-control statement, with the present NP-completeness theorem as their main application here.

1 Introduction

1.1 Background

Communication complexity studies the minimum number of bits two parties must exchange to compute a function when their inputs are distributed. In the standard two-party model, Alice holds $x \in X$ and Bob holds $y \in Y$, and they wish to compute $f(x, y) \in \{0, 1\}$ by exchanging bits according to a predetermined protocol. The deterministic communication complexity $D(f)$ is the minimum worst-case number of bits exchanged. When Alice sends a bit, we will call it a *row bit*; when Bob sends a bit, we will call it a *column bit*, matching the matrix viewpoint used throughout the paper. The model has found applications in circuit complexity [6], streaming algorithms [2], data structures [10], and VLSI design [7, 13].

When introducing the model, Yao asked an interesting meta-complexity question: what is the computational complexity of computing communication complexity itself? Given a Boolean function f specified by its truth table and an integer k , how hard is it to decide whether $D(f) \leq k$? Meta-complexity questions of this kind were later studied by Kushilevitz and Weinreb [8]. The problem is easily seen to be in NP: one can guess a protocol tree and verify that it uses at most k bits. Indeed, if $f : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}$ has $D(f) \leq k$, then one may assume $k \leq n + 1$, since Alice can always send all n input bits and Bob can then return the output bit, and a depth- k protocol

*University of New South Wales

[†]University of Massachusetts Amherst

tree has at most $2^{k+1} - 1$ nodes, each carrying only a polynomial amount of data relative to the truth-table input size 2^{2n} . But is it NP-complete? In the protocol-tree-depth formulation adopted here, this question remained open for over 45 years despite sustained interest in communication complexity and its applications.

1.2 Our Results

We prove that computing the deterministic communication complexity of Boolean functions is NP-complete. Throughout this paper, communication complexity is defined as the *depth of the protocol tree* (the maximum number of bits exchanged on any root-to-leaf path), the most common convention in the literature. We use *depth* and *cost* interchangeably for this quantity.

Theorem 1 (NP-completeness of communication complexity). *Given the truth table of a Boolean function $f : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}$ and an integer k in binary, deciding whether $D(f) \leq k$ is NP-complete.*

The reduction is from $\{0, 1\}$ -Vector Bin Packing, a variant of the vector bin packing problem known to be NP-complete. We construct in polynomial time a communication matrix $\mathcal{M}_4$ whose complexity precisely encodes the satisfiability of the bin packing instance, creating a one-bit gap between satisfiable and unsatisfiable instances. Here $\mathcal{M}_4$ is the final matrix in the four-stage chain described in the overview below. A final padding step then pads the row and column alphabets to a common binary length and reads $\mathcal{M}_4$ as the resulting truth table, without changing deterministic communication complexity.

What makes the theorem nontrivial is that these ingredients have to coexist inside a polynomial-time reduction. Classical interlacing naturally gives the lower-bound and protocol-control statements that the reduction wants, but it does so over full Cartesian-product column sets, which are exponentially large. At the same time, hardness alone is not enough: the reduction has to force any budget-tight protocol to spend its early bits in a way that exposes the relevant combinatorial choices of the source instance. The relaxed-interlacing framework is built to make those demands compatible.

Comparison with concurrent work. Independent concurrent work of Hirahara et al. [4] also proves NP-hardness of deterministic communication complexity. Combined with the same trivial NP upper bound recalled above, that route also yields NP-completeness in the protocol-tree-depth formulation considered here. That route leverages a reduction showing NP-hardness of 1-monochromatic rectangles, based on a reduction in [5], and translates it to the language of communication complexity. Our route instead develops a modular framework based on *relaxed interlacing operations*, building on the ideas used to refute the direct sum conjecture in [9]. We view the main contribution of the present paper as this relaxed-interlacing framework and the accompanying lower-bound and protocol-control machinery.

What this paper adds beyond the concurrent theorem. The distinctive contribution of the present paper is the relaxed-interlacing framework itself. The bridge lemma, extension theorem, relaxed near-exact separation theorem, and stage-by-stage certification tools, which show at each stage that enough structured rows and columns survive for the next step, are designed as reusable tools for controlling both lower bounds and protocol structure in polynomial-size interlaced constructions. These ideas extend the interlacing viewpoint already used in the counterexample to the direct sum conjecture in [9]: that paper used the classical interlace and its lower-bound layer, while the present paper adds protocol control and the polynomial-size relaxed setting. The present

NP-completeness theorem is the main application of that framework here. More broadly, the paper is organised so that later reductions can apply the extension and near-exact separation theorems, together with the lemmas that certify each stage of the scaffold, as black boxes: first isolate a global combinatorial choice through staged certification, then solve the surviving local Stage 1 subproblem on that branch.

Related work. Besides the concurrent work already discussed, meta-complexity questions for communication complexity were studied by Kushilevitz and Weinreb [8]. The interlacing operation introduced in [9] and refined here is closely related to pointer-chasing functions and other layered communication problems [3, 11, 12, 14]. In the pointer-chasing framework, Alice and Bob alternate lookups in shared tables; in the interlacing formulation, the selection structure is exposed directly through indexing (i selects y_i) and the analysis tracks families of index sets combinatorially. This makes modifications such as relaxing the column set or nesting multiple stages natural, and it facilitates the fine-grained protocol-control statements that the reduction requires.

Technical contributions. The framework consists of four components:

1. **Relaxed interlacing** (Section 4). A refinement of the classical interlacing operation that uses almost t -wise independent column sets instead of full Cartesian products, producing polynomial-size instances while retaining the lower-bound and protocol-control features needed for the reduction.
2. **Lower-bound engine** (Section 2). Starting from a robust seed matrix, meaning one that stays hard after controlled row and column deletion, an iterated density-amplification lemma gives lower bounds tolerant of severe loss in the surviving column fraction. This refines and extends the lower-bound methods from [9]: in particular, the bundled form of the iterated partition lemma tracks the three neighbouring density levels y , $y/2$, and $y/4$, while the scalar form is its top-rung consequence via an explicit projection step.
3. **Protocol control** (Sections 3 and 4.5). The near-exact separation theorems constrain the structure of protocols that meet the lower bound tightly: if q is the number of interlace copies, then the first $\log q$ bits must be row bits that cleanly isolate the q outer blocks, meaning the top-level row blocks corresponding to those copies. The extension theorem is the lower-bound input that makes the relaxed separation theorem applicable.
4. **Self-similarity.** Because interlacing a robust matrix produces the transpose of a robust matrix, the operation and its analysis can be nested through multiple rounds of communication. This self-similarity was first leveraged in [9] and enables the multi-stage reduction.

The mechanisms of relaxed interlacing, robustness, protocol control, and self-similarity appear reusable in communication complexity beyond the specific NP-completeness result proved here.

Overview

The proof has three layers.

Classical interlace (Sections 2 and 3). We develop lower bounds and protocol-control statements for classically interlaced matrices. The lower-bound engine starts from a *robust* seed matrix, meaning one that remains hard after controlled row and column deletion, and amplifies its complexity through iterated interlacing. The two-copy interlace is the simplest case: a first-bit analysis

couples three neighbouring density levels for the surviving column set and lifts the lower bound by one bit. Iterating this argument via the *one-step partition lemma* (Lemma 18) produces the *iterated density-amplification lemma* (Lemma 21), which gives lower bounds tolerant of extreme loss in the surviving column set. A separate branch of the partition lemma yields the *odd-copy seed* (Lemma 19), a seed statement for interlaces with one extra unpaired copy, which supplies constant-density seeds at higher copy counts; feeding those seeds into the iterated density-amplification lemma produces the hard-seed input later formalised in Lemma 33. On the protocol-control side, the two-copy lower bound rules out row splits that fail to separate the outer blocks, and an inductive argument turns this into the *near-exact separation lemma* (Lemma 26), which forces the first $\log q$ bits of any protocol whose depth matches that lower bound to be row bits that cleanly isolate each outer block.

Relaxed interlace (Section 4). The classical interlace has exponential-size column sets. We replace the full Cartesian product Y^q by a polynomial-size almost t -wise independent column family, that is, a pseudorandom substitute for the full product, obtaining the *relaxed interlace*. A bridge lemma (Lemma 32) recovers classical restrictions of the form needed by the lower-bound arguments from relaxed submatrices that still keep many rows across many outer blocks and enough columns. Combining the bridge with the hard-seed lemma and the column-loss resilience hypothesis, we prove two theorems that the reduction uses as black boxes. The *extension theorem* (Theorem 38) lifts the classical lower bound to the relaxed setting: it shows that the classical lower bound survives inside the canonical relaxed interlace, provided the extracted submatrix keeps enough well-distributed outer coordinates and enough column density. The *near-exact separation theorem* (Theorem 41) lifts the protocol-control statement: under the corresponding optimal depth bound, the protocol’s first $\log q$ bits must be row bits that isolate the outer blocks, exactly as in the classical case.

Reduction (Section 5). We reduce from $\{0,1\}$ -Vector Bin Packing. The reduction builds a chain of four matrices $\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3, \mathcal{M}_4$ on top of a trivial seed $\mathcal{M}_0 = [1 \ 0]$. Stage 1 is a relaxed interlace of $\mathcal{M}_0$ that encodes capacity constraints; Stage 2 is a relaxed interlace of the transpose of $\mathcal{M}_1$ to encode dimension selection; Stage 3 is a classical 4-fold interlace of the transpose of $\mathcal{M}_2$ that creates one outer block per bin; Stage 4 attaches one row per source vector plus local gadgets that link the bin assignment to the source instance.

In the YES case, a feasible bin packing yields exactly the intended protocol scaffold: choose a bin, choose a coordinate, and then solve the resulting local Stage 1 gadget within the target budget.

In the NO case, the protocol-control theorems rule out any cheaper escape. The Stage 3 argument first forces the protocol to isolate one bin branch, the Stage 2 argument then forces one coordinate block within that branch, and the resulting Stage 4 local gadget exposes an overloaded local Stage 1 instance. That local instance already requires one more bit than the remaining budget, giving the final contradiction.

Proof spine. Figure 1 records the dependency chain used by the main theorem.

2 Classical Interlace: The Lower-Bound Engine

We begin with classical interlacing because it is the cleanest setting in which to develop the two ingredients the reduction needs: lower bounds and protocol-control statements. At the same time, the final reduction must be built from a polynomial-size family of functions. The role of this section and the next is therefore to establish both kinds of argument first in the classical setting, where

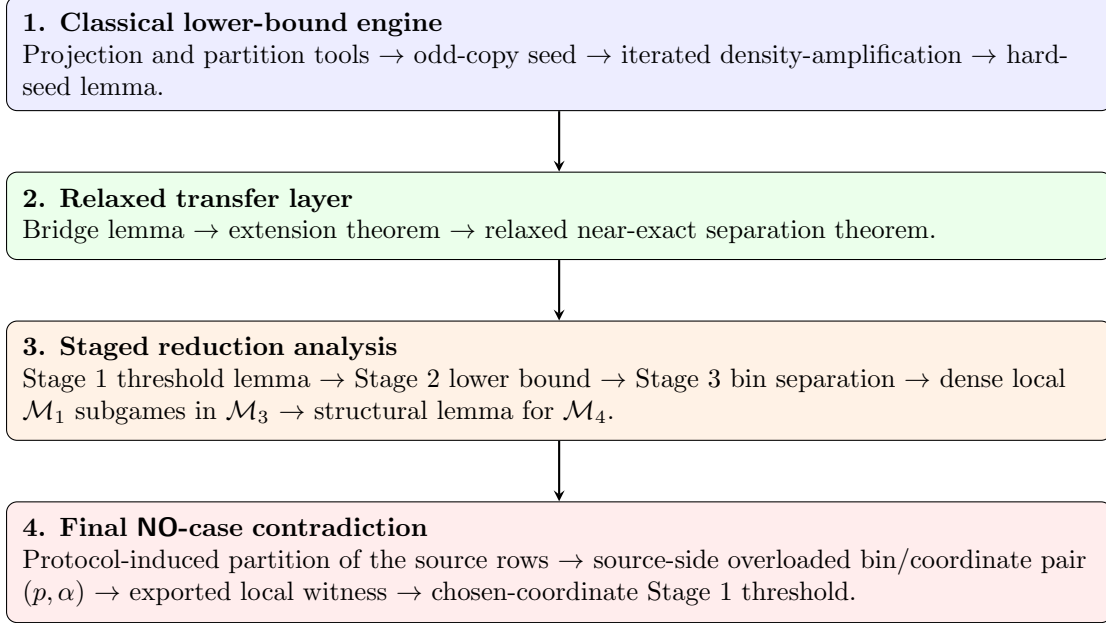


Figure 1: **Proof spine for the main theorem.** Each layer is consumed as a black box by the next one.

the mechanisms are easiest to see. This section builds the lower-bound engine; Section 3 extracts the protocol-control consequences. Afterward we will transfer both to relaxed interlacing, which preserves polynomial size while retaining the same underlying structure.

The first nontrivial case is the two-copy interlace. It is worth isolating for two reasons. First, it is the simplest place where interlacing forces a one-bit increase in communication complexity. Second, the same mechanism will later reappear in the protocol-control arguments, where it rules out row splits that fail to separate the two interlaced copies. So the two-copy case is not just a warm-up: it is the first instance of the pattern that drives the rest of the section.

The discussion in this section has a simple order. We first record the basic projection lemmas, then analyse the two-copy interlace in detail. After introducing the robustness hypothesis as a controlled input condition, we formulate the general one-step move in the partition lemma and study the first examples with one surplus copy beyond a power of two, namely three copies and more generally $2^{k-1} + 1$ copies. Those examples explain how keeping that extra copy through repeated halving yields extra complexity before we turn to the harder task of reducing the column density further.

If Φ is a nonempty family of Boolean matrices, we write

$$D(\Phi) := \min_{N \in \Phi} D(N).$$

Thus $D(\Phi)$ is the least deterministic communication complexity among the matrices in the family Φ .

2.1 Classical Interlace and Bracket Families

We treat a Boolean matrix as a function $M : X \times Y \rightarrow \{0, 1\}$ on finite row and column sets. The classical interlace operation takes p copies of M , lets Alice choose which copy is active, and gives

Bob a column input for every copy in parallel. For any Boolean matrix M , we write $\text{Rows}(M)$ for its row set and $\text{Cols}(M)$ for its column set, so in this paragraph $\text{Rows}(M) = X$ and $\text{Cols}(M) = Y$.

Definition 2 (p -fold interlace). *Let $M : X \times Y \rightarrow \{0, 1\}$ be a Boolean matrix and let $p \geq 1$. The p -fold interlace $M \langle p \rangle$ is the matrix with row set $[p] \times X$, where $[p] := \{1, \dots, p\}$, column set Y^p , and value*

$$M \langle p \rangle ((i, x), (y_1, \dots, y_p)) := M(x, y_i).$$

We will refer to the row subsets $\{i\} \times X$ as the outer row blocks of $M \langle p \rangle$.

To state lower bounds, we will need a standard family of balanced restrictions of $M \langle p \rangle$.

Definition 3 (Restricted submatrix). *Let G be a Boolean matrix with row set X and column set Y . For $R \subseteq X$ and $C \subseteq Y$, we write*

$$G \upharpoonright_{R, C}$$

for the restricted submatrix of G with row set R and column set C .

Definition 4 ((Q, T) -Equipartitioned Row Set). *Let X be a finite set, let $k \geq 1$, and let $Q \subseteq [k]$. For a row set $R \subseteq [k] \times X$ and an integer $T \geq 1$, write*

$$R_q := \{x \in X : (q, x) \in R\} \quad \text{for each } q \in Q.$$

We say that R is (Q, T) -equipartitioned if

$$|R_q| \geq T \quad \text{for every } q \in Q.$$

Definition 5 (Bracket family). *Let M be an $m \times n$ matrix, let $p \geq 1$, and let $0 < x, y \leq 1$. Write*

$$T := \lceil mx \rceil, \quad S := \lceil n^p y \rceil.$$

The family $M[p, x, y]$ consists of all submatrices of $M \langle p \rangle$ whose row set is $([p], T)$ -equipartitioned and whose column set has size at least S .

The bracket notation serves three purposes. First, it tracks protocol degradation: the parameter x measures the fraction of rows surviving in each block and y measures the fraction of columns surviving overall, so small values signal that many bits have been spent. Second, it preserves hardness across blocks: the T -equipartition requirement ensures that every block retains at least $T = \lceil mx \rceil$ rows, excluding rectangles that focus on just a few blocks and thereby ensuring that all p copies of M remain engaged. Third, it provides the language for inductive complexity arguments: the projection and partition lemmas below are all stated as relationships between bracket families at different parameter settings, expressing how complexity evolves when we change p , x , or y .

2.2 Projection and Monotonicity Tools

The next four lemmas are standard background tools that we will repeatedly use. They are not the main theorem-level work of the paper, but the later lower bounds rest on them.

Lemma 6 (Monotonicity). *Let M be a Boolean matrix. If $p' \leq p$, $0 < x' \leq x \leq 1$, and $0 < y' \leq y \leq 1$, then*

$$D(M[p', x', y']) \leq D(M[p, x, y]).$$

Lemma 7 (Maximum projection). *For any matrix M , integers $1 \leq \ell \leq p$, and reals $0 < x, y \leq 1$,*

$$D(M[p, x, y]) \geq D\left(M\left[\ell, x, y^{\ell/p}\right]\right).$$

Lemma 8 (Transpose symmetry). *For every Boolean matrix M ,*

$$D(M^\top) = D(M).$$

Lemma 9 (One-copy transpose). *For every Boolean matrix M and all reals $0 < x, y \leq 1$,*

$$D(M[1, x, y]) = D(M^\top[1, y, x]).$$

Proof. Let $m := |\text{Rows}(M)|$ and $n := |\text{Cols}(M)|$. A member of $M[1, x, y]$ is exactly a submatrix $M \upharpoonright_{R, C}$ with

$$|R| \geq \lceil mx \rceil \quad \text{and} \quad |C| \geq \lceil ny \rceil.$$

Transposition bijects such submatrices with the members of $M^\top[1, y, x]$, and deterministic communication complexity is invariant under transposition by Lemma 8. $\square$

Lemma 10 (Product of projections). *Let $g \in M[p, x, y]$, and let $R = R_1 \cup R_2$ be a partition of the rows of g . Then there exist integers $\ell_1, \ell_2 \geq 0$ and reals $y_1, y_2 \in (0, 1]$ such that*

$$\ell_1 + \ell_2 = p, \quad y_1 y_2 \geq y,$$

and, for each $i \in \{1, 2\}$ with $\ell_i \geq 1$, the child rectangle induced by R_i contains a matrix in $M[\ell_i, x/2, y_i]$.

The only results imported from Mackenzie and Saffidine [9] are these basic structural properties of the interlace operation (monotonicity, maximum projection, product of projections, transpose symmetry) together with the specific rank lower bound for the seed matrix used later in the reduction section (Section 5), specifically in Lemma 49. All subsequent lower-bound and protocol-control arguments in this paper are developed from scratch.

We will also need one simple fixed-matrix consequence of the same idea. Starting from a restriction of $M \langle p \rangle$, we will sometimes keep only a chosen set of coordinates and discard the rest. The next lemma is the formal step that lets us pass from such a restriction to the corresponding smaller bracket family, and it will be used later when we focus on two chosen interlace coordinates inside a larger protocol branch.

Lemma 11 (Coordinate projection). *Let M be an $m \times n$ matrix, let $p \geq 1$, let $Q \subseteq [p]$ with $|Q| = r \geq 1$, let $0 < y \leq 1$, let $T \geq 1$, and let N be a submatrix of $M \langle p \rangle$ whose row set is (Q, T) -equipartitioned and whose column set has size at least $n^p y$. If $0 < x \leq 1$ satisfies $\lceil mx \rceil \leq T$, then N contains a matrix in $M[r, x, y]$.*

Proof. Let $K \subseteq \text{Cols}(M)^p$ be the column set of N , and project it to the coordinates in Q . Each projected column has at most n^{p-r} preimages, so the projected column set $P \subseteq \text{Cols}(M)^r$ has size at least

$$\frac{|K|}{n^{p-r}} \geq n^r y.$$

Since $|P|$ is an integer, this implies

$$|P| \geq \lceil n^r y \rceil.$$

For each $u \in P$, choose one column $\hat{u} \in K$ whose projection is u , and let $\hat{P} \subseteq K$ be the set of these chosen columns. Restrict N to the rows from the outer row blocks in Q and to the columns $\hat{P}$. Choosing a bijection $Q \rightarrow [r]$, this restricted matrix is canonically identified with a member of $M \langle r \rangle$. Since each selected coordinate contributes at least $T \geq \lceil mx \rceil$ rows, the resulting matrix lies in $M[r, x, y]$. $\square$

2.3 The Two-Copy Interlace

We now look at a single classical interlacing step. The right way to read this case is not to jump immediately to the strongest conclusion, but to record the three lower bounds that appear naturally as the column density deteriorates by factors of 2. The underlying mechanism does not depend on robustness: if we already know one-copy lower bounds at densities y , $y/2$, and $y/4$, then a first-bit analysis promotes them to the corresponding two-copy lower bounds at densities y^2 , $y^2/2$, and $y^2/4$.

Lemma 12. *Let M be a Boolean matrix, let $0 < x \leq \frac{1}{2}$, let $0 < y \leq 1$, and let $H \in \mathbb{R}$. Assume*

$$D\left(M\left[1, x, \frac{y}{4}\right]\right) \geq H - 2, \quad D\left(M\left[1, x, \frac{y}{2}\right]\right) \geq H - 1, \quad D(M[1, x, y]) \geq H.$$

Then

$$\begin{aligned} D\left(M\left[2, 2x, \frac{y^2}{4}\right]\right) &\geq H - 1, \\ D\left(M\left[2, 2x, \frac{y^2}{2}\right]\right) &\geq H, \\ D(M[2, 2x, y^2]) &\geq H + 1. \end{aligned}$$

Each rung of this ladder is proved from the one below it by looking only at the protocol's first bit. A row move, meaning that Alice sends the first bit, is handled by Lemma 10, followed if necessary by Lemma 7; a column move, meaning that Bob sends the first bit, lowers the surviving density by a factor of 2 and therefore pushes the protocol down by one rung. This is the first appearance of the pattern that drives the later lower-bound arguments.

Proof. All three rungs follow the same pattern: analyse the protocol's first bit as either a row move or a column move. A row move is handled by Lemma 10: in both the balanced split $(\ell_1, \ell_2) = (1, 1)$ and the degenerate case $\ell_i = 2$, where one child receives all surviving outer copies, one child inherits a one-copy density at least $y/2^j$ (where j indexes the rung), and Lemma 7 reduces the degenerate case to the balanced one. A column move halves the surviving density and drops the argument to the next lower rung. The bottom rung closes directly by maximum projection to the one-copy hypothesis at density $y/4$. We give the top-rung argument in full; the bottom and middle rungs are identical in structure and are recorded in Remark 13 below.

Top rung. Let

$$N_2 \in M[2, 2x, y^2].$$

We show that every deterministic protocol for N_2 uses at least $H + 1$ bits.

If the first bit is a row bit, apply Lemma 10 to the two children. This produces matrices

$$g_i \in M[\ell_i, x, z_i] \quad (i \in \{1, 2\}),$$

with $\ell_1 + \ell_2 = 2$ and $z_1 z_2 \geq y^2$. If $(\ell_1, \ell_2) = (1, 1)$, then one of z_1, z_2 is at least y , so by monotonicity one child has complexity at least

$$D(M[1, x, y]).$$

If instead one child carries both copies, then for that child we have $z_i \geq y^2$, and Lemma 7 gives

$$D(g_i) \geq D(M[1, x, \sqrt{z_i}]) \geq D(M[1, x, y]).$$

Therefore the surviving child has complexity at least H . Hence N_2 has complexity at least $H + 1$.

If the first bit is a column bit, follow the heavier child (the child retaining at least half the columns). It contains a matrix in

$$M \left[2, 2x, \frac{y^2}{2} \right],$$

so the middle rung gives complexity at least H for that child. Again one bit has already been spent, and therefore N_2 has complexity at least $H + 1$. $\square$

Remark 13 (Bottom and middle rungs). *The bottom and middle rungs follow the same case analysis as the top rung.*

Bottom rung ($N_0 \in M [2, 2x, y^2/4]$, target $H - 1$). A row move lands on one-copy density at least $y/2$, giving complexity $\geq H - 1$. A column move halves the density to $y^2/8$; maximum projection gives one-copy density $y/(2\sqrt{2}) \geq y/4$, which by hypothesis has complexity $\geq H - 2$, so with the spent bit the total is $\geq H - 1$.

Middle rung ($N_1 \in M [2, 2x, y^2/2]$, target H). A row move lands on one-copy density at least $y/\sqrt{2} \geq y/2$, giving complexity $\geq H - 1$, hence $\geq H$ with the spent bit. A column move drops to $y^2/4$, which is the bottom rung with target $H - 1$; again the spent bit gives $\geq H$.

2.4 The Robustness Hypothesis

The two-copy ladder does not depend on any special property of M : it lifts arbitrary one-copy lower bounds to two-copy lower bounds. For the rest of the paper we will start from matrices that already remain hard after a controlled amount of row and column deletion. We package those input requirements in the following definition.

Definition 14 ((δ, b) -robust matrix). *Let $b \geq 0$ be real and let $\delta \in (0, \frac{1}{2})$. A Boolean matrix M is (δ, b) -robust if*

$$D(M) \geq 1, \tag{R1}$$

$$D \left(M \left[1, 2^{-b}, \frac{1}{2} + \delta \right] \right) \geq D(M), \tag{R2}$$

$$D \left(M \left[1, 2^{-b}, \frac{1}{8} + \frac{\delta}{4} \right] \right) \geq D(M) - 2, \tag{R3}$$

$$D \left(M \left[1, 2^{-b}, \frac{1}{4} + \frac{\delta}{2} \right] \right) \geq D(M) - 1. \tag{R4}$$

Condition (R2) is the main hardness requirement: even after retaining only a 2^{-b} fraction of the rows and a $\frac{1}{2} + \delta$ fraction of the columns, the matrix is still as hard as M itself. Conditions (R3) and (R4) are the weaker one-copy estimates that the two-copy ladder consumes at densities $y/4$ and $y/2$. Later, when the hard-seed construction (Lemma 33 in Section 4) and the reduction parameters are fixed, we will simply choose δ to be a concrete constant. When later statements quantify over indices up to b , the quantified indices are always required to be integers.

For the robust matrices that will serve as inputs later on, we apply the ladder with $x = 2^{-b}$ and $y = \frac{1}{2} + \delta$. Since the resulting two-copy row threshold is 2^{-b+1} , we will implicitly work in the regime $b \geq 1$ from this point on.

Corollary 15. *Let M be (δ, b) -robust, assume in addition that $b \geq 1$, and write*

$$y := \frac{1}{2} + \delta.$$

Then

$$\begin{aligned} D\left(M\left[2, 2^{-b+1}, \frac{y^2}{4}\right]\right) &\geq D(M) - 1, \\ D\left(M\left[2, 2^{-b+1}, \frac{y^2}{2}\right]\right) &\geq D(M), \\ D\left(M\left[2, 2^{-b+1}, y^2\right]\right) &\geq D(M) + 1. \end{aligned}$$

Proof. Apply Lemma 12 with $x = 2^{-b}$ and $H = D(M)$, and then use (R3), (R4), and (R2). $\square$

Corollary 16. *Let M be (δ, b) -robust, and assume in addition that $b \geq 1$. Then*

$$D\left(M\left[2, 2^{-b+1}, \left(\frac{1}{2} + \delta\right)^2\right]\right) \geq D(M) + 1.$$

Proof. This is the top rung of Corollary 15. $\square$

The next observation is the first protocol-control consequence. It will later reappear inside the separation arguments for the reduction.

Corollary 17. *Let M be (δ, b) -robust, and assume in addition that $b \geq 1$. Consider a protocol on the two-copy interlace of M , and suppose its first bit is a row bit. If the surviving child rectangle still contains two outer copies each with at least a 2^{-b+1} fraction of the rows of M , then that child already contains a matrix of complexity at least $D(M) + 1$.*

Proof. Because the first bit is a row bit, the surviving child keeps the full column set of the two-copy interlace. By hypothesis, it therefore contains a member of

$$M\left[2, 2^{-b+1}, \left(\frac{1}{2} + \delta\right)^2\right]$$

as a restriction. Apply Corollary 16 and then use the fact that deterministic communication complexity is monotone under restriction. $\square$

The two-copy case is still special: the first bit produces only a few branches, and each of them closes immediately. For larger interlaces, the same first-bit analysis does not close on its own. A row move may split the outer copies very unevenly, leaving one child with many more copies than the other, while a column move lowers the surviving density. In the unbalanced row case, the extra copies are not wasted: by maximum projection we can trade them for better column density. The next lemma states the one-step argument in the form we will use later.

2.5 The One-Step Partition Lemma

The precise statement is a first-bit minimum over three possibilities: a degenerate row split keeps all copies in one child, a genuine row split produces two smaller interlaces whose column densities multiply to at least y , and a column split keeps the same number of copies while halving the column parameter. We use this raw one-step lemma, the cleaner row-step and density-amplification consequences will be re-extracted from it afterwards.

Lemma 18 (One-step partition). *Let M be a Boolean matrix, let $p \geq 1$, let $\sigma \in \{0, 1\}$, let $0 < x \leq \frac{1}{2}$, and let $0 < y \leq 1$. If*

$$D(M[2p + \sigma, 2x, y]) \geq 1,$$

then

$$D(M[2p + \sigma, 2x, y]) \geq 1 + \min \left(D(M[2p + \sigma, x, y]), \right. \\ \left. \min_{\substack{0 \leq \ell < p \\ a \in [0,1]}} \max \left(D(M[p + \ell + \sigma, x, y^a]), D(M[p - \ell, x, y^{1-a}]) \right), \right. \\ \left. D(M[2p + \sigma, 2x, y/2]) \right). \quad (1)$$

Proof. Fix

$$g = M \langle 2p + \sigma \rangle \upharpoonright_{R,C} M[2p + \sigma, 2x, y].$$

Since $D(g) \geq 1$, any optimal protocol for g has a first bit.

Assume first that the row player sends the first bit, so the root partitions the rows as $R = R_1 \sqcup R_2$. Apply Lemma 10 to g with this row partition. We obtain integers $\ell_1, \ell_2 \geq 0$ and densities $y_1, y_2 \in (0, 1]$ such that

$$\ell_1 + \ell_2 = 2p + \sigma, \quad y_1 y_2 \geq y,$$

and, for each $i \in \{1, 2\}$ with $\ell_i \geq 1$, the corresponding child contains a restricted submatrix, or subgame,

$$g_i \in M[\ell_i, x, y_i].$$

Relabel if needed so that $\ell_1 \geq \ell_2$.

If $\ell_2 = 0$, then $\ell_1 = 2p + \sigma$, and because $y_2 \leq 1$ we have $y_1 \geq y$. By monotonicity,

$$D(g) \geq 1 + D(M[2p + \sigma, x, y]).$$

Assume henceforth that $\ell_2 \geq 1$. Then

$$\ell_1 = p + \ell + \sigma, \quad \ell_2 = p - \ell$$

for some integer $0 \leq \ell < p$.

If $y = 1$, then $y_1 = y_2 = 1$, so we may choose any $a \in [0, 1]$. Otherwise $0 < y < 1$, and $y_2 \leq 1$ implies $y_1 \geq y$. Define

$$a := \frac{\log y_1}{\log y} \in [0, 1].$$

Then $y_1 = y^a$, and the inequality $y_1 y_2 \geq y$ gives

$$y_2 \geq \frac{y}{y_1} = y^{1-a}.$$

Hence monotonicity gives

$$D(M[\ell_1, x, y_1]) \geq D(M[p + \ell + \sigma, x, y^a]), \quad D(M[\ell_2, x, y_2]) \geq D(M[p - \ell, x, y^{1-a}]).$$

Since one protocol bit has already been spent,

$$D(g) \geq 1 + \max \left(D(M[p + \ell + \sigma, x, y^a]), D(M[p - \ell, x, y^{1-a}]) \right).$$

If instead the first bit is sent by the column player, then one child keeps at least half of the columns and therefore contains a matrix in

$$M[2p + \sigma, 2x, y/2].$$

By monotonicity,

$$D(g) \geq 1 + D(M[2p + \sigma, 2x, y/2]).$$

Since g was arbitrary, taking the minimum over the possible first moves and over the admissible row-split parameters (ℓ, a) yields (1). $\square$

The proof again starts from the protocol's first bit. A column bit lowers the surviving density by a factor of 2, exactly as in the two-copy ladder. A row bit is handled by Lemma 10: each child inherits some number of surviving outer copies and some column parameter, with the product of the two column parameters still at least y . The displayed minimum in (1) records the three resulting cases: a degenerate row-first branch, a genuine row split ranging over the compatible parameters (ℓ, a) , and a column-first branch that halves the surviving column density. In the next steps, we will extract from this raw statement the cleaner bundled consequences that we actually use.

2.6 The Odd-Copy Seed

The three-rung quantity Λ_M . Write

$$\Lambda_M(p, x, y) := \min_{0 \leq j < 3} (j + D(M[p, x, y/2^j])).$$

In the odd-copy regime, the one-step consequence is not a scalar top-rung bound. A first column bit drops the density by a factor of 2, so the proof naturally couples the top density y^2 with the neighbouring rungs $y^2/2$ and $y^2/4$. We therefore state the odd-copy family in the same three-rung quantity Λ_M that is also used in the later iteration. The actual copy-halving step is supplied by the bundled row-step lemma proved later in the appendix, Lemma 70.

Lemma 19. *Let M be (δ, b) -robust, and assume $D(M) \geq 2$. Write*

$$y := \frac{1}{2} + \delta, \quad y_0 := y^2.$$

Then for every integer ℓ with $2 \leq \ell \leq b$,

$$\Lambda_M(2^{\ell-1} + 1, 2^{\ell-b}, y_0) \geq D(M) + \ell.$$

Proof. The proof is given in Appendix B. $\square$

2.7 The $2^{k-1} + 1$ Family

The previous lemma is the odd-copy statement. The familiar scalar lower bound is just its top rung. At each stage one extra copy survives the copy-halving step, so we gain one more bit of complexity while keeping the two-copy density $(\frac{1}{2} + \delta)^2$. This family does *not* reduce the column density below y_0 . Thus it supplies constant-density odd-copy seeds, but not yet the much smaller column densities needed later for the hard-seed construction below.

Corollary 20. *Let M be (δ, b) -robust, and assume $D(M) \geq 2$. Then for every integer k with $2 \leq k \leq b$,*

$$D\left(M\left[2^{k-1} + 1, 2^{k-b}, \left(\frac{1}{2} + \delta\right)^2\right]\right) \geq D(M) + k.$$

Proof. Write $y_0 := (\frac{1}{2} + \delta)^2$. By Lemma 19,

$$\Lambda_M(2^{k-1} + 1, 2^{k-b}, y_0) \geq D(M) + k.$$

Since $\Lambda_M(2^{k-1} + 1, 2^{k-b}, y_0)$ is the minimum of three terms whose first term is

$$D\left(M \left[2^{k-1} + 1, 2^{k-b}, y_0\right]\right),$$

that top rung is at least the minimum. Therefore

$$D\left(M \left[2^{k-1} + 1, 2^{k-b}, y_0\right]\right) \geq D(M) + k.$$

□

Corollary 20 does not amplify the column parameter any further. Rather, it is the top rung of the bundled odd-copy seed: each application of the bundled copy-halving step passes from $2p + 1$ copies to $p + 1$ copies while preserving the three neighbouring density rungs. The extra copy survives through every halving step and yields one extra bit at the end, while the density stays fixed at y_0 .

2.8 The Iterated Density-Amplification Lemma

The two-copy analysis already showed the basic phenomenon we ultimately need: interlacing can raise the lower bound while also forcing the column parameter down from y to y^2 . The odd-copy family above instead uses the bundled copy-halving step: it preserves one extra copy at each halving step while keeping the improved two-copy density y_0 . That gives useful constant-density seeds, but it is still not enough for the hard-seed construction. There we will need lower bounds at column densities that are far smaller than y_0 .

So the next task is to return to the raw one-step partition bound and iterate its density-amplifying consequence. Starting from a seed family $M[p, x, y]$, we repeatedly trade surplus copies for better column density. The difficulty is that a column move can immediately halve the surviving density. In the two-copy proof we handled that with the three rungs at densities y , $y/2$, and $y/4$. The third rung is already needed there: the lowest rung at density $y/4$ is the one that closes directly by maximum projection to a one-copy benchmark. Once that rung is established, it yields the rung at density $y/2$, and that in turn yields the top rung at density y . Once we begin iterating, the same pattern appears at every stage, so an induction that tracks only the top density y is no longer stable.

To make the iteration stable, we therefore track together the lower bounds at those same three nearby densities. The proof still comes from repeated use of Lemma 18; the bundled rungs ensure that the induction closes after column moves without restarting from scratch.

The iterated density-amplification lemma will be stated for an arbitrary seed size p . The parameter p records the input family we start from, while the iteration parameter counts how many times we repeat the one-step argument. Later, the constant-density seeds constructed above, such as the family $2^{k-1} + 1$, will serve as natural choices of p for that theorem.

Recall the three-rung quantity $\Lambda_M(p, x, y)$ from the paragraph above. It records, in one number, the lower bounds at densities y , $y/2$, and $y/4$, with a penalty of one bit or two bits for dropping to the lower rungs. Λ_M tracks the three neighbouring density lower bounds so that the induction closes after repeated column moves. By monotonicity of bracket complexity in the parameters (p, x, y) , the quantity $\Lambda_M(p, x, y)$ is monotone in the same sense.

The iterated density-amplification lemma can now be stated cleanly. The parameter p records the seed family we start from, while k and s count how many times we iterate the one-step argument and how aggressively we trade copies for column density.

Lemma 21 (Iterated density amplification). *Let $2 < \rho$, and set*

$$\beta := \frac{\rho - 1}{\rho - 2}.$$

For any matrix M , non-negative integers $0 \leq s \leq k$, $p \geq 1$, and reals $0 < x \leq 2^{-k}$, $0 < y \leq 1$, if

$$D(M[p, x, y/4]) \geq 1 \quad \text{and} \quad (\rho - 1)^k \leq \rho^{k-s},$$

then

$$\Lambda_M\left(\left\lfloor 2^k \beta^s p \right\rfloor, 2^k x, y^{\rho^s}\right) \geq k + \Lambda_M(p, x, y), \quad (2)$$

$$D\left(M\left[\left\lfloor 2^k \beta^s p \right\rfloor, 2^k x, y^{\rho^s}\right]\right) \geq k + \Lambda_M(p, x, y). \quad (3)$$

The first inequality is the main content. The second is its immediate one-rung consequence, since Λ_M is defined as a minimum whose top rung is the plain bracket lower bound.

The full proof is given in Section A; here we record only the proof roadmap.

Proof roadmap. The proof proceeds as follows.

First, we record the auxiliary one-step bound that underlies Lemma 18 and use it to derive two bundled lemmas. The first is a copy-halving step: it passes from $2p + \sigma$ copies down to $p + \sigma$ copies while preserving the three-rung bounds. The second is a density-amplifying step: it trades surplus copies for lower column density, again at all three rungs simultaneously.

Next, we arrange the target statements on a grid. One coordinate counts how many row-doubling steps have been taken, and the other counts how many density-amplifying stages have been taken. Two elementary arithmetic inequalities on this grid show that the bundled copy-halving and density-amplifying steps produce exactly the predecessor values needed for the induction.

Finally, we verify that the weak terminal lower bound $D(M[p, x, y/4]) \geq 1$ holds throughout the grid. Once that safety check is in place, an induction on the grid gives (2), and (3) follows immediately.

The following scalar consequence is the form we will actually use in later applications.

Corollary 22. *Under the hypotheses of Lemma 21, suppose in addition that for some real H ,*

$$D(M[p, x, y]) \geq H, \quad D(M[p, x, y/2]) \geq H - 1, \quad D(M[p, x, y/4]) \geq H - 2.$$

Then

$$D\left(M\left[\left\lfloor 2^k \beta^s p \right\rfloor, 2^k x, y^{\rho^s}\right]\right) \geq k + H.$$

Proof. The three displayed hypotheses are exactly the statement

$$\Lambda_M(p, x, y) \geq H.$$

Apply (3). □

3 Protocol Control for Classical Interlace

The lower bounds above show that the protocol needs many bits. We now show something stronger: the protocol must spend those bits in a constrained order. Specifically, the first $\log q$ bits must all be row bits that separate the outer blocks, and no column bit can be spent during that phase without exceeding the budget. The key input is the two-copy lower bound from Corollary 16, which rules out row splits that fail to separate.

Corollary 23 (Power-of-two lower bound). *Let M be (δ, b) -robust with $b \geq 1$, assume $D(M) \geq 2$, and write $y_0 := (\frac{1}{2} + \delta)^2$. If $u \geq 2$ is a power of two with $u \leq 2^b$, then*

$$D\left(M\left[u, u2^{-b}, y_0\right]\right) \geq D(M) + \log u.$$

Proof. Write $u = 2^\ell$ with $\ell \geq 1$. Apply Corollary 22 with $p = 2$, $k = \ell - 1$, $s = 0$, $x = 2^{1-b}$, $y = y_0$, $\rho := 3$, and $H = D(M) + 1$. Since $u \leq 2^b$, we have $2^{1-b} \leq 2^{-k}$. The seed hypotheses follow from Corollary 15:

$$D\left(M\left[2, 2^{1-b}, y_0\right]\right) \geq H, \quad D\left(M\left[2, 2^{1-b}, y_0/2\right]\right) \geq H-1, \quad D\left(M\left[2, 2^{1-b}, y_0/4\right]\right) \geq H-2 \geq 1.$$

The numerical side condition is automatic because $s = 0$. Since $2^k p = u$, $2^k x = u2^{-b}$, and $y_0^{\rho^0} = y_0$, Corollary 22 gives $D\left(M\left[u, u2^{-b}, y_0\right]\right) \geq D(M) + \log u$. $\square$

3.1 Row-Only Separation

We begin with a purely row-combinatorial lemma. It does not use any lower-bound machinery; it only records what a row-only protocol partition must look like once no rectangle remains simultaneously heavy in two different outer row blocks. The lower-bound input comes in the following lemma: if a row-only restriction still leaves two outer row blocks above the threshold 2^{-b+1} , and the surviving column density is still at least $(\frac{1}{2} + \delta)^2$, then that rectangle already contains a hard two-copy descendant.

Lemma 24 (No-waste row-only partition). *Let Q be a finite set with $|Q|$ a power of two, let X be a finite set, and let $R^{\text{in}} \subseteq Q \times X$ be a (Q, T_0) -equipartitioned row set. Consider a deterministic protocol that, starting from R^{in} , communicates exactly $\log |Q|$ row bits and zero column bits. Let $\mathcal{P}$ denote the resulting partition into rectangles $R \times C$. Assume that no rectangle $R \times C \in \mathcal{P}$ contains T rows from two distinct outer row blocks. If $|Q|T < T_0$, then every rectangle $R \times C \in \mathcal{P}$ has a unique outer row block $i^* \in Q$ such that, writing*

$$R_i := R \cap (\{i\} \times X),$$

one has

$$|R_{i^*}| \geq T_0 - (|Q| - 1)T, \quad |R_i| < T \text{ for all } i \neq i^*.$$

Proof. Because the protocol uses exactly $\log |Q|$ row bits and zero column bits, it induces a partition of R^{in} into at most $2^{\log |Q|} = |Q|$ nonempty row parts.

Fix $i \in Q$. If every row part contained fewer than T rows from the i -th outer row block, then the total number of rows from that block in R^{in} would be less than $|Q|T < T_0$, contradicting the assumption that R^{in} is (Q, T_0) -equipartitioned. Hence for each $i \in Q$, some rectangle in $\mathcal{P}$ contains at least T rows from the i -th outer row block.

By hypothesis, no rectangle contains T rows from two distinct outer row blocks. Therefore different indices $i \in Q$ must be assigned to different rectangles. Since there are $|Q|$ outer row blocks and at most $|Q|$ nonempty rectangles, this assignment is a bijection. In particular, every rectangle $R \times C \in \mathcal{P}$ has a unique outer row block i^* with $|R_{i^*}| \geq T$, and for every $i \neq i^*$ one has $|R_i| < T$.

Now fix a rectangle $R \times C \in \mathcal{P}$ and let i^* be its unique heavy outer row block (the one with at least T rows). Across the other $|Q| - 1$ rectangles, the i^* -th block contributes fewer than T rows to each. Since the total number of rows from the i^* -th block in R^{in} is at least T_0 , it follows that

$$|R_{i^*}| \geq T_0 - (|Q| - 1)T.$$

This is exactly the claimed conclusion. $\square$

The previous lemma is purely combinatorial. Combined with the two-copy lower bound, it yields the following consequence for row-only protocols on bracket restrictions.

Lemma 25 (Failure of near-exact separation gives a hard child). *Let M be (δ, b) -robust, assume in addition that $b \geq 1$, write $m := |\text{Rows}(M)|$ and $n := |\text{Cols}(M)|$, let q be a power of two, let $0 < x \leq 1$, let*

$$\left(\frac{1}{2} + \delta\right)^2 \leq y \leq 1,$$

and let

$$N \in M[q, x, y].$$

Write

$$T_0 := \lceil mx \rceil \quad \text{and} \quad T := \left\lceil 2^{-b+1}m \right\rceil.$$

Consider a deterministic protocol on N that communicates exactly $\log q$ row bits and zero column bits, and let $\mathcal{P}$ denote the resulting partition into rectangles $R \times C$. Assume $qT < T_0$. If the conclusion of Lemma 24 fails for $\mathcal{P}$, then some rectangle $R \times C \in \mathcal{P}$ satisfies

$$D(N \upharpoonright_{R,C}) \geq D(M) + 1.$$

Proof. If the conclusion of Lemma 24 fails, then by the contrapositive of that lemma some rectangle $R \times C \in \mathcal{P}$ contains at least T rows from two distinct outer row blocks, say $i_1, i_2 \in [q]$. Because the protocol spends zero column bits, every such rectangle has the full column set of N . Let K denote that column set. Since $N \in M[q, x, y]$, we have

$$|K| \geq n^q y.$$

Applying Lemma 11 to $N \upharpoonright_{R,C}$ with $Q = \{i_1, i_2\}$, $T = \lceil 2^{-b+1}m \rceil$, and $x = 2^{-b+1}$ shows that $N \upharpoonright_{R,C}$ contains a matrix in

$$M \left[2, 2^{-b+1}, y \right].$$

Hence restriction monotonicity and Corollary 16 give

$$D(N \upharpoonright_{R,C}) \geq D \left(M \left[2, 2^{-b+1}, y \right] \right) \geq D(M) + 1.$$

$\square$

3.2 The Near-Exact Separation Lemma

We can now record the corresponding separation consequence for bracket restrictions in the restricted regime $\delta \leq \frac{1}{\sqrt{2}} - \frac{1}{2}$.

Lemma 26 (Classical near-exact separation). *Let M be (δ, b) -robust, write $m := |\text{Rows}(M)|$, assume $D(M) \geq 2$, and assume*

$$0 < \delta \leq \frac{1}{\sqrt{2}} - \frac{1}{2},$$

let $q \geq 2$ be a power of two, let $0 < x \leq 1$, and let

$$2\left(\frac{1}{2} + \delta\right)^2 \leq y \leq 1.$$

Assume

$$q2^{-b} \leq x \quad \text{and} \quad q \left\lceil 2^{-b+1}m \right\rceil < \lceil mx \rceil,$$

and let

$$N \in M[q, x, y].$$

Then every deterministic protocol for N of cost at most $D(M) + \log q$ has the following two properties. On every branch, its first $\log q$ bits are row bits.

Let $\mathcal{P}$ denote the resulting partition after those first $\log q$ bits. Then each surviving rectangle $R \times C \in \mathcal{P}$ has a unique outer row block $i^ \in [q]$ such that, writing*

$$R_i := R \cap (\{i\} \times \text{Rows}(M)),$$

one has

$$|R_{i^*}| \geq \lceil mx \rceil - (q-1) \left\lceil 2^{-b+1}m \right\rceil, \quad |R_i| < \left\lceil 2^{-b+1}m \right\rceil \text{ for all } i \neq i^*.$$

Proof sketch. The argument is an induction on the protocol's first $\log q$ bits. Write $q_t := q/2^t$ and $x_t := x/2^t$. The inductive claim is that after the first t bits, every surviving rectangle still contains a restriction in $M[q_t, x_t, y]$.

At each step t , three alternatives are ruled out in order:

- (i) *Termination.* The surviving restriction has complexity at least $D(M) + \log q_t$ (by the power-of-two lower bound), so the rectangle cannot yet be monochromatic.
- (ii) *Column bit.* A column bit halves the surviving column set, leaving a restriction of complexity at least $D(M) + \log q_t$, but only $D(M) + \log q_t - 1$ bits remain in the budget—a contradiction.
- (iii) *Unbalanced row split* (for $q_t \geq 4$). If one child receives more than $q_t/2$ of the assigned outer blocks, then Corollary 20 gives it complexity at least $D(M) + \log q_t$, again exceeding the remaining budget.

When $q_t = 2$, the unbalanced-row alternative disappears: the induction is already at the final two-copy stage, and the existing no-waste closure finishes the proof. Therefore each step must be a balanced row split: each child inherits exactly $q_t/2$ blocks with the full column set, and coordinate projection places it in $M[q_{t+1}, x_{t+1}, y]$.

After all $\log q$ row bits, the partition into rectangles is row-only. If any rectangle still had two outer blocks above the threshold T , then Lemma 25 would produce a two-copy descendant of complexity $D(M) + 1$, contradicting the remaining depth $D(M)$. Hence every rectangle has a unique dominant block, and the quantitative row bounds follow from Lemma 24.

Proof. Write

$$y_0 := \left(\frac{1}{2} + \delta\right)^2, \quad T := \left\lceil 2^{-b+1}m \right\rceil, \quad \text{and} \quad L := D(M) + \log q.$$

For $0 \leq t \leq \log q$, define

$$q_t := q/2^t \quad \text{and} \quad x_t := x/2^t.$$

Since $q2^{-b} \leq x \leq 1$, we have $q \leq 2^b$, and hence $q_t \leq q \leq 2^b$ for every t .

We claim that for each $t \in \{0, \dots, \log q - 1\}$, every surviving rectangle after the first t protocol bits contains a restriction in

$$M[q_t, x_t, y].$$

We prove this by induction on t .

For $t = 0$, the claim is exactly the hypothesis $N \in M[q, x, y]$.

Now fix $t < \log q - 1$, and assume the claim holds at depth t . Let S be any surviving rectangle after the first t bits, and choose a restriction

$$N_S \in M[q_t, x_t, y]$$

contained in S .

First, S cannot already be a leaf of the protocol tree. Indeed, q_t is a power of two, and

$$x_t = \frac{x}{2^t} \geq \frac{q2^{-b}}{2^t} = q_t2^{-b}.$$

Since q_t is a power of two with $q_t \leq 2^b$, Corollary 23 gives

$$D\left(M[q_t, q_t2^{-b}, y_0]\right) \geq D(M) + \log q_t.$$

Since $y \geq y_0$, monotonicity therefore gives

$$D(N_S) \geq D\left(M[q_t, q_t2^{-b}, y_0]\right) \geq 1.$$

So S is not monochromatic, and the protocol must query another bit at S .

We next show that this next bit cannot be a column bit. If it were, then one of the two children would retain at least half of the columns of N_S , and hence would contain a restriction in

$$M[q_t, x_t, y/2].$$

Since q_t is a power of two with $q_t \leq 2^b$, Corollary 23 gives

$$D\left(M[q_t, q_t2^{-b}, y_0]\right) \geq D(M) + \log q_t.$$

Because $y/2 \geq y_0$, monotonicity would then give that child communication complexity at least

$$D\left(M[q_t, q_t2^{-b}, y_0]\right)$$

But after $t + 1$ bits, the remaining depth available under the bound $L = D(M) + \log q$ is

$$L - (t + 1) = D(M) + \log q - (t + 1) = D(M) + \log q_t - 1,$$

a contradiction. So the next bit at S must be a row bit.

Consider the split of the row set of N_S into the two children of that row bit. Each of the q_t outer row blocks contributing to N_S has at least $\lceil mx_t \rceil$ rows, so one child contains at least $\lceil \lceil mx_t \rceil / 2 \rceil \geq \lceil mx_t / 2 \rceil$ rows from that block. Assign each outer row block to such a child. Because this is a row split, each child inherits the full column set of N_S , and hence still has at least $n^{q_t}y$ columns.

Suppose one child receives more than $q_t/2$ assigned blocks. Since q_t is a power of two, that child then receives at least $q_t/2 + 1$ assigned blocks. Applying Lemma 11 to that child with $T = \lceil mx_t/2 \rceil$ and $x = x_t/2$, and then reindexing the selected outer row blocks by $\lfloor q_t/2 + 1 \rfloor$, we see that it contains a submatrix canonically identified with an element of

$$M \lfloor q_t/2 + 1, x_t/2, y \rfloor.$$

Let $k := \log q_t$. Since $t < \log q - 1$, we have $q_t \geq 4$, so $k \geq 2$. Then $q_t/2 + 1 = 2^{k-1} + 1$, and

$$q \lfloor 2^{-b+1}m \rfloor \leq \lceil mx \rceil - 1 < mx,$$

so $x > q2^{-b+1}$. Therefore

$$\frac{x_t}{2} = \frac{x}{2^{t+1}} > \frac{q2^{-b+1}}{2^{t+1}} = 2^{k-b}.$$

Since $y \geq y_0$, monotonicity and Corollary 20 give that child communication complexity at least

$$D \left(M \left[2^{k-1} + 1, 2^{k-b}, y_0 \right] \right) \geq D(M) + k = D(M) + \log q_t.$$

Again this contradicts the remaining depth $L - (t + 1) = D(M) + \log q_t - 1$. Therefore neither child can receive more than $q_t/2$ assigned blocks.

Since there are q_t assigned blocks in total, each child receives exactly $q_t/2$ of them. Applying Lemma 11 once more with $T = \lceil mx_t/2 \rceil$ and $x = x_t/2$, and reindexing the selected outer row blocks by $\lfloor q_t/2 \rfloor$, each child contains a submatrix canonically identified with an element of

$$M \lfloor q_t/2, x_t/2, y \rfloor = M \lfloor q_{t+1}, x_{t+1}, y \rfloor.$$

This proves the inductive step, and hence the claim.

In particular, every surviving rectangle after the first $\log q - 1$ bits contains a restriction in

$$M \lfloor 2, x_{\log q-1}, y \rfloor.$$

Now take $t = \log q - 1$. At this final stage there is no longer an unbalanced-split case to exclude; it remains only to rule out termination and a column query. First,

$$x_{\log q-1} = \frac{x}{q/2} \geq 2^{1-b},$$

so monotonicity and Corollary 16 show that no surviving rectangle at this stage can already be a leaf. Fix one chosen restriction

$$N_S \in M \lfloor 2, x_{\log q-1}, y \rfloor$$

inside the current surviving rectangle. If the next bit were a column bit, then one of the two children would retain at least half of the columns of N_S , and hence would contain a restriction in

$$M \lfloor 2, x_{\log q-1}, y/2 \rfloor.$$

Because $y/2 \geq y_0$, monotonicity and Corollary 16 would then give that child communication complexity at least

$$D\left(M\left[2, 2^{1-b}, y_0\right]\right) \geq D(M) + 1.$$

But after $\log q$ bits, the remaining depth allowed under the bound $L = D(M) + \log q$ is only $D(M)$, a contradiction. Therefore the $\log q$ -th bit is also a row bit on every branch. The first $\log q$ bits of the protocol are therefore row bits on every branch. Let $\mathcal{P}$ be the partition into rectangles after those first $\log q$ bits.

Since $N \in M[q, x, y]$, its row set is $([q], [mx])$ -equipartitioned. If the conclusion of the lemma failed for $\mathcal{P}$, then Lemma 25 would yield a rectangle $R \times C \in \mathcal{P}$ with

$$D(N|_{R,C}) \geq D(M) + 1.$$

But after $\log q$ bits, the remaining depth allowed under the bound $L = D(M) + \log q$ is only $D(M)$, a contradiction. Therefore every rectangle in $\mathcal{P}$ satisfies the stated separation conclusion. $\square$

This completes the classical theory. We now have two tools: the iterated density-amplification lemma (Lemma 21), which gives lower bounds tolerant of extreme column degradation, and the separation lemma (Lemma 26), which constrains the protocol's bit-spending pattern to row-first separation of the outer blocks. Both are stated for classical interlace, which has exponential column sets. The next section transfers them to polynomial-size column sets via relaxed interlacing.

4 Relaxing the Interlace Operation

In the classical interlace $M\langle q \rangle$, Bob's column set is the full Cartesian product Y^q . That gives the clean lower bounds and protocol-control statements proved in the previous section, but it is far too large for the reduction, where we need polynomial-size instances. The relaxed interlace replaces Y^q by a much smaller family of columns chosen so that every small set of coordinates still looks close to uniform.

This replacement is useful only if the classical arguments survive the loss of columns. The key point is that the classical proofs do not need the entire product structure all at once. They use only selected sets of coordinates, and they apply only after the row side has been organised into a sufficiently balanced form. In the relaxed setting we will therefore prove that, starting from a relaxed submatrix that still keeps enough rows across enough outer blocks and enough columns, one can pass to a selected set of coordinates and obtain a classical restriction of the form needed by the lower-bound arguments, with a controlled loss in the column parameter.

This transfer is the first tool we need. We therefore begin by defining the balanced column families and the relaxed interlace operation, and then prove the structural lemmas that recover the needed classical restrictions from such relaxed submatrices.

4.1 Balanced Column Families and the Relaxed Interlace

We begin by formalising the smaller column families that will replace the full product Y^q . The requirement is that every projection onto at most t coordinates is ε -close to uniform.

Definition 27 ((q, t)-Balanced column family). *Let $q, t \geq 1$, let Y be a finite set, and let $\varepsilon \in (0, 1)$. A finite indexed family $\mathcal{S} = (\mathbf{y}^{(1)}, \dots, \mathbf{y}^{(N)})$ of elements of Y^q is (q, t)-balanced with accuracy ε if for every $J \subseteq [q]$ with $|J| \leq t$ and every pattern $\mathbf{a} \in Y^J$,*

$$\Pr_{\mathbf{y} \sim \mathcal{S}}[\mathbf{y}|_J = \mathbf{a}] = \frac{1}{|Y|^{|J|}} \pm \frac{\varepsilon}{|Y|^{|J|}}.$$

Here $\mathbf{y} \sim \mathcal{S}$ means that one chooses an index uniformly from $[N]$ and returns the corresponding tuple $\mathbf{y}^{(j)}$. We write $S_{q,t}(Y)$ for such a family and $\varepsilon_{q,t}$ for its accuracy parameter. Repeated tuples are allowed and count as distinct columns.

Remark 28 (Explicit balanced families). *Standard explicit constructions of almost t -wise independent sample spaces over finite alphabets, such as the one due to Alon et al. [1], yield (q, t) -balanced column families of size at most*

$$\text{poly}(q) |Y|^{O(t)} \varepsilon^{-O(1)}.$$

Moreover, such a family can be constructed deterministically in time polynomial in this bound, and hence in time

$$\text{poly}(q) |Y|^{O(t)} \varepsilon^{-O(1)}.$$

Throughout the paper we will take

$$\varepsilon_{q,t} = (2qt)^{-c}$$

for a sufficiently large absolute constant $c > 0$, so that

$$|S_{q,t}(Y)| = \text{poly}(q, |Y|^t).$$

In particular, for this choice of $\varepsilon_{q,t}$, the family $S_{q,t}(Y)$ is computable in time $\text{poly}(q, |Y|^t)$.

With such a balanced column family fixed, the relaxed interlace is obtained by restricting Bob's domain to that family.

Definition 29 (Relaxed interlace). *Let $M : X \times Y \rightarrow \{0, 1\}$ be a Boolean matrix, let $q \geq t \geq 1$, and let*

$$\mathcal{S} = (\mathbf{y}^{(1)}, \dots, \mathbf{y}^{(L)}) = S_{q,t}(Y)$$

be a (q, t) -balanced column family in Y^q , where $\mathbf{y}^{(j)} = (y_1^{(j)}, \dots, y_q^{(j)})$. The relaxed interlace is the matrix

$$M \langle q, \mathcal{S} \rangle : ([q] \times X) \times [L] \rightarrow \{0, 1\}$$

defined by

$$M \langle q, \mathcal{S} \rangle ((i, x), j) := M(x, y_i^{(j)}).$$

Thus the columns are indexed by the members of $\mathcal{S}$, with multiplicity.

Lemma 30 (Projection closure of balanced families). *Let $q, t \geq 1$, let Y be finite, let*

$$\mathcal{S} = (\mathbf{y}^{(1)}, \dots, \mathbf{y}^{(N)}) \subseteq Y^q$$

be (q, t) -balanced with accuracy ε , and let $J \subseteq [q]$. Then the projected indexed family

$$\pi_J(\mathcal{S}) := (\mathbf{y}^{(1)}|_J, \dots, \mathbf{y}^{(N)}|_J) \subseteq Y^{|J|}$$

is $(|J|, t)$ -balanced with the same accuracy ε .

Proof. Let $K \subseteq [|J|]$ with $|K| \leq t$, and let $\mathbf{a} \in Y^K$. Identify K with the corresponding subset $J_K \subseteq J \subseteq [q]$. Then

$$\Pr_{\mathbf{z} \sim \pi_J(\mathcal{S})} [\mathbf{z}|_K = \mathbf{a}] = \Pr_{\mathbf{y} \sim \mathcal{S}} [\mathbf{y}|_{J_K} = \mathbf{a}].$$

Since $\mathcal{S}$ is (q, t) -balanced with accuracy ε , the right-hand side equals

$$\frac{1}{|Y|^{|K|}} \pm \frac{\varepsilon}{|Y|^{|K|}}.$$

This is exactly the $(|J|, t)$ -balanced condition for $\pi_J(\mathcal{S})$. □

4.2 From Relaxed to Classical Interlace

The classical lower bounds and protocol-control statements from the previous section are stated for the bracket families $M[t, x, y]$, which live inside the full classical interlace $M\langle t \rangle$. To apply them in the relaxed setting, we need a bridge: given a subgame of the relaxed interlace whose row set is (Q, T) -equipartitioned for some $Q \subseteq [q]$ with $|Q| = t$, and whose surviving column set is still large, we want to extract a classical bracket witness with a controlled column-density loss.

The transfer consists of two lemmas, which we state separately. The first is purely combinatorial: it extracts balanced row subsets from arbitrary row selections. The second is specific to relaxed interlacing: once the relevant t coordinates have been fixed, it projects the balanced column family onto those coordinates and uses the almost-independence guarantee to control the density loss.

Lemma 31 (Balancing-by-blocks). *Let M be an $m \times n$ matrix, let $q \geq 1$, let $x \in (0, 1)$, and set $T := \lceil mx \rceil$. Call block $\{i\} \times \text{Rows}(M)$ in the interlaced row set $[q] \times \text{Rows}(M)$ full if it contains at least T rows from a given subset. Let $R' \subseteq [q] \times \text{Rows}(M)$ be any row subset with density $\beta := |R'|/(qm) \geq x$, and set $k := \lceil q(\beta - x)/(1 - x) \rceil$. Then there exists $J \subseteq [q]$ with $|J| = k$ such that $R' \cap (J \times \text{Rows}(M))$ is (J, T) -equipartitioned.*

Proof. For each block $i \in [q]$, let $r_i := |R' \cap (\{i\} \times \text{Rows}(M))|$. A block is deficient if $r_i < T$, so each deficient block contributes at most $T - 1$ rows. Let Q' be the set of deficient blocks and let $J_{\text{full}} := [q] \setminus Q'$ be the set of full blocks. Counting total rows gives

$$qm\beta = |R'| \leq |J_{\text{full}}| \cdot m + |Q'| \cdot (T - 1) = qm - |Q'|(m - T + 1).$$

Since $T = \lceil mx \rceil \leq mx + 1$, we have $m - T + 1 \geq m(1 - x)$, so

$$|Q'| \leq \frac{q(1 - \beta)}{1 - x} \quad \text{and therefore} \quad |J_{\text{full}}| \geq \frac{q(\beta - x)}{1 - x}.$$

Since $|J_{\text{full}}|$ is an integer, this implies $|J_{\text{full}}| \geq k$. Pick any k -element subset $J \subseteq J_{\text{full}}$. $\square$

This is a general extraction principle for interlaced matrices, classical or relaxed. In particular, if a row set has total density $\beta \geq x$, then the lemma produces $\lceil q(\beta - x)/(1 - x) \rceil$ blocks on which the surviving rows are individually dense. This is the form in which we will use it later.

Lemma 32 (Relaxed-to-classical transfer). *Let M be an $m \times n$ matrix, let $q \geq t \geq 1$, and let*

$$\mathcal{S} = (\mathbf{y}^{(1)}, \dots, \mathbf{y}^{(L)}) = S_{q,t}(\text{Cols}(M))$$

be (q, t) -balanced with accuracy $\varepsilon := \varepsilon_{q,t}$, where $\mathbf{y}^{(j)} = (y_1^{(j)}, \dots, y_q^{(j)})$. Write

$$N := M\langle q, \mathcal{S} \rangle.$$

Fix $Q \subseteq [q]$ with $1 \leq |Q| \leq t$, write $u := |Q|$, let $x \in (0, 1]$, set $T := \lceil mx \rceil$, and let $R' \subseteq [q] \times \text{Rows}(M)$ and $C' \subseteq [L]$ satisfy:

- (a) R' is (Q, T) -equipartitioned, and
- (b) $|C'|/L \geq y$ for some $y \in (0, 1]$.

Then there exist $R^ \subseteq R'$ and $C^* \subseteq C'$ such that the submatrix $N \downarrow_{R^*, C^*}$ is canonically identified with an element of $M[u, x, y/(1 + \varepsilon)]$.*

Proof. Step 1: trim the rows. Discard all rows outside Q , and from each remaining block $\{q'\} \times \text{Rows}(M)$ with $q' \in Q$, keep exactly T rows. Call the surviving set R^* . Then R^* is (Q, T) -equipartitioned and $|R^*| = uT$.

Step 2: project the columns. Write $Q = \{q_1 < \dots < q_u\}$, and define

$$\pi_Q : [L] \rightarrow \text{Cols}(M)^u, \quad \pi_Q(j) := (y_{q_1}^{(j)}, \dots, y_{q_u}^{(j)}).$$

Since $u \leq t$, balancedness means that every pattern $\mathbf{a} \in \text{Cols}(M)^u$ has probability at most $(1+\varepsilon)/n^u$, so every such pattern has at most

$$\frac{1+\varepsilon}{n^u} L$$

preimages under π_Q . Since $|C'| \geq yL$, the number of distinct patterns appearing in $\pi_Q(C')$ is at least

$$\frac{|C'|}{(1+\varepsilon)L/n^u} \geq \frac{y}{1+\varepsilon} n^u.$$

Since the number of distinct patterns is an integer, there are at least $\lceil (y/(1+\varepsilon)) n^u \rceil$ of them. Pick one index from C' for each distinct pattern and trim to a set $C^* \subseteq C'$ of exact size $\lceil (y/(1+\varepsilon)) n^u \rceil$.

Step 3: identify the subgame. Relabel the ordered set $Q = \{q_1, \dots, q_u\}$ by $[u]$. Under this relabelling, R^* becomes a row set in $[u] \times \text{Rows}(M)$ with exactly T rows in each block, and each index $j \in C^*$ is identified with the projected tuple $\pi_Q(j) \in \text{Cols}(M)^u$. The submatrix $N \upharpoonright_{R^*, C^*}$ is therefore canonically identified with a restriction of the classical interlace $M \langle u \rangle$ whose row set is $([u], T)$ -equipartitioned and whose column set has size at least $\lceil (y/(1+\varepsilon)) n^u \rceil$. Hence it is canonically identified with an element of $M[u, x, y/(1+\varepsilon)]$. $\square$

The only quantitative loss is the factor $y \mapsto y/(1+\varepsilon)$; with the choice $\varepsilon_{q,t} = (2qt)^{-c}$, this factor is subsumed in the parameter ranges used later.

A classical seed for the extension theorem. The bridge lemma tells us how to recover a classical bracket witness from a balanced relaxed submatrix. To turn that structural fact into a relaxed lower bound, we still need a classical witness that is already hard enough to tolerate the column-density loss that will appear later in the extension argument. The next lemma records exactly that input.

This lemma combines the two classical mechanisms already developed into the form used in the relaxed section. The bundled odd-copy seed from Lemma 19 provides the required three density rungs slightly above the $t/2$ threshold, while Lemma 21 reduces the column parameter to the much smaller density needed later. The scalar family *Corollary 20* is simply the top rung of that seed. The parameter j measures the initial surplus: the seed starts from the $2^{j-1} + 1$ -copy family and is then iterated until it reaches scale t .

Lemma 33. *Fix an integer $j \geq 2$. Then for all sufficiently large powers of two t , the following implication holds: if M is (δ, b) -robust with $D(M) \geq 3$ and $t \leq 2^b$, then*

$$D\left(M\left[\frac{(2^{j-1}+2)t}{2^j}, 2^{-b}t, 2^{-2^{0.49}\sqrt{\log t}}\right]\right) \geq D(M) + \log t.$$

This is a specialised application of Lemma 21. In Appendix B we first prove Lemma 19, which supplies the three density rungs required by Corollary 22, and then apply that corollary with $p = 2^{j-1} + 1$. The full proof is deferred to Appendix B.

4.3 Lower Bounds After Column Loss

Fix relaxed-interlace parameters $q \geq t \geq 1$ and write $\varepsilon_{q,t}$ for the corresponding balanced-family error term. Suppose a relaxed subgame starts with column density at least h , and that a branch of the protocol has already spent c column bits. Then the surviving relaxed column density is still at least $h 2^{-c}$. If we now apply the bridge on one chosen coordinate, the resulting one-copy classical density is

$$y_c(h) := \left(\frac{h 2^{-c}}{1 + \varepsilon_{q,t}} \right)^{1/t}.$$

This quantity appears in the one-copy lower bounds later on. We collect these lower bounds into one named condition.

Definition 34 (Column-loss resilient). *Let M be a Boolean matrix, and let $b \geq 1$. Fix powers of two $q \geq t \geq 1$ with $t \leq 2^b$, and let $h \in (0, 1]$. Using the densities $y_c(h)$ defined above, we say that (M, b) is (q, t, h) -column-loss resilient if the following two conditions hold:*

(i)

$$D \left(M \left[1, 2^{-b}, y_{\log q + D(M)}(h) \right] \right) \geq 1;$$

(ii) *for every pair of integers $0 \leq k \leq D(M)$ and $0 \leq c \leq \log t + k$,*

$$\Lambda_M(1, 2^{-b}, y_c(h)) \geq D(M) - k.$$

These are the lower-bound conditions we will need later. The first clause is the one-copy bound used at the terminal leaf of the extension argument. The second clause says that after any allowed amount of column loss, the resulting one-copy classical game is still hard enough for the remaining depth. The next lemma then upgrades that one-copy information to the two-copy form needed in the separation argument.

Lemma 35. *Assume $b \geq 1$, and let $0 < y \leq 1$. If*

$$\Lambda_M(1, 2^{-b}, y) \geq H,$$

then

$$\Lambda_M(2, 2^{-b+1}, y^2) \geq H + 1.$$

Proof. The hypothesis means exactly that

$$\begin{aligned} D \left(M \left[1, 2^{-b}, y \right] \right) &\geq H, \\ D \left(M \left[1, 2^{-b}, y/2 \right] \right) &\geq H - 1, \\ D \left(M \left[1, 2^{-b}, y/4 \right] \right) &\geq H - 2. \end{aligned}$$

Apply Lemma 12 with $x = 2^{-b}$. This gives

$$\begin{aligned} D \left(M \left[2, 2^{-b+1}, y^2 \right] \right) &\geq H + 1, \\ D \left(M \left[2, 2^{-b+1}, y^2/2 \right] \right) &\geq H, \end{aligned}$$

and

$$D\left(M\left[2, 2^{-b+1}, y^2/4\right]\right) \geq H - 1.$$

By the definition of Λ_M , these three inequalities say exactly that

$$\Lambda_M(2, 2^{-b+1}, y^2) \geq H + 1.$$

□

Corollary 36. *Let M be a Boolean matrix, let $b \geq 1$, fix powers of two $q \geq t \geq 1$ with $t \leq 2^b$, and let $h \in (0, 1]$. If (M, b) is (q, t, h) -column-loss resilient, then for every pair of integers $0 \leq k \leq D(M)$ and $0 \leq c \leq \log t + k$,*

$$\Lambda_M(2, 2^{-b+1}, y_c(h)^2) \geq D(M) - k + 1.$$

Proof. Column-loss resilience gives

$$\Lambda_M(1, 2^{-b}, y_c(h)) \geq D(M) - k$$

for every pair k, c in the stated range. Apply Lemma 35 with $y = y_c(h)$. □

4.4 The Extension Theorem

Notation summary. The table below collects the main symbols used in the extension and separation theorems. All items have been defined above; the table is provided for quick reference.

Symbol	Meaning
$D(f)$	deterministic communication complexity of f
$M \langle p \rangle$	p -fold classical interlace of M
$M \langle q, \mathcal{S} \rangle$	relaxed interlace with balanced column family $\mathcal{S}$
$M[p, x, y]$	bracket family: restrictions with $\geq \lceil mx \rceil$ rows per block, $\geq \lceil n^p y \rceil$ columns
$\Lambda_M(p, x, y)$	three-rung lower-bound quantity $\min_{0 \leq j < 3} (j + D(M[p, x, y/2^j]))$
$M \upharpoonright_{R, C}$	restriction of M to rows R , columns C
(δ, b) -robust	input hypothesis: M stays hard after controlled row/column deletion
$S_{q,t}(Y)$	(q, t) -balanced column family over alphabet Y
$\varepsilon_{q,t}$	accuracy parameter of the balanced column family
$y_c(h)$	one-copy classical density after c column bits: $(h 2^{-c} / (1 + \varepsilon_{q,t}))^{1/t}$

We can now state the relaxed lower-bound theorem. The one-copy lower bounds after column loss are bundled into Definition 34. The classical seed remains separate: in one application it may come from Lemma 33, while in another it may come from a stronger stage-specific lower bound. The statement below keeps that seed input abstract and does not mention any protocol budget.

Remark 37. *Only clause (i) of Definition 34 is used in the proofs of Theorem 38 and Corollary 39. We nevertheless state the surrounding transfer package with the full two-clause hypothesis because the Stage 2 certification lemmas in Section 5 are written uniformly in that form and are reused later without splitting the two clauses apart.*

Theorem 38 (Extension theorem). *Let M be a Boolean matrix, write $m := |\text{Rows}(M)|$, let $b \geq 1$, and let $t \leq 2^b$ and $r \geq 1$ be powers of two. Set*

$$q := rt, \quad \varepsilon := \varepsilon_{q,t}, \quad \widehat{M} := M \langle q, S_{q,t}(\text{Cols}(M)) \rangle.$$

Fix an integer p_{seed} and reals

$$2^{-b} \leq x_{\text{seed}} \leq 1, \quad h, h_{\text{seed}} \in (0, 1].$$

Assume

$$\frac{t}{2} \leq p_{\text{seed}} \leq t,$$

that (M, b) is (q, t, h) -column-loss resilient,

$$D(M[p_{\text{seed}}, x_{\text{seed}}, h_{\text{seed}}]) \geq D(M) + \log t,$$

and

$$\frac{h 2^{-(\log t + D(M))}}{1 + \varepsilon} \geq h_{\text{seed}}.$$

Let

$$N := \widehat{M} \upharpoonright_{R', C'}$$

be any submatrix such that

(i) R' is $(Q, \lceil r x_{\text{seed}} m \rceil)$ -equipartitioned for some $Q \subseteq [q]$ with $|Q| = r p_{\text{seed}}$,

(ii) $\frac{|C'|}{|\text{Cols}(\widehat{M})|} \geq h$.

Then

$$D(N) \geq D(M) + \log q.$$

Proof. Suppose for contradiction that N is computed by a deterministic protocol Π of depth at most

$$L - 1, \quad L := D(M) + \log q.$$

We build a root-to-leaf chain in the protocol tree together with, at each node on that chain, a further restriction of the current protocol rectangle that still retains many outer row blocks and many columns.

Write

$$Q_0 := Q, \quad T_0 := \lceil r x_{\text{seed}} m \rceil, \quad R_0 := R', \quad C_0 := C', \quad s_0 := 0, \quad c_0 := 0.$$

Starting from the root rectangle $\widehat{M} \upharpoonright_{R_0, C_0} = N$, define inductively a chain

$$\widehat{M} \upharpoonright_{R_0, C_0} \supseteq \widehat{M} \upharpoonright_{R_1, C_1} \supseteq \cdots \supseteq \widehat{M} \upharpoonright_{R_D, C_D}$$

ending at a leaf of Π , together with integers $s_i, c_i \geq 0$, subsets $Q_i \subseteq [q]$, and thresholds $T_i \geq 1$, as follows.

If the current node asks a column question, choose a child whose column set has size at least half the current one, and keep the same row restriction:

$$Q_{i+1} := Q_i, \quad T_{i+1} := T_i, \quad s_{i+1} := s_i, \quad c_{i+1} := c_i + 1.$$

If the current node asks a row question, then for each $q \in Q_i$, at least one child contains at least $\lceil T_i/2 \rceil$ rows from the block $\{q\} \times \text{Rows}(M)$. One of the two children therefore has this property

for at least $\lceil |Q_i|/2 \rceil$ values of q . Choose that child, choose a subset $Q_{i+1} \subseteq Q_i$ of exactly $\lceil |Q_i|/2 \rceil$ such outer blocks, and restrict further to those blocks:

$$T_{i+1} := \lceil T_i/2 \rceil, \quad s_{i+1} := s_i + 1, \quad c_{i+1} := c_i.$$

In either case we obtain a further restriction of the chosen child rectangle. By construction, R_i is (Q_i, T_i) -equipartitioned for every i , and since each step is either a row step or a column step, we have $i = s_i + c_i$ for every i .

By induction on i , every rectangle on this chain satisfies

$$|Q_i| \geq \left\lceil \frac{r p_{\text{seed}}}{2^{s_i}} \right\rceil, \quad T_i \geq \left\lceil \frac{r x_{\text{seed}} m}{2^{s_i}} \right\rceil, \quad \frac{|C_i|}{|\text{Cols}(\widehat{M})|} \geq h 2^{-c_i}. \quad (4)$$

We now distinguish two cases.

Case 1: the chain reaches a leaf before $\log r$ row bits have been spent. Thus $s_D < \log r$. Since r is a power of two,

$$|Q_D| \geq \left\lceil \frac{r p_{\text{seed}}}{2^{s_D}} \right\rceil \geq \left\lceil \frac{r p_{\text{seed}}}{2^{\log r - 1}} \right\rceil = \lceil 2p_{\text{seed}} \rceil \geq t,$$

where the last inequality uses $p_{\text{seed}} \geq t/2$. Also

$$T_D \geq \left\lceil \frac{r x_{\text{seed}} m}{2^{s_D}} \right\rceil \geq \left\lceil 2^{-b} m \right\rceil,$$

since $x_{\text{seed}} \geq 2^{-b}$ and $s_D < \log r$. Choose any t -element subset $J \subseteq Q_D$. The leaf rectangle $\widehat{M} \upharpoonright_{R_D, C_D}$ is monochromatic, hence every restriction of it has communication complexity 0.

Apply Lemma 32 to the leaf rectangle with $Q = J$, $u = t$, $x = 2^{-b}$, and

$$y := \frac{|C_D|}{|\text{Cols}(\widehat{M})|} \geq h 2^{-c_D}.$$

This yields a submatrix g of the leaf rectangle lying in

$$M \left[t, 2^{-b}, \frac{h 2^{-c_D}}{1 + \varepsilon} \right].$$

Since g is a restriction of a monochromatic leaf rectangle, we have $D(g) = 0$. On the other hand,

$$D(g) \geq D \left(M \left[t, 2^{-b}, \frac{h 2^{-c_D}}{1 + \varepsilon} \right] \right) \geq D \left(M \left[1, 2^{-b}, \left(\frac{h 2^{-c_D}}{1 + \varepsilon} \right)^{1/t} \right] \right),$$

where the second inequality is Lemma 7. Since $D = s_D + c_D \leq L - 1$, we have

$$c_D \leq L - 1 = D(M) + \log q - 1,$$

so

$$\left(\frac{h 2^{-c_D}}{1 + \varepsilon} \right)^{1/t} \geq y_{\log q + D(M)}(h).$$

Clause (i) of Definition 34 and Lemma 6 therefore imply

$$D \left(M \left[1, 2^{-b}, \left(\frac{h 2^{-c_D}}{1 + \varepsilon} \right)^{1/t} \right] \right) \geq D \left(M \left[1, 2^{-b}, y_{\log q + D(M)}(h) \right] \right) \geq 1.$$

Hence $D(g) \geq 1$, contradicting $D(g) = 0$.

Case 2: the chain spends at least $\log r$ row bits. Let i be the first index with $s_i = \log r$. Then

$$|Q_i| \geq \left\lceil \frac{r p_{\text{seed}}}{2^{\log r}} \right\rceil = p_{\text{seed}}, \quad T_i \geq \left\lceil \frac{r x_{\text{seed}} m}{2^{\log r}} \right\rceil = \lceil x_{\text{seed}} m \rceil.$$

Choose any p_{seed} -element subset $J \subseteq Q_i$. Since $i \leq D \leq L - 1$ and $s_i = \log r$, we also have

$$c_i = i - \log r \leq L - 1 - \log r = D(M) + \log t - 1.$$

Apply Lemma 32 to the rectangle $\widehat{M} \upharpoonright_{R_i, C_i}$ with $Q = J$, $u = p_{\text{seed}}$, $x = x_{\text{seed}}$, and

$$y := \frac{|C_i|}{|\text{Cols}(\widehat{M})|} \geq h 2^{-c_i}.$$

This yields a submatrix of the current rectangle lying in

$$M \left[p_{\text{seed}}, x_{\text{seed}}, \frac{h 2^{-c_i}}{1 + \varepsilon} \right].$$

Since $c_i \leq D(M) + \log t - 1$, the seed-comparison hypothesis gives

$$\frac{h 2^{-c_i}}{1 + \varepsilon} \geq \frac{h 2^{-(D(M) + \log t)}}{1 + \varepsilon} \geq h_{\text{seed}}.$$

Hence Lemma 6 and the seed lower bound imply that $\widehat{M} \upharpoonright_{R_i, C_i}$ contains a submatrix of communication complexity at least $D(M) + \log t$.

The remaining depth below node i is at most

$$L - 1 - i \leq L - 1 - \log r = D(M) + \log t - 1,$$

so the subtree below node i cannot compute that submatrix. This final contradiction completes the proof. $\square$

Corollary 39 (Localised extension). *Assume the hypotheses of Theorem 38. Let $a \in \{0, 1\}$, let r' be a power of two with $1 \leq r' \leq r$, and let*

$$N^\# := \widehat{M} \upharpoonright_{R^\#, C^\#}$$

be a submatrix such that

(i) $R^\#$ is $(Q^\#, \lceil r' x_{\text{seed}} m \rceil)$ -equipartitioned for some $Q^\# \subseteq [q]$ with $|Q^\#| = r' p_{\text{seed}}$, and

$$(ii) \frac{|C^\#|}{|\text{Cols}(\widehat{M})|} \geq h 2^{-a}.$$

Then

$$D(N^\#) \geq D(M) + \log(r't).$$

Proof. Suppose for contradiction that $N^\sharp$ is computed by a deterministic protocol of depth at most

$$L' - 1, \quad L' := D(M) + \log(r't).$$

Starting from $N^\sharp$, follow the same row/column-halving chain as in the proof of Theorem 38. This yields nested rectangles together with sets $Q_i \subseteq [q]$, thresholds T_i , and counts $s_i, c_i \geq 0$ such that $i = s_i + c_i$ and

$$|Q_i| \geq \left\lceil \frac{r' p_{\text{seed}}}{2^{s_i}} \right\rceil, \quad T_i \geq \left\lceil \frac{r' x_{\text{seed}} m}{2^{s_i}} \right\rceil, \quad \frac{|C_i|}{|\text{Cols}(\widehat{M})|} \geq h 2^{-(a+c_i)}.$$

Case 1: the chain reaches a leaf before $\log r'$ row bits have been spent. Then exactly as in Theorem 38, the leaf still contains t outer blocks with at least $\lceil 2^{-b} m \rceil$ rows each, so Lemmas 7 and 32 produce a one-copy witness in

$$M \left[1, 2^{-b}, \left(\frac{h 2^{-(a+c_D)}}{1 + \varepsilon} \right)^{1/t} \right].$$

Since $a \leq 1$, $r't \leq q$, and $c_D \leq L' - 1$, we have

$$a + c_D \leq a + L' - 1 \leq D(M) + \log q.$$

Clause (i) of Definition 34 therefore gives communication complexity at least 1, contradicting the fact that every subgame of a monochromatic leaf rectangle has complexity 0.

Case 2: the chain spends at least $\log r'$ row bits. Let i be the first index with $s_i = \log r'$. Exactly as in Theorem 38, the current rectangle still contains p_{seed} outer blocks, each with at least $\lceil x_{\text{seed}} m \rceil$ rows, and

$$c_i \leq L' - 1 - \log r' = D(M) + \log t - 1.$$

Applying Lemma 32 on any such p_{seed} blocks yields a witness in

$$M \left[p_{\text{seed}}, x_{\text{seed}}, \frac{h 2^{-(a+c_i)}}{1 + \varepsilon} \right].$$

Since $a \in \{0, 1\}$,

$$a + c_i \leq D(M) + \log t.$$

The theorem-level seed comparison therefore gives

$$\frac{h 2^{-(a+c_i)}}{1 + \varepsilon} \geq h_{\text{seed}},$$

so the seed lower bound implies that the current rectangle contains a subgame of communication complexity at least $D(M) + \log t$. But the remaining depth below node i is at most

$$L' - 1 - i \leq L' - 1 - \log r' = D(M) + \log t - 1,$$

contradiction. □

Remark 40. The main input to Theorem 38 is the canonical relaxed interlace $\widehat{M}$ itself. Later projected-family uses do not require a separate, stronger theorem statement. Instead, Lemma 30 keeps the relevant balancedness and accuracy parameter after projection, and Lemma 32 then supplies the same bridge inside the projected family. Thus the proof of Corollary 39 carries over unchanged once the projected family is relabelled by its surviving coordinates.

4.5 The Near-Exact Separation Theorem

Theorem 38 gives the relaxed lower bound. For the reduction we also need protocol control: under the tight depth bound, the protocol must spend its first $\log q$ bits isolating the outer row block. The theorem below proves this directly in the (t, r, q, h) -parameterised setting. The proof has three parts: an outer r -phase using the extension theorem, an unbalanced-row continuation that follows the larger child until it satisfies the hypotheses of the bridge lemma, and an inner t -phase handled by the classical near-exact separation theorem.

Theorem 41 (Relaxed Near-Exact Separation). *Let M be (δ, b) -robust, write $m := |\text{Rows}(M)|$, assume $D(M) \geq 2$, and assume*

$$0 < \delta \leq \frac{1}{\sqrt{2}} - \frac{1}{2}.$$

Let $b \geq 1$, let $t \geq 2$ and $r \geq 1$ be powers of two with $t \leq 2^b$, and set

$$q := rt, \quad \varepsilon := \varepsilon_{q,t}, \quad \widehat{M} := M \langle q, S_{q,t}(\text{Cols}(M)) \rangle.$$

Fix an integer p_{seed} and reals

$$2^{-b} \leq x_{\text{seed}} \leq \frac{1}{r}, \quad h, h_{\text{seed}} \in (0, 1].$$

Assume

$$\frac{t}{2} \leq p_{\text{seed}} \leq t,$$

that (M, b) is (q, t, h) -column-loss resilient,

$$D(M[p_{\text{seed}}, x_{\text{seed}}, h_{\text{seed}}]) \geq D(M) + \log t,$$

and

$$\frac{h 2^{-(\log t + D(M))}}{1 + \varepsilon} \geq h_{\text{seed}}.$$

Assume moreover that

$$2\left(\frac{1}{2} + \delta\right)^2 \leq \frac{h}{1 + \varepsilon} \quad \text{and} \quad q \left\lceil 2^{-b+1}m \right\rceil < m.$$

Let $C' \subseteq \text{Cols}(\widehat{M})$ satisfy

$$\frac{|C'|}{|\text{Cols}(\widehat{M})|} \geq h,$$

and write

$$N := \widehat{M} \upharpoonright_{\text{Rows}(\widehat{M}), C'}.$$

Then every deterministic protocol for N of cost at most $D(M) + \log q$ has the following two properties. On every branch, its first $\log q$ bits are row bits.

Let $\mathcal{P}$ denote the resulting partition after those first $\log q$ bits. Then every rectangle $R \times C \in \mathcal{P}$ has a unique dominant outer block $i^ \in [q]$ such that, writing*

$$R_i := R \cap (\{i\} \times \text{Rows}(M)),$$

one has

$$|R_{i^*}| \geq m - (q - 1) \left\lceil 2^{-b+1}m \right\rceil, \quad |R_i| < \left\lceil 2^{-b+1}m \right\rceil \quad \text{for all } i \neq i^*.$$

Proof sketch. The argument splits into three phases.

- (i) *Outer r -phase* (first $\log r$ bits). An induction on $s = 0, \dots, \log r$ shows that after the first s bits, every surviving rectangle still contains a relaxed witness (a submatrix of the relaxed interlace satisfying the balanced-family and density hypotheses) with $q_s = q/2^s$ equipartitioned outer blocks and column density $\geq h$. At each step, Corollary 39 rules out termination and column bits.
- (ii) *Unbalanced-row continuation.* If a child receives more than $q_s/2$ blocks, we trace it through further forced row bits until it accumulates enough blocks to reduce to a classical witness that already exceeds the remaining budget.
- (iii) *Inner t -phase* (next $\log t$ bits). After the outer phase, each surviving rectangle contains a classical t -copy bracket witness obtained via the bridge lemma. The classical near-exact separation lemma (Lemma 26) then forces the next $\log t$ bits to be row bits that cleanly isolate the outer blocks.

After all $\log q = \log r + \log t$ row bits, a final application of Lemma 24 and the two-copy lower bound (Corollary 16) shows that every rectangle has a unique dominant block with the claimed row bounds.

Proof. Write

$$T := \left\lceil 2^{-b+1}m \right\rceil, \quad y := \frac{h}{1+\varepsilon}, \quad \text{and} \quad L := D(M) + \log q.$$

We keep the proof in the advertised three phases: an outer r -phase, the unbalanced-row continuation that rules out uneven row splits, and an inner t -phase.

Phase 1: outer r -phase. For $0 \leq s \leq \log r$, set

$$r_s := \frac{r}{2^s}, \quad q_s := \frac{q}{2^s} = r_s t, \quad T_s := \left\lceil \frac{m}{2^s} \right\rceil.$$

Phase 1 invariant. For each $s \in \{0, \dots, \log r\}$, every surviving rectangle after the first s protocol bits contains a relaxed submatrix whose row set is (Q, T_s) -equipartitioned for some $Q \subseteq [q]$ with $|Q| = q_s$, and whose column density is at least h .

We prove this invariant by induction on s . For $s = 0$, the claim is immediate: N itself has row set $([q], m)$ -equipartitioned and column density at least h .

Now fix $s < \log r$, and assume the claim holds at depth s . Let S be any surviving rectangle after the first s bits, and choose a relaxed witness

$$N_S \subseteq S$$

whose row set is (Q, T_s) -equipartitioned with $|Q| = q_s$ and whose column density is at least h .

Step 1: S is not already a leaf. Choose any subset $J \subseteq Q$ of size $r_s p_{\text{seed}}$, and trim each block in J down to exactly $\lceil r_s x_{\text{seed}} m \rceil$ rows. This is possible because

$$T_s = \left\lceil \frac{m}{2^s} \right\rceil \geq \lceil r_s x_{\text{seed}} m \rceil,$$

by the theorem-level bound $x_{\text{seed}} \leq 1/r$. The resulting submatrix satisfies the hypotheses of Corollary 39 with $a = 0$ and $r' = r_s$, so it has communication complexity at least

$$D(M) + \log(r_s t) = D(M) + \log q_s \geq 1.$$

Therefore S is not a leaf.

Step 2: the next bit at S cannot be a column bit. If it were, one of the two children would retain at least half of the columns of N_S . The trimmed witness from Step 1 would then survive in that child with the same row structure and with column density at least $h/2$. Applying Corollary 39 with $a = 1$ and $r' = r_s$ would show that this child has communication complexity at least

$$D(M) + \log q_s.$$

But after $s + 1$ bits, the remaining depth allowed under the bound $L = D(M) + \log q$ is only

$$L - (s + 1) = D(M) + \log q_s - 1,$$

a contradiction. Hence the $(s + 1)$ -st bit is a row bit.

Phase 2: the unbalanced-row continuation. The goal in this phase is to show that an unbalanced child would force a contradiction before the protocol can hand off to the classical t -copy argument. It remains to rule out the case where the next row split sends strictly more than half of the surviving outer blocks to one child. Each block in Q contributes at least T_s rows to N_S , so one of the two children contains at least $\lceil T_s/2 \rceil \geq T_{s+1}$ rows from that block. Assign each block of Q to such a child.

Suppose one child receives more than $q_s/2$ assigned blocks. Write

$$r_s = 2^\ell, \quad \text{so} \quad q_s = 2^\ell t$$

with $\ell \geq 1$. Then that child receives at least

$$\frac{q_s}{2} + 1 = 2^{\ell-1}t + 1$$

assigned blocks. Let S_0 denote that child.

We now trace the subtree below S_0 . We claim that for each $j \in \{0, \dots, \ell\}$, there is a descendant S_j of S_0 , reached after exactly j further protocol bits, such that:

- (a) all those j further bits are row bits;
- (b) S_j contains a relaxed submatrix whose row set is $(Q_j, \lceil m/2^{s+1+j} \rceil)$ -equipartitioned;
- (c) the following hold:

$$|Q_j| \geq 2^{\ell-j-1}t + 1 \quad \text{for } j < \ell, \quad |Q_\ell| \geq \frac{t}{2} + 1.$$

For $j = 0$, this is exactly the construction of S_0 . Now fix $j < \ell$, and assume S_j has been constructed. Choose any subset of Q_j of size $2^{\ell-j-1}p_{\text{seed}}$, and trim each selected block down to exactly

$$\left\lceil 2^{\ell-j-1}x_{\text{seed}}m \right\rceil$$

rows. This is possible because

$$\left\lceil \frac{m}{2^{s+1+j}} \right\rceil \geq \left\lceil 2^{\ell-j-1}x_{\text{seed}}m \right\rceil,$$

again by $x_{\text{seed}} \leq 1/r$, since

$$2^{\ell-j-1} = \frac{r_s}{2^{j+1}} = \frac{r}{2^{s+j+1}}.$$

Corollary 39 with $a = 0$ and $r' = 2^{\ell-j-1}$ shows that S_j is not a leaf, and Corollary 39 with $a = 1$ and the same r' shows that its next bit cannot be a column bit. Hence the next bit is a row bit.

Assign each block in Q_j to a child that contains at least $\lceil m/2^{s+2+j} \rceil$ rows from that block. Since

$$|Q_j| \geq 2^{\ell-j-1}t + 1,$$

one child receives at least

$$\left\lceil \frac{2^{\ell-j-1}t + 1}{2} \right\rceil = 2^{\ell-j-2}t + 1$$

assigned blocks. Let that child be S_{j+1} . This proves the claim.

Applying the claim with $j = \ell$, we obtain a descendant S_ℓ at depth

$$s + 1 + \ell = s + 1 + \log r_s = \log r + 1$$

that contains at least $t/2 + 1$ outer blocks, each contributing at least

$$\left\lceil \frac{m}{2^{s+1+\ell}} \right\rceil = \left\lceil \frac{m}{2r} \right\rceil$$

rows, while the column density is still at least h .

Choose any $J \subseteq Q_\ell$ of size $t/2 + 1$, and from each block in J keep exactly $\lceil m/(2r) \rceil$ rows. Applying Lemma 32 to the resulting restriction gives a classical witness

$$G \in M[t/2 + 1, 1/(2r), y].$$

Since

$$T = \left\lceil 2^{-b+1}m \right\rceil \geq 2^{-b+1}m,$$

the theorem-level condition $qT < m$ implies

$$q 2^{-b+1} < 1,$$

and therefore

$$\frac{1}{2r} > t 2^{-b}.$$

If $t = 2$, this becomes

$$\frac{1}{2r} > 2^{1-b}.$$

Because $y \geq 2(\frac{1}{2} + \delta)^2 \geq (\frac{1}{2} + \delta)^2$, monotonicity and Corollary 16 give

$$D(G) \geq D\left(M\left[2, 2^{1-b}, \left(\frac{1}{2} + \delta\right)^2\right]\right) \geq D(M) + 1 = D(M) + \log t.$$

Now assume $t \geq 4$, and write $u := \log t \geq 2$. Then

$$\frac{t}{2} + 1 = 2^{u-1} + 1, \quad \frac{1}{2r} > t 2^{-b} = 2^{u-b}.$$

Since $y \geq 2(\frac{1}{2} + \delta)^2 \geq (\frac{1}{2} + \delta)^2$, monotonicity and Corollary 20 give

$$D(G) \geq D\left(M\left[2^{u-1} + 1, 2^{u-b}, \left(\frac{1}{2} + \delta\right)^2\right]\right) \geq D(M) + u = D(M) + \log t.$$

In either case,

$$D(G) \geq D(M) + \log t.$$

But S_ℓ is already at depth $\log r + 1$, so the remaining depth below it is at most

$$L - (\log r + 1) = D(M) + \log t - 1,$$

a contradiction. Therefore no child of the row split at S can receive more than $q_s/2$ assigned blocks.

Since there are q_s assigned blocks in total, each child receives exactly $q_s/2 = q_{s+1}$ of them. Restricting to those assigned blocks yields, in each child, a relaxed witness with row set (Q', T_{s+1}) -equipartitioned and column density at least h . This proves the inductive step.

Hence the claim holds for every $s \in \{0, \dots, \log r\}$. In particular, after the first $\log r$ protocol bits, every surviving rectangle contains a relaxed submatrix whose row set is

$$(Q, \lceil m/r \rceil)\text{-equipartitioned}$$

for some $Q \subseteq [q]$ with $|Q| = t$, and whose column density is at least h .

Phase 3: inner t -phase. This is the handoff point: the relaxed witness retained through the outer phase is now converted, via the bridge lemma, into a classical t -copy witness to which the classical near-exact separation lemma applies. Fix any surviving rectangle S after the first $\log r$ bits, and choose such a witness inside S . Applying Lemma 32 to those t blocks gives

$$G_S \in M[t, 1/r, y].$$

The inequality $qT < m$ and the bound $T = \lceil 2^{-b+1}m \rceil \geq 2^{-b+1}m$ imply $q2^{-b+1} < 1$, hence

$$t2^{-b} < \frac{1}{r}.$$

Since t is a power of two, $t \leq 2^b$, and $y \geq 2(\frac{1}{2} + \delta)^2 \geq (\frac{1}{2} + \delta)^2$, Corollary 23 gives

$$D(G_S) \geq D\left(M\left[t, t2^{-b}, \left(\frac{1}{2} + \delta\right)^2\right]\right) \geq D(M) + \log t.$$

Also,

$$tT < \frac{m}{r} \leq \left\lceil \frac{m}{r} \right\rceil.$$

Hence G_S satisfies the row-side hypotheses of Lemma 26, and the theorem-level density assumption gives

$$2\left(\frac{1}{2} + \delta\right)^2 \leq y.$$

By construction in Lemma 32, the witness G_S is not merely a projected image: the lemma first trims to actual row and column sets R^* and C^* , choosing one representative column for each projected pattern, and only then canonically relabels that resulting submatrix. Thus the subtree below S , restricted to those surviving inputs, induces a deterministic protocol for G_S with the same cost.

Consider the protocol subtree below S , restricted to inputs from G_S . Its cost is at most $D(M) + \log t$, since that is exactly the depth remaining below S . Therefore Lemma 26 applies to that restricted subtree with copy parameter t , row-density parameter $1/r$, and column-density parameter $h/(1 + \varepsilon)$, and its first conclusion shows that the next $\log t$ bits below S are row bits.

Since S was arbitrary, the first

$$\log r + \log t = \log q$$

bits of the original protocol are row bits on every branch.

Let $\mathcal{P}$ be the partition after those first $\log q$ bits.

Final row-only contradiction. We now pass from the relaxed witness back to the row-only partition promised in the theorem statement. Apply Lemma 24 with

$$Q = [q], \quad X = \text{Rows}(M), \quad T_0 = m, \quad T = \lceil 2^{-b+1}m \rceil,$$

and $R^{\text{in}} = \text{Rows}(N) = [q] \times \text{Rows}(M)$. The side condition $|Q|T < T_0$ is exactly the theorem-level hypothesis $qT < m$. So it suffices to rule out a rectangle $R \times C \in \mathcal{P}$ containing T rows from two distinct outer blocks.

Suppose such a rectangle exists, with two heavy blocks $i_1 \neq i_2$. Because the first $\log q$ bits are all row bits, this rectangle still has the full column set C' , so its column density inside $\widehat{M}$ is at least h . Trim each heavy block down to exactly T rows and apply Lemma 32. This yields

$$G \in M \left[2, 2^{-b+1}, y \right].$$

Since $y \geq 2(\frac{1}{2} + \delta)^2 \geq (\frac{1}{2} + \delta)^2$, monotonicity and Corollary 16 give

$$D(G) \geq D(M) + 1.$$

By restriction monotonicity,

$$D(N \upharpoonright_{R,C}) \geq D(M) + 1.$$

But after $\log q$ bits the remaining depth under the bound $L = D(M) + \log q$ is only $D(M)$, a contradiction.

Hence no rectangle in $\mathcal{P}$ contains T rows from two distinct outer blocks. Lemma 24 now yields, for every $R \times C \in \mathcal{P}$, a unique $i^* \in [q]$ such that

$$|R_{i^*}| \geq m - (q-1)T, \quad |R_i| < T \text{ for all } i \neq i^*.$$

Since $T = \lceil 2^{-b+1}m \rceil$, this is exactly the claimed near-exact separation. $\square$

The main downstream use of Theorems 38 and 41 is the Stage 2 certification argument in Section 5. Table 1 records where those later applications verify the relevant hypotheses, so that the reduction proofs can refer back to a single checklist rather than repeat the full bookkeeping in prose. It is meant only as a lookup table and can be skipped on a first reading.

Theorem	Hypothesis	Chosen parameter or object	Where it is verified
Theorem 38	power-of-two data, $q = rt$, and $t \leq 2^b$	Stage 2 takes $M = \mathcal{M}_1^\top$, $b = b_1$, $t = t_2$, $r = r_2$, $q = q_2$	The Stage 2 choices are written out in Lemma 58, and the global parameter relations are set in Section 5.2.
Theorem 38	$\frac{t}{2} \leq p_{\text{seed}} \leq t$	$p_2^{\text{seed}} = \frac{9}{16}t_2$	Checked in the Stage 2 setup at the start of Lemma 58.
Theorem 38	column-loss resilience	$(\mathcal{M}_1^\top, b_1)$ in Stage 2	Supplied by Lemma 54; the projected dense-row variant is handled in Corollary 59.
Theorem 38	seed lower bound and seed-density comparison	Stage 2 hard seed with $h_{\text{seed}} = h'_2$	Supplied by Lemma 55; the density comparison is checked in the proof of Lemma 58.
Theorem 41	(δ, b) -robust and $D(M) \geq 2$	again $M = \mathcal{M}_1^\top$ in Stage 2	Verified in the proof of Lemma 58 using Lemmas 8 and 52 and corollary 51.
Theorem 41	the extension-theorem inputs reused during the proof	the same Stage 2 parameters, plus projected dense-row restrictions	The reuse of those inputs is summarised in Corollary 39; the projected version is checked in Corollary 59.
Theorem 41	extra separation inequalities	$2(\frac{1}{2} + \delta)^2 \leq h/(1 + \varepsilon)$ and $q \lceil 2^{-b+1}m \rceil < m$	Checked directly in the proofs of Lemma 58 and corollary 59; the corresponding large- d bounds are collected again in Appendix C.

Table 1: Checklist for the transfer theorems.

5 Reduction from $\{0, 1\}$ -Vector Bin Packing

5.1 Source Problem and Preprocessing

We now turn to the reduction. We work with the following fixed source problem.

$\{0, 1\}$ - d -DIMENSION VECTOR BIN PACKING

Input: A set of n vectors $\{v_1, \dots, v_n\} \subseteq \{0, 1\}^d$, a capacity parameter c , and a number of bins m (where m is a power of two).

Question: Does there exist a partition of the vectors into m bins such that every bin B_i satisfies $\|\sum_{v \in B_i} v\|_\infty \leq c$?

In the reduction we only use the case $c = 1$ and $m = 4$.

Proposition 42. *The restricted problem $\{0, 1\}$ - d -DIMENSION VECTOR BIN PACKING with parameters $c = 1$ and $m = 4$ is NP-complete. This remains true under the promise that*

$$\sum_{i=1}^n v_i(\alpha) \leq 4 \quad \text{for every } \alpha \in [d].$$

Proof. Membership in NP is immediate. For hardness, reduce from 4-COLOURING. Given a graph $G = (V, E)$, create one vector $v_u \in \{0, 1\}^E$ for each vertex $u \in V$, indexed by the edge set. For an

edge $e = \{u, w\}$, set the e -coordinate of v_u and v_w to 1, and set that coordinate to 0 for all other vertices. Then a partition of the vectors into 4 bins satisfies the per-coordinate capacity bound $c = 1$ iff no edge has both endpoints in the same bin, i.e., if and only if G is properly 4-colourable. Moreover, every coordinate corresponds to an edge and therefore appears in exactly two vectors. In particular, the produced instance satisfies

$$\sum_{i=1}^n v_i(\alpha) \leq 2 \leq 4 \quad \text{for every } \alpha \in [d],$$

so the same reduction already proves hardness under the stated promise. $\square$

From now on we may therefore restrict attention to source instances satisfying

$$\sum_{i=1}^n v_i(\alpha) \leq 4 \quad \text{for every } \alpha \in [d].$$

Before applying the reduction, we record the immediate source-side rejection criterion that will be used throughout Stage 4. If some coordinate $\alpha \in [d]$ satisfies

$$\sum_{i=1}^n v_i(\alpha) > 4,$$

then the instance is immediately a **NO**-instance, since across four bins of capacity 1 that coordinate can accommodate at most four 1-entries in total.

Lemma 43. *Let*

$$I = (v_1, \dots, v_n), \quad v_i \in \{0, 1\}^d,$$

be an instance of $\{0, 1\}$ - d -DIMENSION VECTOR BIN PACKING with $c = 1$ and $m = 4$. For each slab $t \in [5]$, let

$$\iota_t : \{0, 1\}^d \hookrightarrow \{0, 1\}^{5d}$$

be the embedding that places a vector in the t -th block of d coordinates and pads all other coordinates with zeros. Define the replicated padded instance

$$I^\circ := (\iota_t(v_i))_{t \in [5], i \in [n]} \cup \{z_1, z_2, z_3, z_4\}, \quad z_1 = z_2 = z_3 = z_4 := 0^{5d}.$$

Then I° is computable from I in polynomial time, and the following hold:

1. *I is a **YES**-instance if and only if I° is a **YES**-instance.*
2. *Whenever I° is a **YES**-instance, it has a feasible packing*

$$(B_1, B_2, B_3, B_4)$$

such that $z_p \in B_p$ for every $p \in [4]$.

3. *Whenever I is a **NO**-instance, every partition*

$$[5n + 4] = B_1 \sqcup B_2 \sqcup B_3 \sqcup B_4$$

of the vectors of I° contains a bin B_p and a coordinate $\beta \in [5d]$ such that

$$|\{u \in B_p : u(\beta) = 1\}| \geq 2 \quad \text{and} \quad |\{u \in B_p : u(\beta) = 0\}| \geq 1.$$

Proof. The transformation is polynomial-time: we make five disjoint-coordinate copies of the instance and append four distinguished zero vectors.

If I has a feasible packing (A_1, A_2, A_3, A_4) , then

$$\left(\bigcup_{t=1}^5 \iota_t(A_1) \cup \{z_1\}, \bigcup_{t=1}^5 \iota_t(A_2) \cup \{z_2\}, \bigcup_{t=1}^5 \iota_t(A_3) \cup \{z_3\}, \bigcup_{t=1}^5 \iota_t(A_4) \cup \{z_4\} \right)$$

is a feasible packing of I° , because the five slabs use disjoint coordinates and adding 0^{5d} does not change any coordinate load. This proves completeness and the canonical-packing claim.

Conversely, if I° has a feasible packing, then restricting that packing to any one slab t and projecting away the zero coordinates yields a feasible packing of the original instance I . This proves soundness.

For the final item, assume I is a NO-instance and fix any partition of the vectors of I° into four bins. For each slab $t \in [5]$, the vectors $\iota_t(v_1), \dots, \iota_t(v_n)$ form a copy of I on its own coordinate block, so this restricted partition cannot be feasible on slab t . Hence for each $t \in [5]$ there exist a bin $p_t \in [4]$ and a coordinate $\alpha_t \in [d]$ such that at least two vectors of slab t lying in B_{p_t} have a 1 in the coordinate $\beta_t := (t-1)d + \alpha_t \in [5d]$.

Since there are 5 slabs but only 4 bins, two slabs $s \neq t$ must satisfy $p_s = p_t$. Fix such a pair and set $p := p_s = p_t$. Then B_p contains at least two vectors from slab t with value 1 at β_t . Every vector from slab s has value 0 at β_t because the slabs have disjoint supports, and B_p contains at least one such vector by the choice of s . Therefore B_p contains both at least two vectors with a 1 at β_t and at least one vector with a 0 at β_t , exactly as required. $\square$

From now on we replace the source instance by its replicated padded version from Lemma 43, relabel its ambient dimension again by d , and relabel its vectors again as $v_1, \dots, v_n$. We reserve distinguished indices $z_1, \dots, z_4 \in [n]$ for the four zero anchors. Thus

$$v_{z_p} = 0^d \quad \text{for every } p \in [4].$$

Whenever the source instance is a YES-instance, we may moreover fix a *canonical feasible packing*

$$\sigma : [n] \rightarrow [4] \quad \text{with} \quad \sigma(z_p) = p \quad \text{for every } p \in [4].$$

Power-of-two normalisation. Fix once and for all a power of two $d_\star$ large enough that every asymptotic inequality used later in Stages 1–4 holds for every power of two $d \geq d_\star$. Given the current preprocessed instance in ambient dimension d , set

$$D := \lceil \max\{d, d_\star\} \rceil_2.$$

Here $\lceil z \rceil_2$ denotes the least power of two that is at least z . Pad every vector by $D - d$ trailing zero coordinates, continue to denote the padded vectors by $v_1, \dots, v_n$, and relabel the ambient dimension by $d := D$. This preserves feasibility of the source instance, preserves the distinguished zero anchors $v_{z_p} = 0^d$, and preserves the final item of Lemma 43, since all added coordinates are identically zero.

5.2 Scaffold

For the remainder of the reduction we fix the audited parameter regime.

Global.

$$\delta := 0.1, \quad \Delta := 5.$$

Here Δ is the fixed surplus-copy parameter j used in the Stage 2 application of the hard-seed lemma.

Stage 1.

$$\begin{aligned} q_1 &:= \lceil 2 \log^2 d \rceil_2 - 2, & 2^a &:= q_1 + 2, & t_1 &:= \lceil 64 \log d \rceil_2, \\ r_1 &:= (q_1 + 2)/t_1, & b_0 &:= 64 \log(64 \log d), \\ b'_1 &:= 3 \log d, & b_1 &:= 2 \log d. \end{aligned}$$

Here b_0 is the base Stage 1 robustness margin used later to initialise the passage from $\mathcal{M}_0$ to $\mathcal{M}_1$.

Stage 2.

$$\begin{aligned} q_2 &:= \lceil d \rceil_2, & t_2 &:= \lceil 3 \log d / \log \log d \rceil_2, & r_2 &:= q_2/t_2, \\ b'_2 &:= 8 \log \log d, & b_2 &:= 3 \log \log d, \\ h_2 &:= 2^{-b_2} = (\log d)^{-3}, & h'_2 &:= 2^{-b'_2} = (\log d)^{-8}. \end{aligned}$$

For all sufficiently large powers of two d , the quantities t_1, t_2, r_1, r_2, q_2 are powers of two, and

$$q_1 + 2 = r_1 t_1, \quad q_2 = r_2 t_2, \quad q_1 + 2 < 4 \log^2 d, \quad q_2 < 2d.$$

Since d has already been normalised to a power of two, we have $q_2 = d$. From this point on we therefore regard the source coordinates as indexed by $[q_2]$.

Stage 0: the seed matrix $\mathcal{M}_0$. We start from the seed matrix

$$\mathcal{M}_0 := [1 \ 0].$$

Stage 1: the capacity scaffold $\mathcal{M}_1$. The first stage keeps the full $q_1 + 5 = 2^a + 3$ -coordinate balanced subgame and uses only the first q_1 outer row blocks as the ambient Stage 1 template. Set

$$C_1 := S_{q_1+5, t_1}(\text{Cols}(\mathcal{M}_0)), \quad \widehat{\mathcal{M}}_1 := \mathcal{M}_0 \langle q_1 + 5, C_1 \rangle.$$

Write

$$R_1^{\text{full}} := [q_1 + 5] \times \text{Rows}(\mathcal{M}_0), \quad R_1 := [q_1] \times \text{Rows}(\mathcal{M}_0) \subseteq R_1^{\text{full}},$$

and define

$$\mathcal{M}_1 := \widehat{\mathcal{M}}_1 \upharpoonright_{R_1, C_1}.$$

Thus $\mathcal{M}_1$ is the row restriction of the full Stage 1 relaxed interlace to its first $q_1 = 2^a - 2$ outer blocks, while the final five subgame coordinates stay available for the local Stage 4 gadget. For $\gamma = (c_1, \dots, c_{q_1+5}) \in C_1$, write

$$\text{tail}(\gamma) \in [2^5]$$

for the 5-bit pattern encoded by the last five coordinates $(c_{q_1+1}, \dots, c_{q_1+5})$. Because C_1 is (q_1+5, t_1) -balanced with $\varepsilon_{q_1+5, t_1} < 1$, every pattern in $\text{Cols}(\mathcal{M}_0)^{t_1} = \{0, 1\}^{t_1}$ occurs on any fixed t_1 -subset of coordinates. Hence

$$|C_1| \geq \frac{2^{t_1}}{1 + \varepsilon_{q_1+5, t_1}} \geq 2^{t_1-1} \geq \frac{1}{2} d^{64}$$

for all sufficiently large d .

Stage 2: the dimension gadget $\mathcal{M}_2$. Next we interlace the transpose of $\mathcal{M}_1$ with order q_2 :

$$\mathcal{M}_2 := \mathcal{M}_1^\top \langle q_2, S_{q_2, t_2}(R_1) \rangle.$$

The q_2 outer row blocks are now identified with the padded source coordinates in $[q_2]$. Thus Stage 2 is the gadget that lets the protocol isolate one dimension. Set

$$R_2 := \text{Rows}(\mathcal{M}_2) = [q_2] \times C_1, \quad C_2 := \text{Cols}(\mathcal{M}_2).$$

Stage 3: the bin gadget $\mathcal{M}_3$. We now take the m -fold classical interlace of $\mathcal{M}_2^\top$, where $m = 4$:

$$\mathcal{M}_3 := \mathcal{M}_2^\top \langle m \rangle.$$

The m -fold interlace creates four outer blocks, one for each bin. Write

$$R_3 := \text{Rows}(\mathcal{M}_3) = [m] \times C_2, \quad C_3 := \text{Cols}(\mathcal{M}_3) = R_2^m.$$

Because $C_2 = S_{q_2, t_2}(R_1)$ is (q_2, t_2) -balanced with $\varepsilon_{q_2, t_2} < 1$, fixing any t_2 coordinates shows

$$|C_2| \geq \frac{|R_1|^{t_2}}{1 + \varepsilon_{q_2, t_2}} \geq \frac{1}{2} |R_1|^{t_2} \geq \frac{1}{2} d^6$$

for all sufficiently large d , since $|R_1| = q_1 \geq \log^2 d$ and $t_2 \geq 3 \log d / \log \log d$. In particular, $h_2 |C_2| \rightarrow \infty$, so

$$3 \left\lceil 2^{-b_2+1} |C_2| \right\rceil \leq 8h_2 |C_2|$$

for all sufficiently large d . We use this estimate below when the Stage 3 row loss is rewritten in the cleaner density form $(1 - 8h_2) |C_2|$.

Stage 4: attach the source instance and obtain $\mathcal{M}_4$. We now add one row for each vector of the preprocessed source instance, including the four distinguished zero anchors from Lemma 43. The row set becomes

$$R_4 := R_3 \cup [n].$$

The column set is

$$C_4 := [2^5] \times R_2^4.$$

We write a column $j \in C_4$ as

$$j = (k, (j_1, j_2, j_3, j_4)), \quad k \in [2^5], \quad (j_1, \dots, j_4) \in R_2^4.$$

We call the rows in $R_3 = [4] \times C_2$ the *template rows* (they carry the Stage 3 scaffold) and the rows in $[n]$ the *vector rows* (they encode the bin-packing instance). We first copy the Stage 3 template on the template rows. For every $r \in R_3$ and every column $j = (k, (j_1, \dots, j_4)) \in C_4$, set

$$\mathcal{M}_4(r, j) := \mathcal{M}_3(r, (j_1, j_2, j_3, j_4)).$$

For each dimension $\alpha \in [q_2]$, define the active support set

$$A_\alpha := \{i \in [n] : v_i(\alpha) = 1\}.$$

By the source-side screen above, $|A_\alpha| \leq 4$ for every $\alpha \in [q_2]$, so we may fix an injection

$$\pi_\alpha : A_\alpha \hookrightarrow [4] \quad \text{for each } \alpha \in [q_2].$$

Parameter	Value / size	Role
q_1	$\Theta(\log^2 d)$	Stage 1 outer-coordinate count; determines the local capacity threshold and the Stage 1 ambient row set.
t_1	$\Theta(\log d)$	Stage 1 balance parameter; controls the size of C_1 and the Stage 1 hard-seed input.
r_1	$(q_1 + 2)/t_1 = \Theta(\log d)$	Stage 1 grouping parameter; appears in the residual Stage 1 lower bound used inside Stage 2 and Stage 4.
b_0	$\Theta(\log \log d)$	Base robustness needed to initialise the Stage 1 relaxed interlace from $\mathcal{M}_0$.
b'_1, b_1	$3 \log d, 2 \log d$	Stage 1 density margins; feed the Stage 1 threshold and the Stage 2 certification lemmas.
q_2	d	Stage 2 outer-coordinate count; this is the source dimension after preprocessing.
t_2	$\Theta(\log d / \log \log d)$	Stage 2 balance parameter; controls the size of C_2 and the Stage 2 seed width.
r_2	q_2/t_2	Stage 2 grouping parameter; appears in the Stage 2 extension and separation statements.
b'_2, b_2	$8 \log \log d, 3 \log \log d$	Stage 2 density margins; tuned so Stage 3 row loss is below the block structure of R_1 .
h_2, h'_2	$(\log d)^{-3}, (\log d)^{-8}$	Stage 2 density thresholds; certify the Stage 2 lower-bound and protocol-control inputs.
B_{cap}	$a+1$ (later shown to equal $D(\mathcal{M}_1)$)	Local Stage 1 capacity budget on a successful branch.
B_{yes}	$\log 4 + \lceil \log q_2 \rceil + B_{\text{cap}}$	Total budget on a successful branch: choose a bin, choose a coordinate, then solve the local Stage 1 sub-problem.
k	B_{yes}	Final target depth in the reduction.

Table 2: Reduction parameters and their roles in the staged scaffold.

Now define the new vector rows. For $i \in [n]$ and $j = (k, (j_1, j_2, j_3, j_4)) \in C_4$, set

$$\mathcal{M}_4(i, j) := \begin{cases} \mathcal{M}_0 \langle 5 \rangle (\pi_\alpha(i), k), & \text{if there exists } \alpha \in [q_2] \text{ such that } v_i(\alpha) = 1 \text{ and, writing each } \\ & j_m \in R_2 = [q_2] \times C_1 \text{ as } j_m = (\alpha_m, \gamma_m), \text{ we have } \alpha_1 = \alpha_2 = \\ & \alpha_3 = \alpha_4 = \alpha, \\ \mathcal{M}_0 \langle 5 \rangle (5, k), & \text{otherwise.} \end{cases}$$

Thus row i is neutral by default, and on the diagonal outer-block slice $\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = \alpha$ it becomes the $\pi_\alpha(i)$ -th active row of the 5-row gadget exactly when $v_i(\alpha) = 1$. In particular, every zero-anchor row remains neutral on every branch because $v_{z_p} = 0^d$.

We set the target protocol budget to

$$B_{\text{cap}} := a + 1, \quad B_{\text{yes}} := \log 4 + \lceil \log q_2 \rceil + B_{\text{cap}}.$$

This is the YES-case target budget.

So a successful branch is supposed to choose a bin, then choose a dimension, and then fall into a local Stage 1 capacity gadget with budget B_{cap} . The full staged scaffold is summarised in Figure 2.

How to read a branch. Reading Figure 2 from top to bottom, a branch of the intended protocol has the following meaning. Fix a canonical feasible packing $\sigma : [n] \rightarrow [4]$, which we think of as placing vector i into bin $\sigma(i)$.

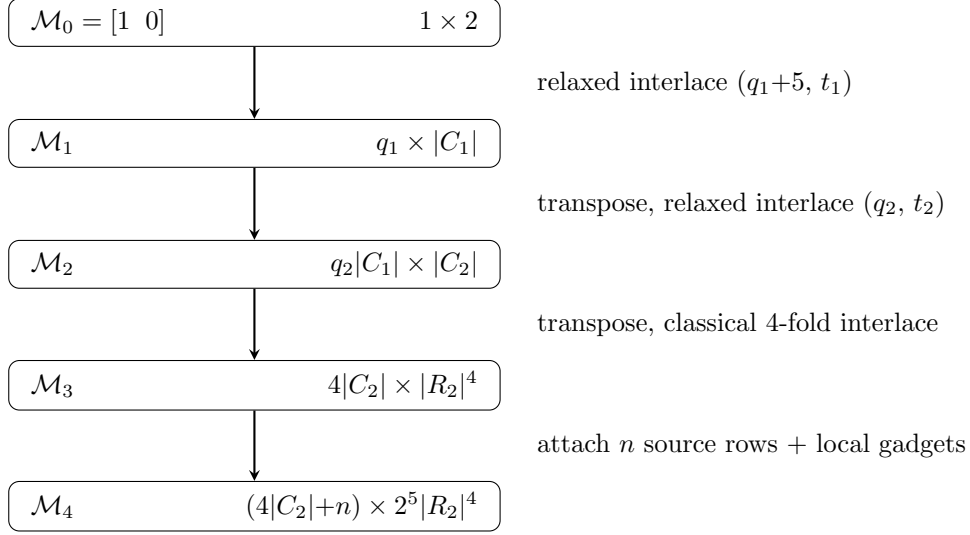


Figure 2: The reduction chain. Each box shows the stage matrix (left) and its row $\times$ column dimensions (right). Arrows indicate the operation applied to the previous stage.

1. the first $\log m$ bits choose a bin p ;
2. the next $\lceil \log q_2 \rceil$ bits choose a dimension α ;
3. inside that (p, α) -branch, we further focus on the *diagonal slice* (the subset of columns in which all four Stage-2 outer-block indices equal α); on that slice, every vector i with $\sigma(i) = p$ and $v_i(\alpha) = 1$ contributes one active local row indexed by $\pi_\alpha(i)$, while every other vector row is neutral.

Formally, if we let

$$A_{p,\alpha} := \{i \in [n] : \sigma(i) = p \text{ and } v_i(\alpha) = 1\}, \quad \ell_{p,\alpha} := |A_{p,\alpha}|,$$

then the diagonal (p, α) -slice contains the q_1 template blocks that define $\mathcal{M}_1$, together with $\ell_{p,\alpha}$ active local rows and one neutral local row. Equivalently, it is a local restriction of the full $(q_1 + 5)$ -coordinate Stage 1 subgame over the common column family C_1 . A feasible packing gives $\ell_{p,\alpha} \leq 1$ for every (p, α) , while an infeasible packing produces some (p, α) with $\ell_{p,\alpha} \geq 2$.

Correctness roadmap. In the YES case, the intended protocol follows the scaffold and reduces every branch to an admissible Stage 1 gadget. In the structured NO case, Stage 3 control isolates a bin branch, Stage 2 control isolates one coordinate block inside that branch, and the resulting local Stage 4 gadget contains one extra active copy of $\mathcal{M}_0$, yielding the final one-bit gap. If a protocol does not spend its early bits in this prescribed way, Theorems 38 and 41 and lemma 26 already force it to exceed the target budget.

Lemma 44. *Suppose the preprocessed $\{0, 1\}$ - d -DIMENSION VECTOR BIN PACKING instance admits a canonical feasible packing $\sigma : [n] \rightarrow [4]$, so that $\sigma(z_p) = p$ for every $p \in [4]$. Let*

$$B_{\text{yes}} := \log 4 + \lceil \log q_2 \rceil + B_{\text{cap}}.$$

Then $\mathcal{M}_4$ has a deterministic protocol of cost at most B_{yes} .

Proof. Fix a canonical feasible packing σ . We describe a protocol in three phases.

Phase 1: choose a bin. Alice first communicates one of the 4 bin labels. If her input row is a template row $(p, c) \in R_3 = [4] \times C_2$, she sends p . If her input row is a vector row $i \in [n]$, she sends $\sigma(i)$. This costs $\log 4$ bits and partitions the rows according to the intended bin.

Phase 2: choose a dimension. After the bin p is fixed, only the p -th component of Bob's tuple is relevant to the template part. Writing $j_p \in R_2 = [q_2] \times C_1$ as $j_p = (\alpha, \gamma)$, Bob sends the outer index $\alpha \in [q_2]$. This costs $\lceil \log q_2 \rceil$ bits and fixes the dimension.

Phase 3: solve the residual capacity gadget. Consider the branch indexed by (p, α) . Let

$$A_{p,\alpha} := \{ i \in [n] : \sigma(i) = p \text{ and } v_i(\alpha) = 1 \}, \quad \ell_{p,\alpha} := |A_{p,\alpha}|.$$

Because σ is a feasible packing and $c = 1$, every bin contains at most one vector with a 1 in any fixed dimension, so

$$\ell_{p,\alpha} \leq 1 \quad \text{for every } (p, \alpha) \in [4] \times [q_2].$$

On this branch there are at most

$$q_1 + 2 = (2^a - 2) + 2 = 2^a$$

distinct row behaviours: the q_1 template blocks of $\widehat{\mathcal{M}}_1$ that define $\mathcal{M}_1$, together with at most one active vector behaviour and one neutral behaviour for all remaining vector rows. Alice can therefore identify her row behaviour using a bits, after which Bob, who knows the full column j , returns the output bit. Thus the residual local gadget has cost at most $a + 1 = B_{\text{cap}}$.

Summing the three phases, the total communication cost is at most

$$\log 4 + \lceil \log q_2 \rceil + B_{\text{cap}} = B_{\text{yes}}.$$

□

5.3 Protocol Control and the NO Case

We now prove the soundness direction. The argument has four parts: a local Stage 1 threshold, the Stage 2 control lemmas, a Stage 3 bin separation statement, and finally the local Stage 4 gadget that converts an overloaded bin/coordinate pair into a hard Stage 1 instance.

More explicitly, the reduction first proves the local Stage 1 threshold that separates admissible branches from overloaded ones. It then certifies the Stage 2 input conditions needed to force the dimension-isolation scaffold, uses the classical separation lemma once more to isolate the Stage 3 bin branch, and finally turns the resulting local overloaded bin/coordinate pair (p, α) into a hard Stage 1 gadget.

Local Stage 1 threshold. For any column family $T \subseteq \text{Cols}(\mathcal{M}_0)^{2^a+3}$ and any integer $1 \leq r \leq 2^a + 3$, write

$$\mathcal{H}_r(T) := \mathcal{M}_0 \langle 2^a + 3, T \rangle \upharpoonright_{[r] \times \text{Rows}(\mathcal{M}_0), T}.$$

In words, $\mathcal{H}_r(T)$ is the restriction of the full $(2^a + 3)$ -coordinate relaxed interlace to its first r outer row blocks.

Lemma 45 (Stage-1 threshold). *Let*

$$S_1 := S_{2^a+3, t_1}(\text{Cols}(\mathcal{M}_0)).$$

Then

$$D(\mathcal{H}_{2^a}(S_1)) = a + 1, \quad D(\mathcal{H}_{2^a+1}(S_1)) = a + 2.$$

Proof. The proof is given in Appendix C. It proceeds directly by bounding the size of monochromatic rectangles in $\mathcal{H}_r(S_1)$ using the balancedness of S_1 , and then applying a heavy-path argument to any putative protocol of depth a (for $r = 2^a$) or $a + 1$ (for $r = 2^a + 1$). The appendix shows that the leaf reached along a heavy path would retain too many rows and too many columns to be monochromatic, yielding the claimed equalities. $\square$

Remark 46. Writing $B_{\text{cap}} := D(\mathcal{H}_{2^a}(S_1))$ where $S_1 := S_{2^a+3,t_1}(\text{Cols}(\mathcal{M}_0))$, the threshold lemma gives: for $G_\ell := \mathcal{H}_{2^a-1+\ell}(S_1)$, $\ell \leq 1 \implies D(G_\ell) \leq B_{\text{cap}}$ (by subgame monotonicity, since G_ℓ is a restriction of $\mathcal{H}_{2^a}(S_1)$), and $\ell \geq 2 \implies D(G_\ell) \geq B_{\text{cap}} + 1$ (since $\mathcal{H}_{2^a+1}(S_1)$ is a subgame of G_ℓ).

Corollary 47. Let

$$S_1 := S_{2^a+3,t_1}(\text{Cols}(\mathcal{M}_0)), \quad \varepsilon_1 := \varepsilon_{2^a+3,t_1}.$$

Fix $S' \subseteq S_1$ with

$$|S'| \geq (1 - \rho)|S_1|$$

for some $\rho \in [0, 1)$ satisfying

$$\rho < \frac{1 - \varepsilon_1}{2}.$$

For $0 \leq \ell \leq 4$, write

$$G'_\ell := \mathcal{H}_{2^a-1+\ell}(S').$$

Then

$$\ell \leq 1 \implies D(G'_\ell) \leq B_{\text{cap}}, \quad \ell \geq 2 \implies D(G'_\ell) \geq B_{\text{cap}} + 1.$$

Proof. The proof is given in Appendix C. It repeats the Stage 1 threshold argument on a dense subfamily S' , showing that a heavy leaf still retains too many rows and columns to be monochromatic. $\square$

Corollary 48 (Chosen-coordinate dense Stage-1 threshold). Let

$$S_1 := S_{2^a+3,t_1}(\text{Cols}(\mathcal{M}_0)), \quad \varepsilon_1 := \varepsilon_{2^a+3,t_1}.$$

Fix $S' \subseteq S_1$ with

$$|S'| \geq (1 - \rho)|S_1|$$

for some $\rho \in [0, 1)$ satisfying

$$\rho < \frac{1 - \varepsilon_1}{2}.$$

Let $Q \subseteq [2^a + 3]$ satisfy

$$|Q| \geq 2^a + 1.$$

Then

$$D(\mathcal{M}_0 \langle 2^a + 3, S' \rangle \upharpoonright_{Q \times \text{Rows}(\mathcal{M}_0), S'}) \geq B_{\text{cap}} + 1.$$

Proof. The proof is given in Appendix C. The appendix trims to the chosen outer coordinates and then applies the dense Stage 1 threshold to the resulting local subgame. $\square$

Stage 1 properties.

The next two Stage 1 lemmas are the pieces that later get reused inside the Stage 2 and final NO-case arguments. The first gives the residual Stage 1 lower bound that survives at the smaller Stage 2 column densities, and the second turns that lower bound into the exact Stage 1 complexity statement.

Lemma 49. *For $\mathcal{M}_0 = [1 \ 0]$, any positive integer p , and reals $0 < x, y \leq 1$ satisfying $p + \log y > 0$, one has*

$$D(\mathcal{M}_0[p, x, y]) \geq \lceil \log(p + \log y) \rceil + 1.$$

Proof. This is [9, Lemma 2.5]: the rank of the classical interlace of $\mathcal{M}_0 = [1 \ 0]$ gives the stated bound. $\square$

Lemma 50 (Residual Stage-1 hardness at Stage-2 densities). *Let*

$$h_2^\downarrow := 2^{-(b_1 + \log r_2)}, \quad h_2'^\downarrow := \frac{2^{-b'_1}}{16t_1}.$$

For every power of two $1 \leq r' \leq r_1$, any extraction

$$N = \mathcal{M}_1 \upharpoonright_{R', C'}$$

such that R' is a $(Q, 1)$ -equipartition with

$$|Q| = \frac{9}{16} r' t_1$$

and

$$\frac{|C'|}{|C_1|} \geq h_2^\downarrow$$

satisfies

$$D(N) \geq D(\mathcal{M}_0) + \log t_1 + \log r'.$$

Proof. The proof is given in Appendix C. It combines the rank lower bound for $\mathcal{M}_0$ with the Stage 1 parameter choices to show that the reduced Stage 2 column densities still retain the full residual Stage 1 hardness. $\square$

Corollary 51.

$$D(\mathcal{M}_1) = D(\mathcal{M}_0) + \log t_1 + \log r_1 = a + 1 = B_{\text{cap}}.$$

Proof. The upper bound is the direct protocol that sends the active Stage 1 row index using $\lceil \log q_1 \rceil = a$ bits and then lets Bob return the output bit. For the lower bound, apply Lemma 50 with $r' = r_1$, any $Q \subseteq [q_1]$ of size $\frac{9}{16} r_1 t_1$, and the full column set C_1 . Since $1 \geq h_2^\downarrow$, this gives

$$D(\mathcal{M}_1) \geq D(\mathcal{M}_0) + \log t_1 + \log r_1 = 1 + \log(q_1 + 2) = a + 1.$$

The first and last displayed equalities use $q_1 + 2 = r_1 t_1 = 2^a$. $\square$

Lemma 52. $\mathcal{M}_1^\top$ is (δ, b_1) -robust.

Proof. The proof is given in Appendix C. The appendix checks the four robustness clauses separately for $\mathcal{M}_1^\top$, using the Stage 1 threshold package and the residual Stage 1 hardness established above. $\square$

Lemma 53. *Let*

$$y_2^{\text{term}} := \left(\frac{h_2 2^{-(\log q_2 + D(\mathcal{M}_1))}}{1 + \varepsilon_{q_2, t_2}} \right)^{1/t_2}.$$

Then

$$D\left(\mathcal{M}_1 \left[1, y_2^{\text{term}}, 2^{-b_1}\right]\right) \geq D(\mathcal{M}_1) - \log r_1.$$

Proof. The proof is given in Appendix C. It estimates the terminal one-copy density seen in Stage 2 and compares it against the residual Stage 1 lower bound from Lemma 50. $\square$

Stage 2 certification.

These are the Stage 2 certification lemmas. They verify the Stage 2 column-loss resilience and hard-seed hypotheses for $\mathcal{M}_1^\top$, then derive the resulting complexity and near-exact separation statements for the dimension gadget $\mathcal{M}_2$. Clause (i) of the Stage 2 column-loss resilience hypothesis is the part reused by the extension theorem route, but we keep the fuller two-clause hypothesis here because the later Stage 2 lemmas and dense row variants repeatedly use the same assumptions. Thus Lemma 54 certifies the base matrix $M = \mathcal{M}_1^\top$ used to build $\mathcal{M}_2$, not $\mathcal{M}_2$ itself.

For convenience, Table 1 summarises the theorem-level bookkeeping reused throughout the Stage 2 arguments below.

Lemma 54 (Stage-2 column-loss resilience). *Let*

$$M := \mathcal{M}_1^\top, \quad q := q_2, \quad t := t_2, \quad h := h_2.$$

Then (M, b_1) is (q, t, h) -column-loss resilient.

Proof. The proof is given in Appendix C. The appendix verifies both clauses of column-loss resilience for $M = \mathcal{M}_1^\top$: the terminal one-copy bound and the residual three-rung bounds after every allowed amount of Stage 2 column loss. $\square$

Lemma 55 (Stage-2 hard seed). *Let*

$$p_2^{\text{seed}} := \frac{9}{16} t_2, \quad x_2^{\text{seed}} := 2^{-b_1} t_2.$$

Then

$$D\left(\mathcal{M}_1^\top \left[p_2^{\text{seed}}, x_2^{\text{seed}}, h_2'\right]\right) \geq D(\mathcal{M}_1) + \log t_2.$$

Proof. Apply Lemma 33 with $M := \mathcal{M}_1^\top$, $j := \Delta = 5$, $t := t_2$, $b := b_1$. Since $\mathcal{M}_1^\top$ is (δ, b_1) -robust by Lemma 52, $D(\mathcal{M}_1) \geq 3$, and $t_2 \leq 2^{b_1}$, the hard-seed lemma gives the required bound at density $2^{-2^{0.49}\sqrt{\log t_2}}$. Because $h_2' = (\log d)^{-8}$ dominates this for large d , monotonicity yields the claim. $\square$

Corollary 56.

$$D(\mathcal{M}_2) = D(\mathcal{M}_1) + \lceil \log q_2 \rceil = B_{\text{cap}} + \lceil \log q_2 \rceil.$$

Proof. The proof is given in Appendix C. It combines the direct upper-bound protocol with the Stage 2 lower-bound input delivered by the extension-theorem route at full density. $\square$

Lemma 57. $\mathcal{M}_2^\top$ is (δ, b_2) -robust.

Proof. The proof is given in Appendix C. The appendix checks the four robustness clauses for $\mathcal{M}_2^\top$ using the Stage 2 lower bound and the structure of the relaxed interlace. $\square$

Stage 2 separation.

Verification checkpoint. We apply Theorem 41 with

$$\begin{aligned} M &:= \mathcal{M}_1^\top, & b &:= b_1, & t &:= t_2, & r &:= r_2, \\ h &:= 1, & h_{\text{seed}} &:= h'_2, & p_{\text{seed}} &:= \frac{9}{16}t_2, & x_{\text{seed}} &:= 2^{-b_1}t_2. \end{aligned}$$

For this choice, the relaxed interlace $\widehat{M}$ appearing in the theorem statement is exactly $\mathcal{M}_2$. The proof below checks the theorem hypotheses in the same order in which they are used; see also Table 1 for the full Stage 2 checklist.

Lemma 58 (Stage-2 near-exact separation). *Let Π be any deterministic protocol for $\mathcal{M}_2$ of total depth at most $D(\mathcal{M}_1) + \log q_2$. After the first $\log q_2$ bits of Π , which are necessarily row questions, each surviving rectangle $R \times C$ satisfies*

$$\begin{aligned} \exists i^* \in [q_2] \text{ such that } & |R \cap (\{i^*\} \times C_1)| \geq |C_1| - (q_2 - 1) \left\lceil 2^{-b_1+1}|C_1| \right\rceil, \\ & |R \cap (\{i\} \times C_1)| < \left\lceil 2^{-b_1+1}|C_1| \right\rceil \quad \text{for all } i \neq i^*. \end{aligned}$$

In particular, every protocol meeting the tight $D(\mathcal{M}_1) + \log q_2$ bound must spend its first $\log q_2$ bits isolating the Stage 2 outer block.

Proof. Set

$$\varepsilon_2 := \varepsilon_{q_2, t_2}, \quad p_2^{\text{seed}} := \frac{9}{16}t_2, \quad x_2^{\text{seed}} := 2^{-b_1}t_2.$$

We apply Theorem 41 with

$$M := \mathcal{M}_1^\top, \quad b := b_1, \quad t := t_2, \quad r := r_2, \quad h := 1, \quad h_{\text{seed}} := h'_2.$$

The relaxed interlace $\widehat{M}$ in the theorem statement is exactly $\mathcal{M}_2$.

The base matrix is (δ, b) -robust by Lemma 52, and $D(M) = D(\mathcal{M}_1) \geq 2$ by Corollary 51 and lemma 8. By Lemma 54, the matrix $(\mathcal{M}_1^\top, b_1)$ is (q_2, t_2, h_2) -column-loss resilient. Since the density thresholds in Definition 34 are the quantities $y_c(h)$, replacing h_2 by $h = 1$ only increases the required column density, so the same one-copy and residual lower bounds remain valid by Lemma 6. The hard-seed hypothesis is Lemma 55. The seed row threshold satisfies

$$2^{-b_1} \leq x_2^{\text{seed}} \leq \frac{t_2}{q_2} = \frac{1}{r_2},$$

since $t_2 \geq 1$ and $q_2 < 2d \leq d^2 = 2^{b_1}$ for all sufficiently large d . The required seed comparison is

$$\frac{2^{-(\log t_2 + D(\mathcal{M}_1))}}{1 + \varepsilon_2} \geq h'_2,$$

which holds for all sufficiently large d by the same calculation as in the proof of Corollary 56. The density assumption $2(\frac{1}{2} + \delta)^2 \leq 1/(1 + \varepsilon_2)$ holds for all sufficiently large d , since the left-hand side is 0.72 and $\varepsilon_2 = o(1)$. Finally,

$$q_2 \left\lceil 2^{-b_1+1}|C_1| \right\rceil \leq q_2(2^{-b_1+1}|C_1| + 1) \leq \frac{4}{d}|C_1| + q_2 < |C_1|$$

for all sufficiently large d , because $q_2 < 2d$ and $|C_1| \geq \frac{1}{2}d^{64}$ as noted in the Stage 1 construction above. The conclusion of Theorem 41 is exactly the displayed dominant-block statement. $\square$

Corollary 59 (Stage-2 separation on dense row restrictions). *Let $S \subseteq C_2$ satisfy*

$$|S| \geq (1 - 8h_2) |C_2|.$$

Let Π be any deterministic protocol for

$$\mathcal{M}_2^\top \upharpoonright_{S, [q_2] \times C_1}$$

of total depth at most $D(\mathcal{M}_1) + \log q_2$. Then on every branch, the first $\log q_2$ bits are column bits.

Let $\mathcal{P}$ denote the resulting partition after those first $\log q_2$ bits. Then every rectangle $R \times C \in \mathcal{P}$ satisfies the dominant-block conclusion of Lemma 58, with rows and columns interchanged.

Proof. Transpose Π . This gives a deterministic protocol for

$$\widehat{M}_S := \mathcal{M}_2 \upharpoonright_{[q_2] \times C_1, S}$$

of the same depth. Set

$$\sigma := \frac{|S|}{|C_2|} \geq 1 - 8h_2.$$

For all sufficiently large d , we have

$$2\left(\frac{1}{2} + \delta\right)^2 \leq \frac{\sigma}{1 + \varepsilon_{q_2, t_2}}.$$

The matrix $\widehat{M}_S$ is exactly the submatrix N from Theorem 41 obtained by taking $C' = S$ inside the same Stage 2 relaxed interlace used to define $\mathcal{M}_2$. We therefore apply Theorem 41 to $\widehat{M}_S$ with the same Stage 2 parameters as in Lemma 58,

$$M := \mathcal{M}_1^\top, \quad b := b_1, \quad t := t_2, \quad r := r_2, \quad q := q_2,$$

but with density parameter $h := \sigma$ in place of $h = 1$. The robustness, complexity, and seed lower-bound inputs are exactly the same ones verified in the proof of Lemma 58. Since $\sigma \geq 1 - 8h_2 \geq h_2$, every threshold $y_c(\sigma)$ is at least the corresponding threshold from the Stage 2 certification package. Hence the same column-loss-resilience lower bounds remain valid by Lemma 6, and the seed comparison only improves at the larger density σ . The displayed large- d inequality above supplies the only new separation side condition. Therefore Theorem 41 yields that the first $\log q_2$ bits of the transposed protocol are row bits, together with the same dominant-block conclusion. Transposing back gives exactly the stated column-bit and dominant-block conclusions. $\square$

Stage 3: isolate the bin branch.

Lemma 60 (Stage-3 bin separation). *Let Π be any deterministic protocol for $\mathcal{M}_3$ of total depth at most $D(\mathcal{M}_2) + 2$. Then on every branch, the first $2 = \log 4$ bits are row bits.*

Let $\mathcal{P}$ denote the resulting partition after those first $2 = \log 4$ bits. Then every rectangle $R \times C \in \mathcal{P}$ has a unique block $i^ \in [4]$ such that*

$$|R \cap (\{i^*\} \times C_2)| \geq |C_2| - 3 \left\lceil 2^{-b_2+1} |C_2| \right\rceil,$$

and

$$|R \cap (\{i\} \times C_2)| < \left\lceil 2^{-b_2+1} |C_2| \right\rceil \quad \text{for all } i \neq i^*.$$

In particular,

$$D(\mathcal{M}_3) = D(\mathcal{M}_2) + 2.$$

Proof. The upper bound is the direct protocol that chooses one of the four outer blocks and then solves $\mathcal{M}_2^\top$.

For the matching lower bound, write

$$y_0 := \left(\frac{1}{2} + \delta\right)^2.$$

Apply Corollary 22 with

$$M := \mathcal{M}_2^\top, \quad p := 2, \quad k := 1, \quad s := 0, \quad x := 2^{1-b_2}, \quad y := y_0, \quad H := D(\mathcal{M}_2) + 1.$$

By Lemma 57 and Corollary 15, the three seed inequalities needed for Corollary 22 hold. Hence

$$D\left(\mathcal{M}_2^\top \left[4, 2^{2-b_2}, y_0\right]\right) \geq D(\mathcal{M}_2) + 2.$$

Since $2^{2-b_2} \leq 1$ and $y_0 \leq 1$, monotonicity gives

$$D(\mathcal{M}_3) = D\left(\mathcal{M}_2^\top [4, 1, 1]\right) \geq D(\mathcal{M}_2) + 2.$$

Therefore $D(\mathcal{M}_3) = D(\mathcal{M}_2) + 2$.

Now apply Lemma 26 with

$$M := \mathcal{M}_2^\top, \quad q := 4, \quad x := 1, \quad y := 1.$$

The matrix $\mathcal{M}_2^\top$ is (δ, b_2) -robust by Lemma 57, and $D(\mathcal{M}_2^\top) = D(\mathcal{M}_2) \geq 2$. Also

$$4 \cdot 2^{-b_2} \leq \frac{4}{(\log d)^3} \leq 1$$

for large d , and

$$4 \left\lceil 2^{-b_2+1} |C_2| \right\rceil < |C_2|$$

for all sufficiently large d . Thus all hypotheses of Lemma 26 are satisfied, and its conclusion is exactly the stated first-bit and dominant-block conclusions. $\square$

Stage 4: local gadgets.

At this point the scaffold has isolated a Stage 3 bin and a Stage 2 coordinate. The remaining task is local: identify the precise Stage 1 subgame that survives on that diagonal slice, and then show that an overloaded bin/coordinate pair (p, α) contributes one extra local coordinate and therefore one extra bit of complexity. The local geometry is summarised in Figure 3. The notation used in that figure is formalised in the lemmas immediately below; in particular, $\hat{X}_{p,\alpha}$, $A_\alpha(B_p)$, $Q_{p,\alpha}(B_p)$, $\ell_\alpha(B_p)$, and $\nu_\alpha(B_p)$ are defined there, and the figure is only a roadmap for that local argument.

Lemma 61. *Fix a bin label $p \in [4]$ and a dimension $\alpha \in [q_2]$. Define the diagonal outer-block family*

$$D_{p,\alpha} := \{(k, (j_1, j_2, j_3, j_4)) \in C_4 : j_m = (\alpha, \gamma_m) \in \{\alpha\} \times C_1 \text{ for all } m \in [4]\}.$$

For each 5-bit shift $s \in [2^5]$, define the shifted compatibility slice

$$X_{p,\alpha}^{(s)} := \{(k, (j_1, j_2, j_3, j_4)) \in D_{p,\alpha} : k = \text{tail}(\gamma_p) \oplus s\}.$$

Then $D_{p,\alpha}$ is the disjoint union of the 2^5 equal slices $\{X_{p,\alpha}^{(s)}\}_{s \in [2^5]}$. Consequently, if $C \subseteq D_{p,\alpha}$ satisfies

$$|C| \geq (1 - \eta) |D_{p,\alpha}|,$$

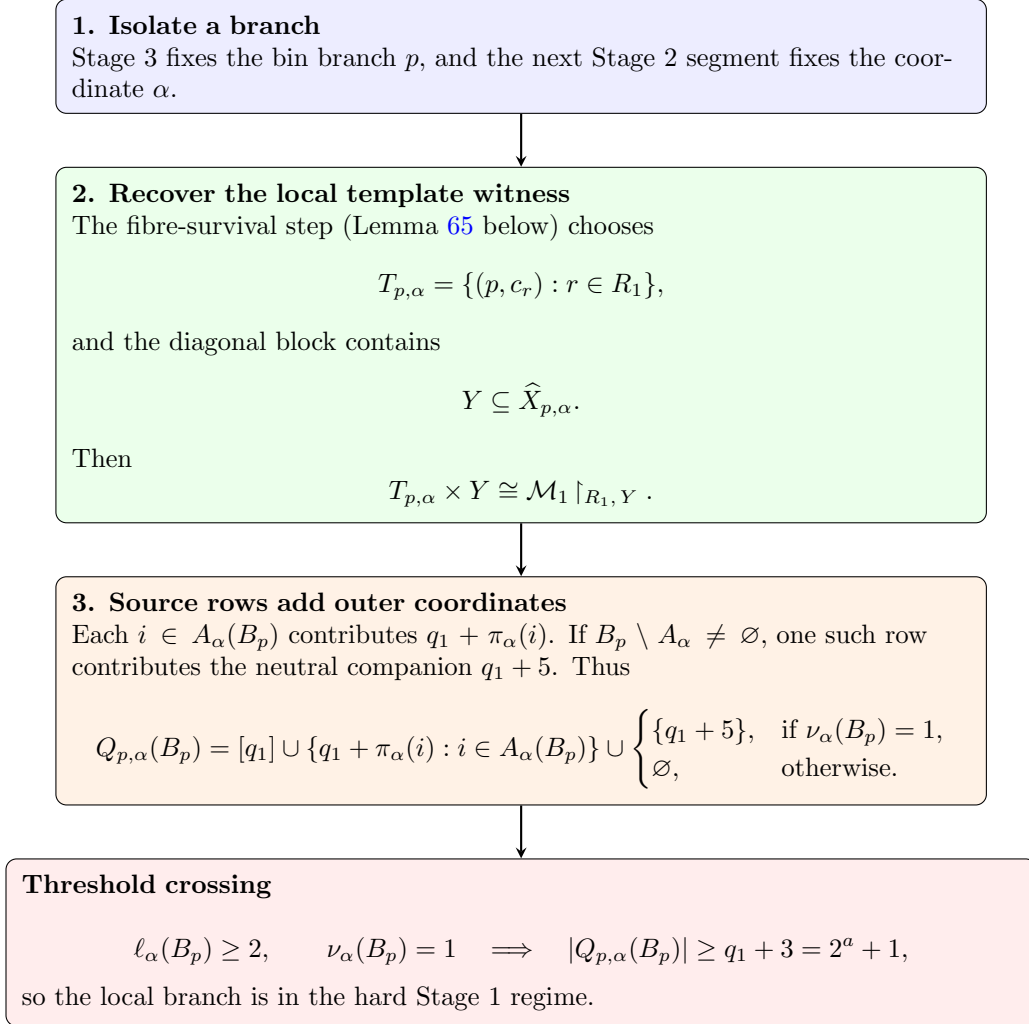


Figure 3: Local picture of the Stage 4 gadget. After Stage 3 isolates a bin p and coordinate α , the template side yields a dense local copy of $\mathcal{M}_1$. The rows in B_p then add active outer coordinates, and a row with $v_i(\alpha) = 0$ supplies the neutral companion. Hence $\ell_\alpha(B_p) \geq 2$ and $\nu_\alpha(B_p) = 1$ force the local branch into the hard Stage 1 regime.

then some shift $s \in [2^5]$ obeys

$$|C \cap X_{p,\alpha}^{(s)}| \geq (1 - 2^5 \eta) |X_{p,\alpha}^{(s)}|.$$

Moreover, replacing k by $k \oplus s$ is a column permutation of the 5-row local gadget, so a subgame extracted on $X_{p,\alpha}^{(s)}$ is isomorphic to the corresponding subgame on $X_{p,\alpha}^{(0)}$.

Proof. For fixed $(\gamma_1, \dots, \gamma_4) \in C_1^4$, exactly one value of $k \in [2^5]$ lands in each slice $X_{p,\alpha}^{(s)}$, namely $k = \text{tail}(\gamma_p) \oplus s$. Hence the slices are disjoint, cover $D_{p,\alpha}$, and all have the same cardinality $|D_{p,\alpha}|/2^5$. If every slice met C in fewer than $(1 - 2^5 \eta)|X_{p,\alpha}^{(s)}|$ columns, then summing over the 2^5 slices would give $|C| < (1 - \eta)|D_{p,\alpha}|$, contradiction. The final claim is immediate because $k \mapsto k \oplus s$ permutes the columns of $\mathcal{M}_0 \langle 5 \rangle$. $\square$

Lemma 62. Fix a bin label $p \in [4]$ and let $B \subseteq [n]$ be the vector rows retained on the corresponding bin branch. Fix a dimension $\alpha \in [q_2]$ and set

$$A_\alpha(B) := B \cap A_\alpha, \quad \ell_\alpha(B) := |A_\alpha(B)|, \quad \nu_\alpha(B) := \begin{cases} 1, & \text{if } B \setminus A_\alpha \neq \emptyset, \\ 0, & \text{otherwise.} \end{cases}$$

Restrict further to the diagonal compatibility slice

$$X_{p,\alpha} := \left\{ (k, (j_1, j_2, j_3, j_4)) \in C_4 : \begin{array}{l} j_m = (\alpha, \gamma_m) \in \{\alpha\} \times C_1 \text{ for all } m \in [4], \\ k = \text{tail}(\gamma_p) \end{array} \right\}.$$

Let

$$Q_{p,\alpha}(B) := [q_1] \cup \{q_1 + \pi_\alpha(i) : i \in A_\alpha(B)\} \cup \begin{cases} \{q_1 + 5\}, & \text{if } \nu_\alpha(B) = 1, \\ \emptyset, & \text{if } \nu_\alpha(B) = 0. \end{cases}$$

Then the selected branch of $\mathcal{M}_4$, restricted to the template rows $\{p\} \times C_2$, the vector rows in B , and the columns $X_{p,\alpha}$, contains a subgame obtained from

$$\mathcal{M}_0 \langle 2^a + 3, S_{2^a+3, t_1}(\text{Cols}(\mathcal{M}_0)) \rangle \upharpoonright_{Q_{p,\alpha}(B) \times \text{Rows}(\mathcal{M}_0), S_{2^a+3, t_1}(\text{Cols}(\mathcal{M}_0))}$$

by adding duplicate rows and duplicate columns. In particular, if $\nu_\alpha(B) = 1$ and $\ell_\alpha(B) \geq 2$, then

$$D(\text{branch}) \geq B_{\text{cap}} + 1.$$

Proof. The proof is given in Appendix C. It matches the surviving template rows and vector rows on the compatibility slice with the corresponding outer coordinates of the $(2^a + 3)$ -coordinate Stage 1 interlace, and then reads off the threshold consequence. $\square$

Lemma 63 (Canonical diagonal copy). Fix a bin label $p \in [4]$. For each $\alpha \in [q_2]$, define

$$\widehat{X}_{p,\alpha} := \{(\text{tail}(\gamma), ((\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma))) : \gamma \in C_1\} \subseteq X_{p,\alpha}.$$

Let

$$\widehat{D}_p := \bigcup_{\alpha \in [q_2]} \widehat{X}_{p,\alpha}.$$

Then, after identifying each column of $\widehat{X}_{p,\alpha}$ with the Stage 2 column $(\alpha, \gamma) \in [q_2] \times C_1 = R_2$, the restriction of $\mathcal{M}_4$ to the template rows $\{p\} \times C_2$ and the columns $\widehat{D}_p$ is exactly $\mathcal{M}_2^\top$.

Proof. Fix $\alpha \in [q_2]$ and $\gamma \in C_1$, and write

$$\hat{j}_{\alpha,\gamma} := (\text{tail}(\gamma), ((\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma))) \in \hat{X}_{p,\alpha}.$$

For any template row $(p, c) \in \{p\} \times C_2$, the Stage 4 definition gives

$$\mathcal{M}_4((p, c), \hat{j}_{\alpha,\gamma}) = \mathcal{M}_3((p, c), ((\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma))).$$

On the p -th Stage 3 branch, the template part depends only on the p -th Stage 2 component j_p , so the right-hand side equals

$$\mathcal{M}_2^\top(c, (\alpha, \gamma)) = \mathcal{M}_2((\alpha, \gamma), c).$$

Thus the restricted template matrix is exactly $\mathcal{M}_2^\top$. $\square$

Corollary 64 (Canonical dense local gadget). *In the setup of Lemma 62, fix the canonical diagonal block $\hat{X}_{p,\alpha} \subseteq X_{p,\alpha}$ from Lemma 63. If $Y \subseteq \hat{X}_{p,\alpha}$ satisfies*

$$|Y| \geq (1 - \rho) |\hat{X}_{p,\alpha}|$$

for some $\rho \in [0, 1)$, then there exists $S' \subseteq S_{2^a+3,t_1}(\text{Cols}(\mathcal{M}_0))$ with

$$|S'| \geq (1 - \rho) |S_{2^a+3,t_1}(\text{Cols}(\mathcal{M}_0))|$$

such that the selected branch of $\mathcal{M}_4$, restricted to the template rows $\{p\} \times C_2$, the vector rows in B , and the columns Y , contains a subgame obtained from

$$\mathcal{M}_0 \langle 2^a + 3, S' \rangle \upharpoonright_{Q_{p,\alpha}(B) \times \text{Rows}(\mathcal{M}_0), S'}$$

by adding duplicate rows and duplicate columns. Consequently, if $\nu_\alpha(B) = 1$, $\ell_\alpha(B) \geq 2$, and

$$\rho < \frac{1 - \varepsilon_{2^a+3,t_1}}{2},$$

then

$$D(\text{branch}) \geq B_{\text{cap}} + 1.$$

Proof. The proof is given in Appendix C. It pulls back a dense subset of the canonical diagonal block to a dense Stage 1 column family and then applies the chosen-coordinate dense Stage 1 threshold. $\square$

Lemma 65 (Fibre survival after Stage-3 row loss). *Let*

$$C_2 = S_{q_2,t_2}(R_1) \quad \text{and} \quad \varepsilon_2 := \varepsilon_{q_2,t_2}.$$

For each $\beta \in [q_2]$ and $r \in R_1$, define the coordinate fibre

$$F_{\beta,r} := \{c \in C_2 : c_\beta = r\}.$$

Then

$$\frac{1 - \varepsilon_2}{|R_1|} |C_2| \leq |F_{\beta,r}| \leq \frac{1 + \varepsilon_2}{|R_1|} |C_2|.$$

Consequently, if $S \subseteq C_2$ satisfies

$$|S| > \left(1 - \frac{1 - \varepsilon_2}{|R_1|}\right) |C_2|,$$

then for every $\beta \in [q_2]$,

$$\{c_\beta : c \in S\} = R_1.$$

Proof. Because C_2 is (q_2, t_2) -balanced and $\varepsilon_2 < 1$, every value $r \in R_1$ occurs in coordinate β with frequency $\frac{1}{|R_1|} \pm \frac{\varepsilon_2}{|R_1|}$, which is exactly the displayed fibre bound.

Now suppose $S \subseteq C_2$ has the stated density and fix $\beta \in [q_2]$. If some $r \in R_1$ were missing from the coordinate set $\{c_\beta : c \in S\}$, then $F_{\beta,r} \subseteq C_2 \setminus S$, hence

$$|C_2 \setminus S| \geq |F_{\beta,r}| \geq \frac{1 - \varepsilon_2}{|R_1|} |C_2|,$$

contradicting the hypothesis on $|S|$. Therefore $\{c_\beta : c \in S\} = R_1$. $\square$

Stopping-time leaves on $\mathcal{M}_3$. Before bringing the vector rows back into the argument, we analyse the pure template matrix $\mathcal{M}_3$ by itself. The next lemma is the stopping-time statement (the point at which all Stage 3 and Stage 2 bits have been spent) that we use later: after those forced bits, every budget-tight branch already contains the exact Stage 1 row types together with a dense Stage 1 column set.

Lemma 66 (Budget-tight $\mathcal{M}_3$ leaves are approximate $\mathcal{M}_1$ leaves). *Write*

$$\eta_2 := q_2 2^{-b_1+1}.$$

Let Π be any deterministic protocol for $\mathcal{M}_3$ of total depth at most $D(\mathcal{M}_2) + 2$. After the first two bits, the surviving rectangles can be relabelled as

$$\{R_p \times C_p : p \in [4]\}$$

so that, for

$$S_p := \{c \in C_2 : (p, c) \in R_p\},$$

one has

$$|S_p| \geq |C_2| - 3 \left\lceil 2^{-b_2+1} |C_2| \right\rceil \geq (1 - 8h_2) |C_2|$$

for every $p \in [4]$.

Fix $p \in [4]$, and identify

$$N_p := \mathcal{M}_2^\top \upharpoonright_{S_p, [q_2] \times C_1}$$

with the restriction of $\mathcal{M}_3$ to $\{p\} \times S_p$ and the diagonal column set

$$D^{(3)} := \{((\beta, \gamma), (\beta, \gamma), (\beta, \gamma), (\beta, \gamma)) : (\beta, \gamma) \in [q_2] \times C_1\}.$$

Then, in the protocol induced by Π on N_p , after its first $\log q_2$ bits the surviving rectangles can be relabelled as

$$\{R_{p,\alpha} \times C_{p,\alpha} : \alpha \in [q_2]\}$$

so that, for every $\alpha \in [q_2]$, there is a set $Y_{p,\alpha} \subseteq C_1$ with

$$|Y_{p,\alpha}| \geq (1 - \eta_2) |C_1|$$

and

$$\{((\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma)) : \gamma \in Y_{p,\alpha}\} \subseteq C_{p,\alpha}.$$

Moreover,

$$\{c_\alpha : c \in S_p\} = R_1.$$

Hence, for each $r \in R_1$, one may choose $c_{\alpha,r} \in S_p$ with $(c_{\alpha,r})_\alpha = r$. If

$$T_{p,\alpha} := \{(p, c_{\alpha,r}) : r \in R_1\},$$

then, after identifying the column

$$((\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma))$$

with $\gamma \in C_1$, the restriction of $\mathcal{M}_3$ to $T_{p,\alpha}$ and the above diagonal columns is exactly

$$\mathcal{M}_1 \upharpoonright_{R_1, Y_{p,\alpha}}.$$

Proof. By Lemma 60, after the first two bits the surviving rectangles may be relabelled by their unique dominant Stage 3 block. Distinct two-bit transcripts cannot share the same dominant block, because

$$2^{-b_2+1} \cdot 4 = \frac{8}{(\log d)^3} < \frac{1}{2}$$

for all sufficiently large d . The same counting argument as in the proof of Lemma 60 shows that every $p \in [4]$ occurs. Thus the two-bit rectangles can indeed be relabelled as $\{R_p \times C_p : p \in [4]\}$, and the displayed bound on $|S_p|$ is exactly the Stage 3 dominant-block conclusion.

Fix $p \in [4]$. Let

$$D^{(3)} := \{((\beta, \gamma), (\beta, \gamma), (\beta, \gamma), (\beta, \gamma)) : (\beta, \gamma) \in [q_2] \times C_1\} \subseteq C_3.$$

For any $(p, c) \in \{p\} \times S_p$ and any $((\beta, \gamma), (\beta, \gamma), (\beta, \gamma), (\beta, \gamma)) \in D^{(3)}$, the Stage 3 definition gives

$$\mathcal{M}_3((p, c), ((\beta, \gamma), (\beta, \gamma), (\beta, \gamma), (\beta, \gamma))) = \mathcal{M}_2^\top(c, (\beta, \gamma)).$$

Hence, after identifying the displayed diagonal column with $(\beta, \gamma) \in [q_2] \times C_1$, the restriction of $\mathcal{M}_3$ to $\{p\} \times S_p$ and $D^{(3)}$ is exactly

$$N_p := \mathcal{M}_2^\top \upharpoonright_{S_p, [q_2] \times C_1}.$$

The protocol subtree below $R_p \times C_p$ has depth at most

$$(D(\mathcal{M}_2) + 2) - 2 = D(\mathcal{M}_1) + \log q_2.$$

Therefore the induced protocol on N_p has depth at most the same quantity. Since $|S_p| \geq (1 - 8h_2)|C_2|$, Corollary 59 applies to the induced protocol on N_p . Therefore, after the next $\log q_2$ bits of the induced protocol, every surviving rectangle has a unique dominant Stage 2 block.

We claim that every $\alpha \in [q_2]$ occurs. Fix $\alpha \in [q_2]$ and define the pure diagonal α -block

$$Z_\alpha := \{((\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma)) : \gamma \in C_1\} \subseteq D^{(3)}.$$

The next $\log q_2$ bits of the induced protocol create at most q_2 rectangles. If none of them had dominant block α , then the transposed dominant-block conclusion from Corollary 59 would give

$$|C \cap Z_\alpha| < \left\lceil 2^{-b_1+1} |C_1| \right\rceil$$

for every such rectangle $R \times C$. Summing over at most q_2 rectangles would yield

$$|C_1| = |Z_\alpha| < q_2 \left\lceil 2^{-b_1+1} |C_1| \right\rceil \leq \eta_2 |C_1| + q_2,$$

contradicting $\eta_2|C_1| + q_2 < |C_1|$ for all sufficiently large d , because $\eta_2 \leq 4/d$, $q_2 < 2d$, and $|C_1| \geq \frac{1}{2}d^{64}$ in the Stage 1 construction above. Thus every $\alpha \in [q_2]$ occurs, and since there are at most q_2 such rectangles altogether, there are exactly q_2 of them and each $\alpha \in [q_2]$ labels exactly one rectangle. Hence the rectangles after the next $\log q_2$ bits of the induced protocol can be relabelled as $\{R_{p,\alpha} \times C_{p,\alpha} : \alpha \in [q_2]\}$.

For a fixed α , let

$$Y_{p,\alpha} := \{\gamma \in C_1 : ((\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma)) \in C_{p,\alpha}\}.$$

If $|Y_{p,\alpha}| < (1 - \eta_2)|C_1|$, then

$$|Z_\alpha \setminus C_{p,\alpha}| > \eta_2|C_1|.$$

Every other rectangle contributes fewer than $\lceil 2^{-b_1+1}|C_1| \rceil$ columns from Z_α , and

$$(q_2 - 1) \lceil 2^{-b_1+1}|C_1| \rceil \leq (q_2 - 1)(2^{-b_1+1}|C_1| + 1) = \eta_2|C_1| - 2^{-b_1+1}|C_1| + q_2 - 1$$

is strictly less than $\eta_2|C_1|$ for all sufficiently large d , because

$$2^{-b_1+1}|C_1| = \frac{2}{d^2}|C_1| \geq d^{62} > q_2 - 1.$$

Therefore

$$|C_1| < (1 - \eta_2)|C_1| + \eta_2|C_1| = |C_1|,$$

contradicting the same large- d inequality as above. Hence $|Y_{p,\alpha}| \geq (1 - \eta_2)|C_1|$.

Since $|S_p| \geq (1 - 8h_2)|C_2|$ and

$$8h_2 = \frac{8}{(\log d)^3} < \frac{1 - \varepsilon_{q_2,t_2}}{|R_1|}$$

for all sufficiently large d , Lemma 65 applies and gives

$$\{c_\alpha : c \in S_p\} = R_1.$$

Thus, for each $r \in R_1$, we may choose $c_{\alpha,r} \in S_p$ with $(c_{\alpha,r})_\alpha = r$.

Finally, for any $r \in R_1$ and $\gamma \in Y_{p,\alpha}$,

$$\begin{aligned} & \mathcal{M}_3((p, c_{\alpha,r}), ((\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma))) \\ &= \mathcal{M}_2^\top(c_{\alpha,r}, (\alpha, \gamma)) = \mathcal{M}_2((\alpha, \gamma), c_{\alpha,r}) = \mathcal{M}_1^\top(\gamma, (c_{\alpha,r})_\alpha) = \mathcal{M}_1(r, \gamma). \end{aligned}$$

Therefore the restriction of $\mathcal{M}_3$ to $T_{p,\alpha}$ and the displayed diagonal columns is exactly $\mathcal{M}_1 \upharpoonright_{R_1, Y_{p,\alpha}}$. $\square$

Size and explicitness. Before returning to the main correctness proof, we record the routine size and constructibility facts for the reduction.

Lemma 67 (Polynomial-time constructibility). *The matrix $\mathcal{M}_4$ has polynomial size and can be constructed in polynomial time from the source instance.*

Proof. By Remark 28, $|C_1| = |S_{q_1+5,t_1}(\text{Cols}(\mathcal{M}_0))| \leq \text{poly}(d)$ (since $q_1 = \text{poly}(\log d)$ and $t_1 = \Theta(\log d)$), and $|C_2| = |S_{q_2,t_2}(R_1)| \leq \text{poly}(d)$ (since $q_2 = d$ and $|R_1|^{t_2} = \text{poly}(d)$). Hence

$$|R_4| = 4|C_2| + n = n + \text{poly}(d) \quad \text{and} \quad |C_4| = 32|R_2|^4 = \text{poly}(d).$$

Therefore the truth table of $\mathcal{M}_4$ has size

$$|R_4| |C_4| = (n + \text{poly}(d)) \text{poly}(d),$$

which is polynomial in the size of the source instance. Both balanced families are constructible in polynomial time, and each entry of $\mathcal{M}_4$ is computable in polynomial time from the stage definitions and the source vector bits. $\square$

Lifting the scaffold structure back to $\mathcal{M}_4$. The next lemma transfers the pure template analysis on $\mathcal{M}_3$ to the actual reduction matrix $\mathcal{M}_4$. A budget-tight protocol cannot spend bits that are constant on the template restriction before the scaffold stopping time $2 + \log q_2$, so the vector rows inherit the same Stage 3 partition and are untouched during the Stage 2 isolation segment. Inside the proof we keep those tasks separate. First comes a stopping-time synchronisation claim showing that the actual branch cannot spend bits that are constant on the current template restriction. Only once that synchronisation is in place do we extract the dense local Stage 1 witness on each (p, α) -branch.

Lemma 68 (Scaffold structure on $\mathcal{M}_4$). *Write*

$$B_{\text{yes}} := \log 4 + \lceil \log q_2 \rceil + B_{\text{cap}} \quad \text{and} \quad \eta_2 := q_2 2^{-b_1+1}.$$

Fix any $k_0 \in [2^5]$ and let

$$X_3^{\text{temp}} := \{k_0\} \times R_2^4 \subseteq C_4.$$

Let Π be any deterministic protocol for $\mathcal{M}_4$ of total depth at most B_{yes} . Then there is a partition

$$[n] = B_1 \sqcup B_2 \sqcup B_3 \sqcup B_4$$

such that, for every pair $(p, \alpha) \in [4] \times [q_2]$, there is a partial branch of Π after its first $2 + \log q_2$ actual bits whose surviving vector rows are exactly B_p and whose surviving columns contain a set

$$Y_{p,\alpha} \subseteq \widehat{X}_{p,\alpha}$$

with

$$|Y_{p,\alpha}| \geq (1 - \eta_2) |\widehat{X}_{p,\alpha}|.$$

Writing

$$S'_{p,\alpha} := \{ \gamma \in C_1 : (\text{tail}(\gamma), ((\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma))) \in Y_{p,\alpha} \},$$

there are rows

$$T_{p,\alpha} := \{(p, c_{p,\alpha,r}) : r \in R_1\} \subseteq \{p\} \times C_2$$

such that the selected branch of $\mathcal{M}_4$, restricted to $T_{p,\alpha}$, the vector rows in B_p , and the columns $Y_{p,\alpha}$, contains a subgame obtained from

$$\mathcal{M}_0 \langle 2^a + 3, S'_{p,\alpha} \rangle \upharpoonright_{Q_{p,\alpha}(B_p) \times \text{Rows}(\mathcal{M}_0), S'_{p,\alpha}}$$

by adding duplicate rows and duplicate columns. In particular, the first two actual bits determine the partition $B_1, \dots, B_4$, and the next $\log q_2$ actual bits are column bits, so the vector rows are unchanged during the Stage 2 isolation segment.

Proof. We organise the proof in three steps. First we restrict to the template slice and record the stopping-time structure inherited from $\mathcal{M}_3$. Then we prove the branch-synchronisation claim that keeps the actual protocol aligned with that template branch up to the scaffold stopping time. Finally we extract the dense local Stage 1 witness on the synchronised (p, α) -branch.

Step 1: restrict to the template slice and record the stopping-time structure. This is the induced protocol on the pure template game obtained by fixing the 5-bit Stage 4 coordinate and discarding the vector rows. Restrict Π to the template slice

$$\mathcal{M}_4 \upharpoonright_{R_3, X_3^{\text{temp}}} = \mathcal{M}_3.$$

By Corollary 56 and lemma 60,

$$B_{\text{yes}} = D(\mathcal{M}_2) + 2 = D(\mathcal{M}_3).$$

Apply Lemma 66 to this induced protocol on $\mathcal{M}_3$.

Fix any $(p, \alpha) \in [4] \times [q_2]$. The corresponding template leaf from Lemma 66 contains an exact copy of

$$\mathcal{M}_1 \upharpoonright_{R_1, Y_{p,\alpha}^{\text{temp}}}$$

for some $Y_{p,\alpha}^{\text{temp}} \subseteq C_1$ with

$$|Y_{p,\alpha}^{\text{temp}}| \geq (1 - \eta_2) |C_1|.$$

Fix once and for all a set $Q \subseteq [q_1]$ of size

$$|Q| = \frac{9}{16} r_1 t_1.$$

Since $1 - \eta_2 \geq h_2^\downarrow$ for all sufficiently large d , the row subset $Q \times \text{Rows}(\mathcal{M}_0) \subseteq R_1$ and the column set $Y_{p,\alpha}^{\text{temp}}$ satisfy the hypotheses of Lemma 50 with $r' = r_1$. Here $Q \times \text{Rows}(\mathcal{M}_0)$ is exactly the $(Q, 1)$ -equipartition required by that lemma. Therefore every such template leaf has residual complexity at least

$$D(\mathcal{M}_0) + \log t_1 + \log r_1 = D(\mathcal{M}_1) = B_{\text{cap}}.$$

Write

$$t_0 := 2 + \log q_2.$$

Step 2: synchronise the actual branch with the template branch. This step proves that the actual protocol cannot spend extra queried bits whose values are already fixed on the current template restriction before the scaffold stopping time.

Induced template branch. Fix any branch of the induced protocol on $\mathcal{M}_3$. Along the corresponding actual branch of Π , delete every queried communication bit whose value is fixed on the current restriction to $R_3 \times X_3^{\text{temp}}$; the remaining queried bits are exactly the induced bits on that branch. Let u be the actual node reached immediately after the t_0 -th remaining queried bit, so its current rectangle, after restriction to $R_3 \times X_3^{\text{temp}}$, is exactly the induced rectangle after the first t_0 induced bits. We claim that u occurs at actual depth exactly t_0 .

Every induced bit comes from a genuine queried communication bit of Π , so u occurs at actual depth at least t_0 . Suppose instead that u occurs at actual depth at least $t_0 + 1$. Then, before the first t_0 induced bits are completed, some queried communication bit on the actual branch has value

fixed on the current restriction to $R_3 \times X_3^{\text{temp}}$. By construction, the current rectangle at u , after restriction to $R_3 \times X_3^{\text{temp}}$, is exactly the induced rectangle after those first t_0 bits. By the previous paragraph, that induced rectangle contains a subgame of communication complexity at least B_{cap} . The subtree of Π below u therefore gives a deterministic protocol for that subgame of depth at most

$$B_{\text{yes}} - (t_0 + 1) = B_{\text{cap}} - 1,$$

contradiction.

Actual branch in $\mathcal{M}_4$. Hence u occurs at actual depth exactly t_0 , so the first $2 + \log q_2$ actual bits of every relevant branch are exactly the first $2 + \log q_2$ induced bits of the protocol on $\mathcal{M}_3$. In particular, the first two actual bits are the Stage 3 row bits from Lemma 60. Relabelling the four resulting two-bit branches by their dominant Stage 3 blocks gives a partition

$$[n] = B_1 \sqcup B_2 \sqcup B_3 \sqcup B_4.$$

For each $p \in [4]$, let $S_p \subseteq C_2$ be the corresponding dense template row set from Lemma 66. Since $|S_p| \geq (1 - 8h_2)|C_2|$, Lemma 63 identifies the restriction of $\mathcal{M}_4$ to the rows $\{p\} \times S_p$ and the columns $\widehat{D}_p$ with

$$N_p := \mathcal{M}_2^\top \upharpoonright_{S_p, [q_2] \times C_1}.$$

This is a second synchronisation step, now inside a fixed Stage 3 branch p : the previous argument aligned Π with the induced protocol on $\mathcal{M}_3$, while the present one rules out extra actual queries that become constant only after the further restriction to the template rows $\{p\} \times S_p$ and the columns $\widehat{D}_p$.

Fix $p \in [4]$. The remaining depth below the two-bit p -branch is at most

$$B_{\text{yes}} - 2 = D(\mathcal{M}_1) + \log q_2.$$

Suppose that, before the first $\log q_2$ induced bits of the protocol on N_p are completed, some queried communication bit on the actual p -branch has value fixed on the current restriction to the template rows $\{p\} \times S_p$ and the columns $\widehat{D}_p$. Let v be the actual node reached immediately after the $\log q_2$ -th queried communication bit whose value is not fixed on that restriction, so its current rectangle, after restriction to those template rows and columns, is exactly the induced rectangle after the first $\log q_2$ induced bits on N_p . If no fixed query occurred before those induced bits were completed, then v would occur at actual depth exactly $2 + \log q_2$. The assumed fixed query therefore forces v to occur at actual depth at least $2 + \log q_2 + 1$. By Corollary 59, the corresponding rectangle on N_p has a unique dominant block $\alpha \in [q_2]$, and the surviving columns contain some $Y \subseteq \widehat{X}_{p,\alpha}$ with

$$|Y| \geq |\widehat{X}_{p,\alpha}| - (q_2 - 1) \left\lceil 2^{-b_1+1} |C_1| \right\rceil.$$

Also, Lemma 65 applies to S_p , because

$$8h_2 = \frac{8}{(\log d)^3} < \frac{1 - \varepsilon_{q_2, t_2}}{|R_1|}$$

for all sufficiently large d . Therefore

$$\{c_\alpha : c \in S_p\} = R_1.$$

Choose $c_r \in S_p$ with $(c_r)_\alpha = r$ for each $r \in R_1$, and set

$$T_{p,\alpha} := \{(p, c_r) : r \in R_1\}.$$

By Lemma 63, the restriction of $\mathcal{M}_4$ to $T_{p,\alpha}$ and Y is exactly a column restriction of $\mathcal{M}_1$. Since

$$\frac{|Y|}{|C_1|} \geq 1 - (q_2 - 1) \left(2^{-b_1+1} + \frac{1}{|C_1|} \right) \geq h_2^\downarrow$$

for all sufficiently large d , the same fixed set $Q \times \text{Rows}(\mathcal{M}_0) \subseteq R_1$ and the columns Y satisfy the hypotheses of Lemma 50 with $r' = r_1$, again with the required $(Q, 1)$ -equipartition already built into the chosen row set. Hence the current rectangle at v , after restriction to the template rows $\{p\} \times S_p$ and the columns $\widehat{D}_p$, still contains a subgame of communication complexity at least B_{cap} . The subtree of Π below v therefore gives a deterministic protocol for that subgame of depth at most

$$B_{\text{yes}} - (2 + \log q_2 + 1) = B_{\text{cap}} - 1,$$

contradiction. Therefore the next $\log q_2$ actual bits after the initial Stage 3 split are exactly the Stage 2 column bits on N_p , and the vector rows remain equal to B_p throughout this segment.

Step 3: extract the dense local witness on each (p, α) -branch. This step identifies the dominant Stage 2 rectangle on the synchronised branch and reads off the corresponding local Stage 1 subgame.

Recover the dominant induced Stage 2 rectangle. Fix $(p, \alpha) \in [4] \times [q_2]$. By Lemma 66, the induced protocol on N_p has a unique α -dominant rectangle after its first $\log q_2$ bits. Because the actual Stage 2 segment on the p -branch consists of exactly those $\log q_2$ induced bits, the actual stage-2 rectangles on the p -branch may be labelled the same way. On the template restriction $\{p\} \times S_p$ and $\widehat{D}_p$, each such transcript cuts out exactly the same rectangle as the corresponding transcript of the induced protocol on N_p . Write $R_{p,\alpha}^\star \times C_{p,\alpha}^\star$ for the unique α -dominant one. Every other stage-2 rectangle is non- α -dominant, so the transposed dominant-block conclusion from Corollary 59 gives

$$|C \cap \widehat{X}_{p,\alpha}| < \left\lceil 2^{-b_1+1} |C_1| \right\rceil$$

for each such rectangle $R \times C$.

If

$$|C_{p,\alpha}^\star \cap \widehat{X}_{p,\alpha}| < (1 - \eta_2) |\widehat{X}_{p,\alpha}|,$$

then summing over the unique α -dominant rectangle together with the remaining at most $q_2 - 1$ stage-2 rectangles would yield

$$|\widehat{X}_{p,\alpha}| < (1 - \eta_2) |\widehat{X}_{p,\alpha}| + (q_2 - 1) \left\lceil 2^{-b_1+1} |C_1| \right\rceil.$$

But $|\widehat{X}_{p,\alpha}| = |C_1|$ and

$$(q_2 - 1) \left\lceil 2^{-b_1+1} |C_1| \right\rceil \leq (q_2 - 1) (2^{-b_1+1} |C_1| + 1) < \eta_2 |C_1|$$

for all sufficiently large d , because

$$2^{-b_1+1} |C_1| = \frac{2}{d^2} |C_1| \geq d^{62} > q_2 - 1.$$

Hence the right-hand side is strictly less than $|\widehat{X}_{p,\alpha}|$, contradiction. Therefore

$$Y_{p,\alpha} := C_{p,\alpha}^* \cap \widehat{X}_{p,\alpha}$$

satisfies

$$|Y_{p,\alpha}| \geq (1 - \eta_2) |\widehat{X}_{p,\alpha}|.$$

Translate that rectangle back to the actual $\mathcal{M}_4$ branch. Since Lemma 65 applies to S_p , we again have

$$\{c_\alpha : c \in S_p\} = R_1.$$

Choose $c_{p,\alpha,r} \in S_p$ with $(c_{p,\alpha,r})_\alpha = r$ for each $r \in R_1$, and set

$$T_{p,\alpha} := \{(p, c_{p,\alpha,r}) : r \in R_1\}.$$

Let

$$S'_{p,\alpha} := \{\gamma \in C_1 : (\text{tail}(\gamma), ((\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma))) \in Y_{p,\alpha}\}.$$

After identifying each column of $Y_{p,\alpha}$ with its corresponding $\gamma \in S'_{p,\alpha}$, the restriction of $\mathcal{M}_4$ to $T_{p,\alpha}$ and $Y_{p,\alpha}$ is exactly

$$\mathcal{M}_1 \upharpoonright_{R_1, S'_{p,\alpha}}.$$

With $A_\alpha(B_p)$, $\nu_\alpha(B_p)$, and $Q_{p,\alpha}(B_p)$ as in Lemma 62, every vector row $i \in A_\alpha(B_p)$ agrees on $Y_{p,\alpha}$ with the local row $\mathcal{M}_0 \langle 5 \rangle (\pi_\alpha(i), \text{tail}(\gamma))$, and every vector row in $B_p \setminus A_\alpha$ agrees on $Y_{p,\alpha}$ with the neutral local row $\mathcal{M}_0 \langle 5 \rangle (5, \text{tail}(\gamma))$. Since π_α is injective, keeping the template rows $T_{p,\alpha}$, all rows of $A_\alpha(B_p)$, and at most one row from $B_p \setminus A_\alpha$ when $\nu_\alpha(B_p) = 1$ yields a subgame obtained from

$$\mathcal{M}_0 \langle 2^a + 3, S'_{p,\alpha} \rangle \upharpoonright_{Q_{p,\alpha}(B_p) \times \text{Rows}(\mathcal{M}_0), S'_{p,\alpha}}$$

by adding duplicate rows and duplicate columns. Because the Stage 2 segment consists only of column bits, the vector rows on this branch are still exactly B_p . This proves the local witness extraction claim, and hence the lemma. $\square$

Assembling the NO proof. With the stopping-time description already included in Lemma 68, the final contradiction is short: take the protocol-induced partition of the vector rows, choose the overloaded bin/coordinate pair (p, α) from the source-side NO witness, and then apply the chosen-coordinate Stage 1 threshold to the local subgame extracted by Lemma 68.

Local verification checklist. To verify the final contradiction locally, it is enough to check the following four steps.

1. Apply Lemma 68 to obtain the protocol-induced partition $B_1, \dots, B_4$, together with the dense local branch $(p, \alpha) \mapsto (Y_{p,\alpha}, S'_{p,\alpha}, T_{p,\alpha})$.
2. Use the final item of Lemma 43 to choose a bin p and coordinate α with $\ell_\alpha(B_p) \geq 2$ and $\nu_\alpha(B_p) = 1$.
3. On that branch, read off from Lemma 68 the local Stage 1 subgame indexed by $Q_{p,\alpha}(B_p)$ over the dense column set $S'_{p,\alpha}$.
4. Because $|Q_{p,\alpha}(B_p)| \geq 2^a + 1$, apply Corollary 48 to obtain the residual lower bound $B_{\text{cap}} + 1$, which exceeds the remaining budget.

Proof of Theorem 1. Let

$$k := B_{\text{yes}} = \log 4 + \lceil \log q_2 \rceil + B_{\text{cap}}.$$

The reduction maps the preprocessed $\{0, 1\}$ -Vector Bin Packing instance to the pair $(\mathcal{M}_4, k)$.

YES direction. If the source instance is a YES-instance, then Lemma 44 yields a protocol for $\mathcal{M}_4$ of cost at most k . Hence $D(\mathcal{M}_4) \leq k$.

NO direction. Now assume the source instance is a NO-instance and suppose towards contradiction that some deterministic protocol Π computes $\mathcal{M}_4$ with depth at most k .

1. Apply the scaffold lift. Write

$$\eta_2 := q_2 2^{-b_1+1}.$$

Fix any $k_0 \in [2^5]$, and apply Lemma 68 to Π . This yields a partition

$$[n] = B_1 \sqcup B_2 \sqcup B_3 \sqcup B_4$$

such that, for every $p \in [4]$, the next $\log q_2$ actual bits below the p -branch are exactly the Stage 2 column bits. Hence the vector rows remain equal to B_p throughout that segment.

For every pair $(p, \alpha) \in [4] \times [q_2]$, there is a partial branch of Π after its first $2 + \log q_2$ actual bits whose surviving columns contain a set

$$Y_{p,\alpha} \subseteq \widehat{X}_{p,\alpha}$$

with

$$|Y_{p,\alpha}| \geq (1 - \eta_2) |\widehat{X}_{p,\alpha}|.$$

Writing

$$S'_{p,\alpha} := \{\gamma \in C_1 : (\text{tail}(\gamma), ((\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma))) \in Y_{p,\alpha}\},$$

the same branch, restricted to $T_{p,\alpha} \subseteq \{p\} \times C_2$, the vector rows in B_p , and the columns $Y_{p,\alpha}$, contains a subgame obtained from

$$\mathcal{M}_0 \langle 2^a + 3, S'_{p,\alpha} \rangle \upharpoonright_{Q_{p,\alpha}(B_p) \times \text{Rows}(\mathcal{M}_0), S'_{p,\alpha}}$$

by adding duplicate rows and duplicate columns.

2. Choose the overloaded local pair. Because the source instance is NO, the final item of Lemma 43 gives a bin $p \in [4]$ and a coordinate $\alpha \in [d]$ such that

$$\ell_\alpha(B_p) \geq 2 \quad \text{and} \quad \nu_\alpha(B_p) = 1.$$

Since $q_2 = d$ after the power-of-two normalisation, this same α lies in $[q_2]$. Set

$$\rho := \eta_2.$$

Since $\eta_2 \leq 4/d$, we have

$$\rho < \frac{1 - \varepsilon_{2^a+3, t_1}}{2}$$

for all sufficiently large d . By the definition of $Q_{p,\alpha}(B_p)$ in Lemma 62, this local coordinate set consists of the q_1 template coordinates, one additional coordinate for each active row in $A_\alpha(B_p)$, and one neutral coordinate because $\nu_\alpha(B_p) = 1$. Hence

$$|Q_{p,\alpha}(B_p)| = q_1 + \ell_\alpha(B_p) + 1 = q_1 + \ell_\alpha(B_p) + \nu_\alpha(B_p) = 2^a - 2 + \ell_\alpha(B_p) + \nu_\alpha(B_p) \geq 2^a + 1.$$

3. Read off the local Stage 1 witness. The selected branch of Π already contains the local Stage 1 subgame over $S'_{p,\alpha}$ indexed by $Q_{p,\alpha}(B_p)$, as supplied by Lemma 68. Duplicate rows and duplicate columns preserve deterministic communication complexity, so it is enough to lower-bound that extracted subgame.

4. Cross the Stage 1 threshold. Applying Corollary 48 to the chosen-coordinate subgame over $S'_{p,\alpha}$ above gives lower bound $B_{\text{cap}} + 1$. Duplicate rows and duplicate columns preserve deterministic communication complexity, so the same lower bound holds for the selected branch of Π .

But Lemma 68 produces this selected branch after its first

$$\log 4 + \log q_2$$

actual bits. Since q_2 is a power of two, $\log q_2 = \lceil \log q_2 \rceil$, and therefore only

$$B_{\text{cap}} = k - \log 4 - \log q_2$$

bits remain. This contradicts the residual lower bound $B_{\text{cap}} + 1$. Hence $D(\mathcal{M}_4) > k$ on every NO-instance.

Polynomial time. By Lemma 67, the matrix $\mathcal{M}_4$ is computable in polynomial time from the source instance.

Finally, as noted after the theorem statement, a routine padding step duplicates rows and columns as needed to encode $\mathcal{M}_4$ as the truth table of a Boolean function on equal-length binary inputs. Restricting to one representative of each repeated row or column, and duplicating an existing row or column, does not change deterministic communication complexity. So this final encoding preserves the value being reduced, and the matrix reduction above yields the stated NP-completeness result for truth tables. $\square$

Further directions. The relaxed-interlacing toolkit extends ideas that already proved useful in the counterexample to the direct sum conjecture, and the present paper develops those ideas into a reusable reduction framework. The same framework plausibly extends to stronger hardness and approximation results, including additive approximability, but those questions lie beyond the scope of the present paper.

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A Proof of the Iterated Density-Amplification Lemma

In the body we stated the iterated density-amplification lemma and gave its proof roadmap. We now record the appendix structure for the full proof. The proof has two layers. First, we show that a single application of Lemma 18 extends to Λ_M : one step either halves the number of copies or trades copies for lower column density, and in both cases the three-rung bounds are preserved. Second, we index the iterated applications by two coordinates (how many copy-halving steps and how many density-amplifying steps have been taken) and prove Lemma 21 by induction over that grid.

Λ_M tracks the three neighbouring density lower bounds: a column move lowers the density by a factor of 2, so the induction must track the neighbouring densities y , $y/2$, and $y/4$ simultaneously.

A.1 Auxiliary One-Step Bound

We begin with the same first-bit analysis that underlies the partition lemma (Lemma 18). Here we rewrite it more explicitly, treating the degenerate row split separately, so that it can be applied repeatedly in the proofs below.

Lemma 69. *Let M be a matrix, let $p \in \mathbb{N}$ with $p \geq 1$, let $\sigma \in \{0, 1\}$, and let $0 < x \leq \frac{1}{2}$, $0 < y \leq 1$. If*

$$D(M[2p + \sigma, 2x, y]) \geq 1,$$

then

$$D(M[2p + \sigma, 2x, y]) \geq 1 + \min \left(D(M[2p + \sigma, x, y]), \right. \\ \left. \min_{\substack{0 \leq \ell < p \\ a \in [0, 1]}} \max \left(D(M[p + \ell + \sigma, x, y^a]), D(M[p - \ell, x, y^{1-a}]) \right), \right. \\ \left. D(M[2p + \sigma, 2x, y/2]) \right).$$

Proof. There are three ways the first move can contribute to the lower bound:

$$A := D(M[2p + \sigma, x, y]) \quad (\text{degenerate row split}),$$

$$B := \min_{\substack{0 \leq \ell < p \\ a \in [0, 1]}} \max \left(D(M[p + \ell + \sigma, x, y^a]), D(M[p - \ell, x, y^{1-a}]) \right) \quad (\text{genuine row split}),$$

$$K := D(M[2p + \sigma, 2x, y/2]) \quad (\text{column-first branch}).$$

So it is enough to show that every

$$g = M \langle 2p + \sigma \rangle \upharpoonright_{R, C} M[2p + \sigma, 2x, y].$$

satisfies

$$D(g) \geq 1 + \min(A, B, K).$$

The hypothesis on the bracket family implies $D(g) \geq 1$, so an optimal protocol for g has a first bit.

Column-first case. If the first bit is sent by the column player, one child keeps at least half of the columns and therefore contains a subgame from $M[2p + \sigma, 2x, y/2]$. Hence

$$D(g) \geq 1 + K.$$

Row-first case. Assume now that the first bit is sent by the row player, so the root partitions the row set as $R = R_1 \sqcup R_2$. Apply Lemma 10 to g with this partition. We obtain integers $\ell_1, \ell_2 \geq 0$ and densities $y_1, y_2 \in (0, 1]$ such that

$$\ell_1 + \ell_2 = 2p + \sigma, \quad y_1 y_2 \geq y,$$

and, for each $i \in \{1, 2\}$ with $\ell_i \geq 1$, the child rectangle induced by R_i contains a subgame

$$g_i \in M[\ell_i, x, y_i].$$

Relabel so that $\ell_1 \geq \ell_2$.

We separate the case $\ell_2 = 0$ because then only one child contains any surviving copies, so the generic two-child parameterisation does not apply.

Degenerate row split. If $\ell_2 = 0$, then $\ell_1 = 2p + \sigma$, the first child contains

$$g_1 \in M[2p + \sigma, x, y_1],$$

and, because $y_2 \leq 1$, we have $y_1 \geq y$. By monotonicity,

$$D(g) \geq 1 + A.$$

Genuine row split. Assume from now on that $\ell_2 \geq 1$. Then for a unique integer $0 \leq \ell < p$ we may write

$$\ell_1 = p + \ell + \sigma, \quad \ell_2 = p - \ell.$$

If $0 < y < 1$, set $a := \log(y_1)/\log(y) \in [0, 1]$; then $y_1 = y^a$, and $y_1 y_2 \geq y$ implies $y_2 \geq y^{1-a}$. If $y = 1$, then $y_1 = y_2 = 1$, and any $a \in [0, 1]$ satisfies $y_1 = y^a$ and $y_2 \geq y^{1-a}$. In either case, monotonicity gives

$$D(g_1) \geq D(M[p + \ell + \sigma, x, y^a]), \quad D(g_2) \geq D(M[p - \ell, x, y^{1-a}]).$$

Hence

$$D(g) \geq 1 + \max(D(M[p + \ell + \sigma, x, y^a]), D(M[p - \ell, x, y^{1-a}])) \geq 1 + B.$$

In each of the three root cases we have proved

$$D(g) \geq 1 + \min(A, B, K).$$

Taking the minimum over all $g \in M[2p + \sigma, 2x, y]$ proves the lemma. $\square$

A.2 The Bundled Copy-Halving Step

The next lemma is the bundled copy-halving consequence of the raw one-step partition bound. It passes from $2p + \sigma$ copies down to $p + \sigma$ copies while preserving the three-rung bounds. The only extra bookkeeping is that a column-first move from the j -th rung reaches the $(j + 1)$ -st rung, so the three rung inequalities have to be chained together before taking their minimum.

The required bookkeeping fact is the following implication:

$$\left. \begin{array}{l} X_j \geq 1 + \min(Y_j, X_{j+1}) \quad (j = 0, 1, 2), \\ X_3 \geq Y_2 \end{array} \right\} \implies \min(X_0, 1 + X_1, 2 + X_2) \geq 1 + \min(Y_0, 1 + Y_1, 2 + Y_2). \quad (5)$$

We will refer to (5) as the *three-rung bookkeeping identity*. Indeed, if

$$L := 1 + \min(Y_0, 1 + Y_1, 2 + Y_2),$$

then $X_2 \geq 1 + \min(Y_2, X_3) \geq 1 + Y_2$, so

$$2 + X_2 \geq 3 + Y_2 \geq L.$$

Next,

$$X_1 \geq 1 + \min(Y_1, X_2) \geq 1 + \min(Y_1, 1 + Y_2),$$

hence

$$1 + X_1 \geq 2 + \min(Y_1, 1 + Y_2) \geq L.$$

Finally,

$$X_0 \geq 1 + \min(Y_0, X_1) \geq 1 + \min(Y_0, 1 + Y_1, 2 + Y_2) = L.$$

Taking the minimum of X_0 , $1 + X_1$, and $2 + X_2$ gives (5).

Lemma 70. Let M be a matrix, let $p \in \mathbb{N}$ with $p \geq 1$, let $\sigma \in \{0, 1\}$, and let $0 < x \leq \frac{1}{2}$, $0 < y \leq 1$. Assume that

$$D(M[2p, 2x, y/4]) \geq 1.$$

Then

$$\Lambda_M(2p + \sigma, 2x, y) \geq 1 + \Lambda_M(p + \sigma, x, y).$$

Proof. For $0 \leq j \leq 2$, set

$$R_j := D(M[2p + \sigma, 2x, y/2^j]), \quad S_j := D(M[p + \sigma, x, y/2^j]),$$

and also

$$R_3 := D(M[2p + \sigma, 2x, y/8]).$$

Thus R_j is the source family at rung j , while S_j is the target family on $p + \sigma$ copies.

Since $2p \leq 2p + \sigma$ and $y/2^j \geq y/4$ for $j = 0, 1, 2$, the assumption and monotonicity give

$$R_j \geq 1 \quad (j = 0, 1, 2).$$

Fix $j \in \{0, 1, 2\}$. Applying Lemma 69 with density $y/2^j$ yields

$$R_j \geq 1 + \min(A_j, B_j, R_{j+1}),$$

where

$$A_j := D(M[2p + \sigma, x, y/2^j]),$$

and

$$B_j := \min_{\substack{0 \leq \ell < p \\ a \in [0, 1]}} \max \left(D(M[p + \ell + \sigma, x, (y/2^j)^a]), D(M[p - \ell, x, (y/2^j)^{1-a}]) \right).$$

It therefore suffices to show that both A_j and B_j are at least S_j .

The term A_j . By monotonicity in the first parameter, $A_j \geq S_j$.

The term B_j . For every admissible ℓ and a , projecting from $p + \ell + \sigma$ copies down to $p + \sigma$ copies via Lemma 7 gives

$$D(M[p + \ell + \sigma, x, (y/2^j)^a]) \geq D\left(M\left[p + \sigma, x, (y/2^j)^{a \frac{p + \sigma}{p + \ell + \sigma}}\right]\right).$$

Because

$$0 \leq a \frac{p + \sigma}{p + \ell + \sigma} \leq 1 \quad \text{and} \quad 0 < y/2^j \leq 1,$$

raising the base $y/2^j \in (0, 1]$ to an exponent in $[0, 1]$ can only increase it, so the projected density is at least $y/2^j$. Hence the displayed term is at least S_j , so $B_j \geq S_j$. Therefore

$$R_j \geq 1 + \min(S_j, R_{j+1}) \quad (j = 0, 1, 2).$$

Closing the bottom rung. To close the last rung, apply Lemma 7 to R_3 with projection parameter $p + \sigma$. Writing

$$\alpha := \frac{p + \sigma}{2p + \sigma},$$

we obtain

$$R_3 \geq D(M[p + \sigma, 2x, (y/8)^\alpha]).$$

Since $\alpha \leq 2/3$ and $0 < y \leq 1$,

$$(y/8)^\alpha = y^\alpha 8^{-\alpha}.$$

Because $\alpha \leq 1$ and $0 < y \leq 1$, we have $y^\alpha \geq y$. Because $\alpha \leq 2/3$, we also have $8^{-\alpha} \geq 8^{-2/3}$. Hence

$$(y/8)^\alpha \geq y \cdot 8^{-2/3} = \frac{y}{4}.$$

Monotonicity in the second and third parameters therefore gives

$$R_3 \geq D(M[p + \sigma, x, y/4]) = S_2.$$

Bundling the three rungs. Applying the three-rung bookkeeping identity (5) with $X_j = R_j$ and $Y_j = S_j$ yields

$$\Lambda_M(2p + \sigma, 2x, y) = \min(R_0, 1 + R_1, 2 + R_2) \geq 1 + \min(S_0, 1 + S_1, 2 + S_2) = 1 + \Lambda_M(p + \sigma, x, y).$$

This proves the lemma. $\square$

A.3 The Bundled Density-Amplifying Step

The next lemma is the bundled density-amplifying consequence of the same raw one-step partition bound. It starts from $2p$ copies and trades copy count for a stronger column parameter while preserving the same three-rung bounds. The only nontrivial verification is in the row-first case: after the one-step bound produces a split (ℓ, a) , we show that one child always belongs to one of the two target bundles.

Lemma 71. *Let M be a matrix, let $p \in \mathbb{N}$ with $p \geq 1$, let $0 < x \leq \frac{1}{2}$, let $0 < y \leq 1$, and let $0 < \tau \leq 1$. Write*

$$u = y^{\frac{1}{1+\tau}}, \quad v = y^{\frac{\tau}{1+\tau}}.$$

Assume that

$$D(M[2p, 2x, y/4]) \geq 1.$$

Then

$$\Lambda_M(2p, 2x, y) \geq 1 + \min(\Lambda_M(p, x, u), \Lambda_M(\lfloor p(1 - \tau) \rfloor + 1, x, v)).$$

Proof. Set $q := \lfloor p(1 - \tau) \rfloor + 1$. This is the copy count in the second target family. For $0 \leq j \leq 2$, define

$$\begin{aligned} T_j &:= D(M[2p, 2x, y/2^j]), & U_j &:= D(M[p, x, u/2^j]), \\ V_j &:= D(M[q, x, v/2^j]), & Z_j &:= \min(U_j, V_j), \end{aligned}$$

and also

$$T_3 := D(M[2p, 2x, y/8]).$$

Thus T_j is the source family at rung j , while U_j and V_j are the two target families, and Z_j is their minimum.

Since $y/2^j \geq y/4$ for $j = 0, 1, 2$, the assumption implies

$$T_j \geq 1 \quad (j = 0, 1, 2).$$

Fix $j \in \{0, 1, 2\}$. Applying Lemma 69 with density $y/2^j$ yields

$$T_j \geq 1 + \min(A_j, B_j, T_{j+1}),$$

where

$$A_j := D(M[2p, x, y/2^j]),$$

and

$$B_j := \min_{\substack{0 \leq \ell < p \\ a \in [0,1]}} \max \left(D(M[p + \ell, x, (y/2^j)^a]), D(M[p - \ell, x, (y/2^j)^{1-a}]) \right).$$

It therefore suffices to show that both A_j and B_j are at least Z_j .

The term A_j . For the first term, Lemma 7 with projection parameter p gives

$$A_j \geq D(M[p, x, (y/2^j)^{1/2}]).$$

Because $\tau \leq 1$, we have $1/(1 + \tau) \geq 1/2$, and therefore

$$(y/2^j)^{1/2} = \frac{y^{1/2}}{2^{j/2}} \geq \frac{y^{1/(1+\tau)}}{2^j} = \frac{u}{2^j}.$$

Thus $A_j \geq U_j \geq Z_j$.

The term B_j . Now fix admissible ℓ and a . We claim that the corresponding value inside the maximum is always at least Z_j . The cases below are exhaustive: either $\ell \geq p\tau$, or $\ell < p\tau$; in the second regime, either $a \leq \frac{1}{1+\tau}$ or $a > \frac{1}{1+\tau}$.

Case 1: $\ell \geq p\tau$. If $\ell \geq p\tau$, then $p + \ell \geq p(1 + \tau)$. Projecting the first child down to p copies gives

$$D(M[p + \ell, x, (y/2^j)^a]) \geq D(M[p, x, (y/2^j)^{a \frac{p}{p+\ell}}]).$$

Since

$$a \frac{p}{p + \ell} \leq \frac{1}{1 + \tau},$$

we have

$$(y/2^j)^{a \frac{p}{p+\ell}} \geq (y/2^j)^{\frac{1}{1+\tau}} = \frac{u}{2^{j/(1+\tau)}} \geq \frac{u}{2^j}.$$

So in this case the first child is at least $U_j \geq Z_j$.

Case 2: $\ell < p\tau$ and $a \leq \frac{1}{1+\tau}$. Assume from now on that $\ell < p\tau$. If $a \leq \frac{1}{1+\tau}$, then $p + \ell \geq p$ and

$$(y/2^j)^a \geq (y/2^j)^{\frac{1}{1+\tau}} \geq \frac{u}{2^j},$$

so monotonicity gives

$$D(M[p + \ell, x, (y/2^j)^a]) \geq U_j \geq Z_j.$$

Case 3: $\ell < p\tau$ and $a > \frac{1}{1+\tau}$. If instead $a > \frac{1}{1+\tau}$, then

$$1 - a < \frac{\tau}{1 + \tau}$$

and $p - \ell > p(1 - \tau)$, hence

$$p - \ell \geq \lfloor p(1 - \tau) \rfloor + 1 = q.$$

Moreover,

$$(y/2^j)^{1-a} \geq (y/2^j)^{\frac{\tau}{1+\tau}} = \frac{v}{2^{j\tau/(1+\tau)}} \geq \frac{v}{2^j}.$$

Therefore

$$D(M[p - \ell, x, (y/2^j)^{1-a}]) \geq V_j \geq Z_j.$$

Since every admissible pair (ℓ, a) yields a value at least Z_j , we have $B_j \geq Z_j$. Hence

$$T_j \geq 1 + \min(Z_j, T_{j+1}) \quad (j = 0, 1, 2).$$

Closing the bottom rung. To close the bottom rung, project T_3 from $2p$ copies to p copies:

$$T_3 \geq D\left(M\left[p, 2x, \sqrt{y/8}\right]\right).$$

Because

$$\frac{u}{4} = \frac{y^{1/(1+\tau)}}{4} \leq \frac{\sqrt{y}}{4} \leq \sqrt{\frac{y}{8}},$$

monotonicity in the second and third parameters gives

$$T_3 \geq U_2 \geq Z_2.$$

Bundling the three rungs. Applying the three-rung bookkeeping identity (5) with $X_j = T_j$ and $Y_j = Z_j$ gives

$$\Lambda_M(2p, 2x, y) \geq 1 + \min(Z_0, 1 + Z_1, 2 + Z_2).$$

Because $Z_j = \min(U_j, V_j)$ for each j , the last minimum expands to

$$\min(Z_0, 1 + Z_1, 2 + Z_2) = \min(U_0, 1 + U_1, 2 + U_2, V_0, 1 + V_1, 2 + V_2),$$

which is exactly

$$\min(\Lambda_M(p, x, u), \Lambda_M(q, x, v)).$$

This proves the lemma. □

A.4 Grid Setup and Arithmetic Facts

We now set up the grid on which the induction will run. The horizontal coordinate records how many row-doubling steps have been taken, while the vertical coordinate records the current density-amplification stage. The coordinates (t, r) index the intermediate lower bounds produced by those two kinds of moves.

Write

$$\tau := \frac{1}{\rho - 1}, \quad \Lambda_0 := \Lambda_M(p, x, y), \quad \beta = \frac{1}{1 - \tau} = \frac{\rho - 1}{\rho - 2}.$$

For integers t, r with

$$0 \leq t \leq s, \quad 0 \leq r \leq k - s + t,$$

we call (t, r) an *admissible pair*. The index r measures row depth: after r row-doubling steps, the row parameter has grown from x to $2^r x$. The index t measures column-amplification stage.

For each admissible pair (t, r) , define

$$P_{t,r} := \lfloor 2^r \beta^t p \rfloor.$$

This is the copy count that we target at that grid point. The factor 2^r records the row-doubling steps, while the factor β^t records the extra copy count required to support t density-amplifying stages.

The exponents below are engineered so that a density-amplifying step either stays on the same stage or drops by exactly one stage, which is what the two-parameter induction needs.

Next define the column exponents. At stage 0, there has been no density amplification yet, so we set

$$E_{0,r} := 1 \quad \text{for } 0 \leq r \leq k - s.$$

For stages $1 \leq t \leq s$, define

$$E_{t,r} := \rho^t \left(\frac{\rho - 1}{\rho} \right)^{k-r-(s-t)} \quad \text{for } 0 \leq r \leq k - s + t.$$

These exponents are chosen so that, for stages $t \geq 2$, one output of the bundled density-amplifying step stays on the same stage while the other lands exactly one stage lower; at stage $t = 1$, the second output falls back to stage 0 by monotonicity. That is the numerical feature that makes the two-parameter induction close.

Finally, for each admissible pair (t, r) , write

$$F_{t,r} := \Lambda_M(P_{t,r}, 2^r x, y^{E_{t,r}}).$$

So $F_{t,r}$ is the lower bound we prove at grid point (t, r) .

The endpoint of the theorem is the terminal admissible pair (s, k) :

$$F_{s,k} = \Lambda_M\left(\lfloor 2^k \beta^s p \rfloor, 2^k x, y^{\rho^s}\right).$$

Thus the desired inequality (2) will follow once we prove that

$$F_{t,r} \geq r + \Lambda_0 \quad \text{for every admissible pair } (t, r).$$

Bridge inequalities.

The induction moves across the grid by two elementary floor inequalities. If (t, r) is admissible and $r < k - s + t$, then

$$P_{t,r+1} = \lfloor 2^{r+1} \beta^t p \rfloor \geq 2 \lfloor 2^r \beta^t p \rfloor = 2P_{t,r}.$$

This is the inequality needed for the bundled copy-halving step.

If $1 \leq t \leq s$, then, since $1 - \tau = 1/\beta$,

$$\lfloor P_{t,r}(1 - \tau) \rfloor + 1 = \left\lfloor \frac{P_{t,r}}{\beta} \right\rfloor + 1.$$

Also $P_{t,r} \geq 2^r \beta^t p - 1$, so

$$\frac{P_{t,r}}{\beta} \geq 2^r \beta^{t-1} p - \frac{1}{\beta}.$$

Because $0 < 1/\beta < 1$ and $\lfloor u - c \rfloor + 1 \geq \lfloor u \rfloor$ for every real u and every $0 < c < 1$, we obtain

$$\lfloor P_{t,r}(1 - \tau) \rfloor + 1 \geq \lfloor 2^r \beta^{t-1} p \rfloor.$$

Indeed, if $0 < c < 1$, then $u - c > u - 1$, so $\lfloor u - c \rfloor \geq \lfloor u - 1 \rfloor = \lfloor u \rfloor - 1$, and hence $\lfloor u - c \rfloor + 1 \geq \lfloor u \rfloor$. Therefore

$$\lfloor P_{t,r}(1 - \tau) \rfloor + 1 \geq \lfloor 2^r \beta^{t-1} p \rfloor = P_{t-1,r}.$$

This is the inequality needed for the bundled density-amplifying step.

Side-condition propagation.

We also need to know that the weak side condition at the seed point propagates across the whole grid. For $1 \leq t \leq s$,

$$E_{t,0} = \rho^t \left(\frac{\rho - 1}{\rho} \right)^{k-s+t} = \frac{(\rho - 1)^{k-s+t}}{\rho^{k-s}} \leq \frac{(\rho - 1)^k}{\rho^{k-s}} \leq 1,$$

because $t \leq s$ and the standing assumption $(\rho - 1)^k \leq \rho^{k-s}$ holds. Also,

$$E_{t,r+1} = \frac{\rho}{\rho - 1} E_{t,r} < 2E_{t,r} \quad (1 \leq t \leq s),$$

since $\rho > 2$. Therefore $E_{t,r} \leq 2^r$ for every admissible pair (t, r) ; for stage 0 this is immediate from $E_{0,r} = 1$. For $1 \leq t \leq s$, this follows by induction on r : the base case is $E_{t,0} \leq 1 = 2^0$, and if $E_{t,r} \leq 2^r$, then

$$E_{t,r+1} < 2E_{t,r} \leq 2^{r+1}.$$

Since $\beta > 1$, we have $P_{t,r} \geq 2^r p$, hence

$$\frac{E_{t,r+1} p}{2P_{t,r}} \leq 1 \quad \text{whenever } r < k - s + t.$$

Applying Lemma 7 with projection parameter p gives

$$D \left(M \left[2P_{t,r}, 2^{r+1} x, y^{E_{t,r+1}} / 4 \right] \right) \geq D \left(M \left[p, 2^{r+1} x, (y^{E_{t,r+1}} / 4)^{p/(2P_{t,r})} \right] \right).$$

Because $0 < y \leq 1$, $\frac{E_{t,r+1} p}{2P_{t,r}} \leq 1$, and $\frac{p}{2P_{t,r}} \leq 1$, we have

$$(y^{E_{t,r+1}} / 4)^{p/(2P_{t,r})} = y^{E_{t,r+1} p / (2P_{t,r})} 4^{-p/(2P_{t,r})} \geq y \cdot 4^{-1} = \frac{y}{4}.$$

Since $x \leq 2^{r+1} x$, monotonicity and the hypothesis at the seed point $D(M[p, x, y/4]) \geq 1$ imply

$$D \left(M \left[2P_{t,r}, 2^{r+1} x, y^{E_{t,r+1}} / 4 \right] \right) \geq 1 \quad \text{for every admissible } (t, r) \text{ with } r < k - s + t. \quad (6)$$

A.5 Proof of Lemma 21

Proof of Lemma 21. We now prove by induction on r that

$$F_{t,r} \geq r + \Lambda_0 \quad \text{for every admissible pair } (t, r).$$

Base case $r = 0$. At depth 0 there is no row amplification yet. For $t = 0$, this is exactly $F_{0,0} = \Lambda_0$. For $1 \leq t \leq s$, we have $P_{t,0} \geq p$ and $E_{t,0} \leq 1$, so $y^{E_{t,0}} \geq y$. Monotonicity therefore gives

$$F_{t,0} = \Lambda_M(P_{t,0}, x, y^{E_{t,0}}) \geq \Lambda_M(p, x, y) = \Lambda_0.$$

Inductive step. Fix $0 \leq r < k$ and assume that $F_{t,r} \geq r + \Lambda_0$ for every stage t with $r \leq k - s + t$. Let t satisfy $r + 1 \leq k - s + t$. Since $x \leq 2^{-k}$ and $r < k$, we have

$$2^r x \leq 2^{r-k} \leq \frac{1}{2},$$

so both bundled ladder lemmas are applicable with local row density $2^r x$.

Case $t = 0$: pure row amplification. We are still in the pure row-amplification strip. Using the bridge inequality $P_{0,r+1} \geq 2P_{0,r}$, monotonicity in the first parameter, (6) (the propagated leaf-side condition), and Lemma 70, we get

$$F_{0,r+1} = \Lambda_M(P_{0,r+1}, 2^{r+1}x, y) \geq \Lambda_M(2P_{0,r}, 2^{r+1}x, y) \geq 1 + \Lambda_M(P_{0,r}, 2^r x, y) = 1 + F_{0,r}.$$

Hence $F_{0,r+1} \geq (r + 1) + \Lambda_0$.

Cases $1 \leq t \leq s$: density-amplifying stages. Here one application of the bundled density-amplifying step either stays on stage t or decreases to stage $t - 1$. Since

$$E_{t,r+1} \frac{\rho - 1}{\rho} = E_{t,r},$$

Lemma 71 with parameter $\tau = 1/(\rho - 1)$ and side condition (6) (the propagated leaf-side condition) gives

$$\begin{aligned} \Lambda_M(2P_{t,r}, 2^{r+1}x, y^{E_{t,r+1}}) &\geq 1 + \min\left(\Lambda_M(P_{t,r}, 2^r x, y^{E_{t,r}}), \right. \\ &\quad \left. \Lambda_M(\lfloor P_{t,r}(1 - \tau) \rfloor + 1, 2^r x, y^{E_{t,r+1}/\rho})\right). \end{aligned}$$

Since $P_{t,r+1} \geq 2P_{t,r}$, monotonicity in the first parameter gives $F_{t,r+1} \geq$ the left-hand side.

Case $t = 1$: first density-amplification stage. The second branch reduces to stage 0. Indeed,

$$\frac{E_{1,r+1}}{\rho} = \left(\frac{\rho - 1}{\rho}\right)^{k-r-s} \leq 1,$$

and the second bridge inequality gives $\lfloor P_{1,r}(1 - \tau) \rfloor + 1 \geq P_{0,r}$. Hence monotonicity yields

$$\Lambda_M(\lfloor P_{1,r}(1 - \tau) \rfloor + 1, 2^r x, y^{E_{1,r+1}/\rho}) \geq \Lambda_M(P_{0,r}, 2^r x, y) = F_{0,r}.$$

Therefore

$$F_{1,r+1} \geq 1 + \min(F_{1,r}, F_{0,r}).$$

Case $2 \leq t \leq s$: higher density-amplification stages. The second branch decreases by exactly one stage, because

$$\frac{E_{t,r+1}}{\rho} = E_{t-1,r}.$$

Using again $\lfloor P_{t,r}(1 - \tau) \rfloor + 1 \geq P_{t-1,r}$, monotonicity gives

$$\Lambda_M(\lfloor P_{t,r}(1 - \tau) \rfloor + 1, 2^r x, y^{E_{t,r+1}/\rho}) \geq \Lambda_M(P_{t-1,r}, 2^r x, y^{E_{t-1,r}}) = F_{t-1,r}.$$

Therefore

$$F_{t,r+1} \geq 1 + \min(F_{t,r}, F_{t-1,r}).$$

In all three cases, the induction hypothesis gives both terms inside the minimum at least $r + \Lambda_0$, so

$$F_{t,r+1} \geq (r + 1) + \Lambda_0.$$

This completes the induction.

Taking $(t, r) = (s, k)$ yields

$$\Lambda_M([2^k \beta^s p], 2^k x, y^{\rho^s}) = F_{s,k} \geq k + \Lambda_0,$$

which is (2). Finally, the minimum defining Λ_M contains the $j = 0$ term, so

$$D\left(M\left[\lfloor 2^k \beta^s p \rfloor, 2^k x, y^{\rho^s}\right]\right) \geq \Lambda_M([2^k \beta^s p], 2^k x, y^{\rho^s}).$$

Combining this with (2) gives (3). □

B Proofs of the Odd-Copy Seed and Hard-Seed Lemmas

We now prove the bundled odd-copy seed that will serve as the input to Corollary 22.

Proof of Lemma 19. Write

$$y := \frac{1}{2} + \delta, \quad y_0 := y^2.$$

We first verify a uniform side condition. For every integer $2 \leq \ell \leq b$,

$$D\left(M\left[2^{\ell-1}, 2^{\ell-b}, y_0/4\right]\right) \geq 1.$$

Indeed, by Lemma 7,

$$D\left(M\left[2^{\ell-1}, 2^{\ell-b}, y_0/4\right]\right) \geq D\left(M\left[1, 2^{\ell-b}, (y_0/4)^{1/2^{\ell-1}}\right]\right).$$

Since $\ell \leq b$, we have $2^{-b} \leq 2^{\ell-b}$. Since $\ell \geq 2$,

$$(y_0/4)^{1/2^{\ell-1}} \geq (y_0/4)^{1/2} = y/2 = \frac{1}{4} + \frac{\delta}{2}.$$

Hence monotonicity and (R4) (the one-copy lower bound at density $y/2$) give

$$D\left(M\left[1, 2^{\ell-b}, (y_0/4)^{1/2^{\ell-1}}\right]\right) \geq D\left(M\left[1, 2^{-b}, y/2\right]\right) \geq D(M) - 1 \geq 1.$$

By Corollary 15,

$$\begin{aligned} D\left(M\left[2, 2^{1-b}, y_0\right]\right) &\geq D(M) + 1, \\ D\left(M\left[2, 2^{1-b}, y_0/2\right]\right) &\geq D(M), \\ D\left(M\left[2, 2^{1-b}, y_0/4\right]\right) &\geq D(M) - 1. \end{aligned}$$

Therefore

$$\Lambda_M(2, 2^{1-b}, y_0) \geq D(M) + 1.$$

We now prove by induction on ℓ that

$$\Lambda_M(2^{\ell-1} + 1, 2^{\ell-b}, y_0) \geq D(M) + \ell \quad (2 \leq \ell \leq b).$$

For the base case $\ell = 2$, apply Lemma 70 with

$$p = 1, \quad \sigma = 1, \quad x = 2^{1-b}, \quad y = y_0.$$

The side condition is exactly the displayed uniform bound for $\ell = 2$, so

$$\Lambda_M(3, 2^{2-b}, y_0) \geq 1 + \Lambda_M(2, 2^{1-b}, y_0) \geq D(M) + 2.$$

Now let $3 \leq \ell \leq b$, and assume

$$\Lambda_M(2^{\ell-2} + 1, 2^{\ell-1-b}, y_0) \geq D(M) + \ell - 1.$$

Apply Lemma 70 with

$$p = 2^{\ell-2}, \quad \sigma = 1, \quad x = 2^{\ell-1-b}, \quad y = y_0.$$

The same uniform side condition gives

$$\Lambda_M(2^{\ell-1} + 1, 2^{\ell-b}, y_0) \geq 1 + \Lambda_M(2^{\ell-2} + 1, 2^{\ell-1-b}, y_0) \geq D(M) + \ell.$$

□

Proof of Lemma 33. Fix an integer $j \geq 2$. We show that for all sufficiently large powers of two t , the displayed implication in Lemma 33 holds.

Write

$$y := \frac{1}{2} + \delta, \quad y_0 := y^2, \quad \eta := \log \frac{1}{y_0} > 0.$$

Choose a threshold $T \geq 2^j$ so large that whenever $t \geq T$ is a power of two, and

$$L := \log t, \quad k := L - j, \quad \rho := \sqrt{L}, \quad s := \left\lceil \frac{\sqrt{L}}{\log L} \right\rceil, \quad \beta := \frac{\rho - 1}{\rho - 2},$$

then

$$\begin{aligned} \rho > 2, \quad s \leq k, \quad \beta^s &\leq \frac{2^{j-1} + 2}{2^{j-1} + 1}, \\ \eta 2^{\frac{1}{2}\sqrt{L}} &\geq 2^{0.49\sqrt{L}}, \quad \left(\frac{\sqrt{L}}{\log L} + 1 \right) \sqrt{L} \cdot \frac{\ln 2}{2} \log L < L - j. \end{aligned}$$

Such a threshold exists because j is fixed and $y_0 \in (0, 1)$ is a constant independent of t , while

$$\frac{\left\lceil \sqrt{L}/\log L \right\rceil}{L} \rightarrow 0, \quad \frac{\left\lceil \sqrt{L}/\log L \right\rceil}{\sqrt{L} - 2} \rightarrow 0, \quad \eta 2^{0.01\sqrt{L}} \rightarrow \infty.$$

The restrictive comparison here is

$$\left(\frac{\sqrt{L}}{\log L} + 1 \right) \sqrt{L} \cdot \frac{\ln 2}{2} \log L < L - j,$$

because its left-hand side is $O(L/\log L)$ whereas the right-hand side is linear in L . The other requirements follow in the same large- L regime because j is fixed and y_0 is a constant in $(0, 1)$.

Now let $t \geq T$ be a power of two, and suppose M is (δ, b) -robust with $D(M) \geq 3$ and $t \leq 2^b$. Since $t \geq 2^j$ and $t \leq 2^b$, we have

$$j \leq L \leq b.$$

In particular $k = L - j \in \mathbb{N}$. Set

$$p := 2^{j-1} + 1, \quad x := 2^{j-b}, \quad H := D(M) + j.$$

By Lemma 19 with $\ell = j$,

$$\Lambda_M(p, x, y_0) \geq H.$$

By the definition of Λ_M , this means

$$D(M[p, x, y_0]) \geq H, \quad D(M[p, x, y_0/2]) \geq H - 1, \quad D(M[p, x, y_0/4]) \geq H - 2.$$

In particular,

$$D(M[p, x, y_0/4]) \geq 1.$$

Also,

$$x = 2^{j-b} \leq 2^{j-L} = 2^{-k}.$$

To check the numerical side condition in Lemma 21, note that

$$s\rho \ln \rho \leq \left(\frac{\sqrt{L}}{\log L} + 1 \right) \sqrt{L} \cdot \frac{\ln 2}{2} \log L < L - j = k.$$

Hence

$$\left(1 - \frac{1}{\rho} \right)^k \leq e^{-k/\rho} \leq e^{-s \ln \rho} = \rho^{-s},$$

and therefore

$$(\rho - 1)^k \leq \rho^{k-s}.$$

All hypotheses of Corollary 22 are now satisfied, so

$$D\left(M\left[\left\lfloor 2^k \beta^s p \right\rfloor, 2^k x, y_0^{\rho^s}\right]\right) \geq k + H = D(M) + L = D(M) + \log t.$$

Finally,

$$2^k x = 2^{L-j} 2^{j-b} = 2^{-b} t,$$

and

$$2^k \beta^s p \leq 2^{L-j} \cdot \frac{2^{j-1} + 2}{2^{j-1} + 1} \cdot (2^{j-1} + 1) = \frac{(2^{j-1} + 2)t}{2^j}.$$

Also,

$$\rho^s = (\sqrt{L})^s \geq (\sqrt{L})^{\sqrt{L}/\log L} = 2^{\frac{1}{2}\sqrt{L}},$$

so

$$y_0^{\rho^s} = 2^{-\eta \rho^s} \leq 2^{-\eta 2^{\frac{1}{2}\sqrt{L}}} \leq 2^{-2^{0.49\sqrt{L}}}.$$

By monotonicity,

$$D\left(M\left[\frac{(2^{j-1} + 2)t}{2^j}, 2^{-b}t, 2^{-2^{0.49\sqrt{\log t}}}\right]\right) \geq D(M) + \log t,$$

as claimed. $\square$

C Technical Proofs for the Reduction

We group the deferred reduction proofs into three packages, matching the way the main text later consumes them.

1. **The Stage 1 threshold package.** These proofs establish the local Stage 1 threshold, its dense-column variants, and the chosen-coordinate form used in the final contradiction.
2. **The Stage 1/Stage 2 bootstrap.** These proofs certify the residual Stage 1 hardness, the robustness statements, the Stage 2 column-loss and hard-seed inputs, and the resulting Stage 2 lower-bound consequences.
3. **The Stage 4 local gadget package.** These proofs identify the local branch of $\mathcal{M}_4$ with the corresponding Stage 1 subgame and then propagate the dense chosen-coordinate threshold to that branch.

C.1 Checklist of recurring large- d inequalities

The main text repeatedly invokes a small set of large- d inequalities. The table below gathers the ones that are reused in the transfer layer and in Stages 2–4 of the reduction. Here $\eta_2 := q_2 2^{-b_1+1}$, as in Lemmas 66 and 68.

Inequality	Used for	First use in the main text
$2(\frac{1}{2} + \delta)^2 \leq \frac{1}{1 + \varepsilon_{q_2, t_2}}$ and $q_2 \lceil 2^{-b_1+1} C_1 \rceil < C_1 $	the Stage 2 application of the relaxed near-exact separation theorem	Lemma 58
$2(\frac{1}{2} + \delta)^2 \leq \frac{\sigma}{1 + \varepsilon_{q_2, t_2}}$ for $\sigma \geq 1 - 8h_2$	the projected dense-row Stage 2 application of the relaxed near-exact separation theorem	Corollary 59
$8h_2 < \frac{1 - \varepsilon_{q_2, t_2}}{ R_1 }$	showing that every projected Stage 1 row type survives after the Stage 3 row loss, via Lemma 65	Lemma 66
$1 - \eta_2 \geq h_2^\downarrow$	showing that the surviving Stage 2 columns are dense enough for the Stage 1 lower bound used in the scaffold argument	Lemma 68
$(q_2 - 1)(2^{-b_1+1} C_1 + 1) < \eta_2 C_1 $	showing that the non-dominant Stage 2 rectangles cannot cover too much of the chosen block $\hat{X}_{p, \alpha}$	Lemma 68
$\eta_2 < \frac{1 - \varepsilon_{2^a+3, t_1}}{2}$	the final application of the chosen-coordinate Stage 1 threshold	proof of Theorem 1

C.2 The Stage-1 Threshold Package

Claim 72 (Rectangle bound for $\mathcal{H}_r(S_1)$). *Write*

$$q := 2^a + 3, \quad t := t_1, \quad \varepsilon := \varepsilon_{q, t}.$$

If $A \times B$ is a monochromatic rectangle in $\mathcal{H}_r(S_1)$ and $1 \leq s \leq \min\{|A|, t\}$, then

$$|B| \leq \frac{1 + \varepsilon}{2^s} |S_1|.$$

Proof. Choose $J \subseteq A$ with $|J| = s$. Since $A \times B$ is monochromatic, there is a unique bit pattern $\mathbf{b} \in \{0, 1\}^J$ such that every $\mathbf{c} = (c_1, \dots, c_q) \in B$ satisfies

$$\mathcal{H}_r(S_1)(i, \mathbf{c}) = \mathbf{b}_i \quad \text{for every } i \in J.$$

Equivalently, every column in B realises the same prescribed pattern on the coordinates J . Because S_1 is (q, t) -balanced, the definition of Definition 27 gives

$$|B| \leq \frac{1 + \varepsilon}{2^{|J|}} |S_1| = \frac{1 + \varepsilon}{2^s} |S_1|.$$

□

Proof of Lemma 45. Write

$$q := 2^a + 3, \quad t := t_1, \quad \varepsilon := \varepsilon_{q,t}.$$

Because $\mathcal{M}_0 = [1 \ 0]$ has one row and two columns, every row of $\mathcal{H}_r(S_1)$ is indexed by an outer coordinate $i \in [r]$, every column is a tuple $\mathbf{c} = (c_1, \dots, c_q) \in S_1$, and

$$\mathcal{H}_r(S_1)(i, \mathbf{c}) = 1 \quad \Longleftrightarrow \quad c_i \text{ is the first column of } \mathcal{M}_0.$$

We will also use that t is much larger than a . Indeed, $2^a = \lceil 2 \log^2 d \rceil_2 < 4 \log^2 d$, so $a \leq 2 + 2 \log \log d$, while $t = \lceil 64 \log d \rceil_2 \geq 64 \log d$. Hence

$$a + 2 \leq t. \tag{7}$$

Upper bounds. Alice can send the active copy index $i \in [r]$ using $\lceil \log r \rceil$ bits, after which Bob returns the output bit of $\mathcal{M}_0$. Therefore

$$D(\mathcal{H}_{2^a}(S_1)) \leq a + 1, \quad D(\mathcal{H}_{2^{a+1}}(S_1)) \leq a + 2.$$

Lower bound for $\mathcal{H}_{2^a}(S_1)$. Suppose towards contradiction that some deterministic protocol Π computes $\mathcal{H}_{2^a}(S_1)$ with depth at most a . Follow one root-to-leaf path greedily, called the *heavy path*: at every Alice node move to a child containing at least half of the current rows, and at every Bob node move to a child containing at least half of the current columns. Let $R_\lambda \times C_\lambda$ be the monochromatic leaf rectangle reached this way, and let u and v be the numbers of Alice bits and Bob bits on that path. Then $u + v \leq a$ and

$$|R_\lambda| \geq \left\lceil \frac{2^a}{2^u} \right\rceil = 2^{a-u}, \quad |C_\lambda| \geq 2^{-v} |S_1|.$$

Since $2^k \geq k + 1$ for every integer $k \geq 0$,

$$|R_\lambda| \geq 2^{a-u} \geq a - u + 1 \geq v + 1.$$

By (7), we also have $v + 1 \leq a + 1 \leq t$. Applying Claim 72 with $s = v + 1$ yields

$$|C_\lambda| \leq \frac{1 + \varepsilon}{2^{v+1}} |S_1| < 2^{-v} |S_1|,$$

since $\varepsilon < 1$. This contradicts $|C_\lambda| \geq 2^{-v} |S_1|$. Hence

$$D(\mathcal{H}_{2^a}(S_1)) = a + 1.$$

Lower bound for $\mathcal{H}_{2^a+1}(S_1)$. Now suppose towards contradiction that some deterministic protocol computes $\mathcal{H}_{2^a+1}(S_1)$ with depth at most $a + 1$. Choose a heavy leaf $R_\lambda \times C_\lambda$ exactly as above, and let u and v be the numbers of Alice and Bob bits on that path. Then $u + v \leq a + 1$, and

$$|R_\lambda| \geq \left\lceil \frac{2^a + 1}{2^u} \right\rceil, \quad |C_\lambda| \geq 2^{-v}|S_1|.$$

If $u = a + 1$, then $v = 0$ and $|R_\lambda| \geq 1 = v + 1$. Otherwise $u \leq a$, so

$$|R_\lambda| \geq \left\lceil \frac{2^a + 1}{2^u} \right\rceil \geq 2^{a-u} + 1 \geq a - u + 2 \geq v + 1.$$

Again (7) gives $v + 1 \leq a + 2 \leq t$, so Claim 72 with $s = v + 1$ yields

$$|C_\lambda| \leq \frac{1 + \varepsilon}{2^{v+1}}|S_1| < 2^{-v}|S_1|,$$

contradicting $|C_\lambda| \geq 2^{-v}|S_1|$. Therefore

$$D(\mathcal{H}_{2^a+1}(S_1)) = a + 2.$$

□

For $r \in \mathbb{N}$, let

$$H'_r := \mathcal{H}_r(S').$$

Claim 73 (Rectangle bound for dense Stage-1 restrictions). *If $A \times B$ is a monochromatic rectangle in H'_r and $1 \leq s \leq \min\{|A|, t_1\}$, then*

$$|B| \leq \frac{1 + \varepsilon_1}{1 - \rho} \cdot \frac{|S'|}{2^s}.$$

Proof. Because $B \subseteq S' \subseteq S_1$, the same rectangle is also monochromatic in $\mathcal{H}_r(S_1)$. Hence Claim 72 gives

$$|B| \leq \frac{1 + \varepsilon_1}{2^s}|S_1| \leq \frac{1 + \varepsilon_1}{1 - \rho} \cdot \frac{|S'|}{2^s},$$

using $|S'| \geq (1 - \rho)|S_1|$. □

Proof of Corollary 47. Upper bounds. Alice can still send the active copy index $i \in [r]$ using $\lceil \log r \rceil$ bits and Bob then returns the output bit of $\mathcal{M}_0$. Therefore

$$D(H'_{2^a}) \leq a + 1, \quad D(H'_{2^a+1}) \leq a + 2.$$

Lower bound for H'_{2^a} . Suppose towards contradiction that some deterministic protocol computes H'_{2^a} with depth at most a . Follow a heavy root-to-leaf path as in the proof of Lemma 45, and let $R_\lambda \times C_\lambda$ be the monochromatic leaf rectangle reached this way. If u and v are the numbers of Alice and Bob bits on that path, then $u + v \leq a$,

$$|R_\lambda| \geq 2^{a-u} \geq v + 1, \quad |C_\lambda| \geq 2^{-v}|S'|.$$

By (7), we also have $v + 1 \leq t_1$. Applying Claim 73 with $s = v + 1$ yields

$$|C_\lambda| \leq \frac{1 + \varepsilon_1}{1 - \rho} \cdot \frac{|S'|}{2^{v+1}} < 2^{-v}|S'|,$$

because $\rho < (1 - \varepsilon_1)/2$ is equivalent to $(1 + \varepsilon_1)/(1 - \rho) < 2$. This contradicts the lower bound on $|C_\lambda|$. Therefore

$$D(H'_{2^a}) = a + 1 = B_{\text{cap}}.$$

Lower bound for H'_{2^a+1} . Suppose instead that some deterministic protocol computes H'_{2^a+1} with depth at most $a + 1$. Choose a heavy leaf $R_\lambda \times C_\lambda$ exactly as above, and let u, v be the numbers of Alice and Bob bits on that path. Then $u + v \leq a + 1$,

$$|R_\lambda| \geq \left\lceil \frac{2^a + 1}{2^u} \right\rceil, \quad |C_\lambda| \geq 2^{-v} |S'|.$$

If $u = a + 1$, then $v = 0$ and $|R_\lambda| \geq 1 = v + 1$. Otherwise $u \leq a$, so the same calculation as in Lemma 45 gives $|R_\lambda| \geq v + 1$. Again $v + 1 \leq a + 2 \leq t_1$, so Claim 73 with $s = v + 1$ yields

$$|C_\lambda| \leq \frac{1 + \varepsilon_1}{1 - \rho} \cdot \frac{|S'|}{2^{v+1}} < 2^{-v} |S'|,$$

contradiction. Hence

$$D(H'_{2^a+1}) = a + 2 = B_{\text{cap}} + 1.$$

If $\ell \leq 1$, then G'_ℓ is a subgame of H'_{2^a} , so

$$D(G'_\ell) \leq B_{\text{cap}}.$$

If $\ell \geq 2$, then H'_{2^a+1} is a subgame of G'_ℓ , so

$$D(G'_\ell) \geq B_{\text{cap}} + 1.$$

□

Proof of Corollary 48. Choose $Q' \subseteq Q$ with

$$|Q'| = 2^a + 1.$$

Since deleting rows cannot increase communication complexity, it is enough to show

$$D(G_{Q'}) \geq B_{\text{cap}} + 1,$$

where

$$G_{Q'} := \mathcal{M}_0 \langle 2^a + 3, S' \rangle \upharpoonright_{Q' \times \text{Rows}(\mathcal{M}_0), S'}.$$

Suppose towards contradiction that some deterministic protocol computes $G_{Q'}$ with depth at most $a + 1$. Choose a heavy monochromatic leaf $R_\lambda \times C_\lambda$, and let u, v be the numbers of Alice and Bob bits on that path. Then $u + v \leq a + 1$,

$$|R_\lambda| \geq \left\lceil \frac{2^a + 1}{2^u} \right\rceil, \quad |C_\lambda| \geq 2^{-v} |S'|.$$

If $u = a + 1$, then $v = 0$ and $|R_\lambda| \geq 1 = v + 1$. Otherwise $u \leq a$, so the same calculation as in Lemma 45 gives $|R_\lambda| \geq v + 1$. Again $v + 1 \leq a + 2 \leq t_1$.

The proof of Claim 73 applies to an arbitrary set $J \subseteq A$ of active coordinates of size s , so it does not depend on those coordinates being the first $2^a + 1$ positions. Applying that claim with $s = v + 1$ therefore yields

$$|C_\lambda| \leq \frac{1 + \varepsilon_1}{1 - \rho} \cdot \frac{|S'|}{2^{v+1}} < 2^{-v} |S'|,$$

because $\rho < (1 - \varepsilon_1)/2$ is equivalent to $(1 + \varepsilon_1)/(1 - \rho) < 2$. This contradicts the lower bound on $|C_\lambda|$. Therefore

$$D(G_{Q'}) \geq a + 2 = B_{\text{cap}} + 1.$$

□

C.3 The Stage-1 and Stage-2 Bootstrap

Proof of Lemma 50. Set

$$t := t_1, \quad q' := r't, \quad p_{\text{seed}} := \frac{9}{16}t, \quad b := b_0 + \log r', \quad x_{\text{seed}} := 2^{-b}.$$

Choose $J \subseteq [q_1 + 5]$ of size q' containing Q ; this is possible because

$$q' = r't_1 \leq r_1 t_1 = q_1 + 2 < q_1 + 5.$$

Let

$$\mathcal{S}_J := \pi_J(C_1).$$

Because $C_1 = S_{q_1+5, t_1}(\text{Cols}(\mathcal{M}_0))$ is (q_1+5, t_1) -balanced, Lemma 30 shows that the projected family $\mathcal{S}_J$ is (q', t_1) -balanced with accuracy

$$\varepsilon := \varepsilon_{q', t_1}.$$

Here we use $\varepsilon_{q_1+5, t_1} \leq \varepsilon_{q', t_1}$ to enlarge the allowed accuracy parameter from the projected family to ε . Define

$$\widehat{M}_J := \mathcal{M}_0 \langle q', \mathcal{S}_J \rangle.$$

After relabelling J by $[q']$, the given matrix N is a submatrix of $\widehat{M}_J$ whose row set is $(Q, 1)$ -equipartitioned, with $|Q| = r'p_{\text{seed}}$, and whose column density is at least $h_2^\downarrow$. This is exactly the projected-family reuse recorded in Remark 40: Lemma 30 preserves the required balancedness and accuracy for $\mathcal{S}_J$, and Lemma 32 supplies the same bridge inside $\widehat{M}_J$. Hence the same localized-extension conclusion as in Corollary 39 holds for the projected ambient matrix $\widehat{M}_J$.

Column-loss resilience. For $\mathcal{M}_0$, any one-copy bracket $\mathcal{M}_0[1, x, y]$ has communication complexity 1 as soon as $y > \frac{1}{2}$, because then it contains both columns of $\mathcal{M}_0$. Since $q' < q_1 + 5 < 4 \log^2 d + 3$, $h_2^\downarrow > 1/(2d^3)$, $t_1 \geq 64 \log d$, and $1 + \varepsilon \leq 2$, we have

$$\left(\frac{h_2^\downarrow 2^{-(\log q' + 1)}}{1 + \varepsilon} \right)^{1/t_1} > \frac{1}{2}$$

for all sufficiently large d . Also every admissible pair $0 \leq k \leq D(\mathcal{M}_0) = 1$ and $0 \leq c \leq \log t_1 + k$ satisfies $c \leq \log t_1 + 1$, so the same estimate shows

$$\left(\frac{h_2^\downarrow 2^{-c}}{1 + \varepsilon} \right)^{1/t_1} > \frac{1}{2}.$$

Hence $(\mathcal{M}_0, b_0 + \log r')$ is $(q', t_1, h_2^\downarrow)$ -column-loss resilient.

Hard seed. By Lemma 49,

$$D\left(\mathcal{M}_0 \left[p_{\text{seed}}, x_{\text{seed}}, h_2'^\downarrow \right]\right) \geq \left\lceil \log(p_{\text{seed}} + \log h_2'^\downarrow) \right\rceil + 1.$$

Since $t_1 \geq 64 \log d$ and $\log(16t_1) = o(\log d)$,

$$p_{\text{seed}} + \log h_2'^\downarrow = \frac{9}{16}t_1 - 3 \log d - \log(16t_1) > \frac{1}{2}t_1$$

for all sufficiently large d . Because t_1 is a power of two, this yields

$$D\left(\mathcal{M}_0 \left[p_{\text{seed}}, x_{\text{seed}}, h_2'^\downarrow \right]\right) \geq \log t_1 + 1 = D(\mathcal{M}_0) + \log t_1.$$

Seed comparison. Since

$$\frac{h_2^{\prime\downarrow}}{h_2^\downarrow} = \frac{r_2}{16t_1 d},$$

and $r_2 \leq q_2 < 2d$, while $1 + \varepsilon \leq 2$, we obtain

$$\frac{h_2^{\prime\downarrow}}{h_2^\downarrow} \leq \frac{1}{8t_1} \leq \frac{1}{(1 + \varepsilon) 2t_1} = \frac{2^{-(\log t_1 + D(\mathcal{M}_0))}}{1 + \varepsilon}.$$

Thus

$$\frac{h_2^\downarrow 2^{-(\log t_1 + D(\mathcal{M}_0))}}{1 + \varepsilon} \geq h_2^{\prime\downarrow}.$$

Finally, because $|\text{Rows}(\mathcal{M}_0)| = 1$,

$$\lceil r' x_{\text{seed}} |\text{Rows}(\mathcal{M}_0)| \rceil = \lceil 2^{-b_0} \rceil = 1,$$

exactly matching the assumed $(Q, 1)$ -equipartition.

Applying Corollary 39 with

$$M := \mathcal{M}_0, \quad q := q', \quad t := t_1, \quad r := r', \quad h := h_2^\downarrow, \quad a := 0,$$

together with the already verified parameters

$$p_{\text{seed}} := \frac{9}{16} t_1, \quad x_{\text{seed}} := 2^{-(b_0 + \log r')}, \quad h_{\text{seed}} := h_2^{\prime\downarrow},$$

yields

$$D(N) \geq D(\mathcal{M}_0) + \log q' = D(\mathcal{M}_0) + \log t_1 + \log r'. \quad \square$$

Proof of Lemma 52. By Lemma 9, it suffices to prove the three inequalities

$$\begin{aligned} D\left(\mathcal{M}_1 \left[1, \frac{1}{2} + \delta, 2^{-b_1}\right]\right) &\geq D(\mathcal{M}_1), \\ D\left(\mathcal{M}_1 \left[1, \frac{1}{4} + \frac{\delta}{2}, 2^{-b_1}\right]\right) &\geq D(\mathcal{M}_1) - 1, \\ D\left(\mathcal{M}_1 \left[1, \frac{1}{8} + \frac{\delta}{4}, 2^{-b_1}\right]\right) &\geq D(\mathcal{M}_1) - 2. \end{aligned}$$

Write

$$\beta_0 := \frac{1}{2} + \delta = 0.6, \quad \beta_1 := \frac{1}{4} + \frac{\delta}{2} = 0.3, \quad \beta_2 := \frac{1}{8} + \frac{\delta}{4} = 0.15,$$

and

$$r'_0 := r_1, \quad r'_1 := \frac{r_1}{2}, \quad r'_2 := \frac{r_1}{4}.$$

For $j \in \{0, 1, 2\}$, let

$$N = \mathcal{M}_1 \upharpoonright_{\tilde{R}, \tilde{C}} \in \mathcal{M}_1 \left[1, \beta_j, 2^{-b_1}\right].$$

Since $\text{Rows}(\mathcal{M}_1) = [q_1] \times \text{Rows}(\mathcal{M}_0)$ and $|\text{Rows}(\mathcal{M}_0)| = 1$, the row set $\tilde{R}$ corresponds to a subset $\tilde{Q} \subseteq [q_1]$ with

$$|\tilde{Q}| = |\tilde{R}| \geq \beta_j q_1.$$

Set

$$q_j'' := \frac{9}{16} r'_j t_1.$$

Because

$$\beta_0 > \frac{9}{16}, \quad \beta_1 > \frac{9}{32}, \quad \beta_2 > \frac{9}{64},$$

and $q_1 = r_1 t_1 - 2$, we have

$$\beta_j q_1 \geq q_j''$$

for all sufficiently large d . Choose $Q \subseteq \tilde{Q}$ of size q_j'' , set $R' := Q \times \text{Rows}(\mathcal{M}_0) \subseteq \tilde{R}$, and keep the same column set $\tilde{C}$. Since

$$\frac{|\tilde{C}|}{|C_1|} \geq 2^{-b_1} > 2^{-(b_1 + \log r_2)} = h_2^\downarrow,$$

Lemma 50 with $r' = r'_j$ gives

$$D(N) \geq D(\mathcal{M}_1 \upharpoonright_{R', \tilde{C}}) \geq D(\mathcal{M}_0) + \log t_1 + \log r'_j.$$

By Corollary 51, the right-hand side equals $D(\mathcal{M}_1) - j$. This proves the three robust lower bounds for $\mathcal{M}_1^\top$, and $D(\mathcal{M}_1^\top) = D(\mathcal{M}_1) \geq 1$ follows from Lemma 8. $\square$

Proof of Lemma 53. Since $q_2 < 2d$, $D(\mathcal{M}_1) = a + 1 < 3 \log \log d + 3$, $1 + \varepsilon_{q_2, t_2} \leq 2$, and $t_2 \geq 3 \log d / \log \log d$, we have

$$y_2^{\text{term}} \geq 2^{-\frac{(\log d + 3 \log \log d + 6) \log \log d}{3 \log d}}.$$

For all sufficiently large d , the displayed exponent is at most $\frac{1}{2} \log \log d$, so

$$y_2^{\text{term}} > (\log d)^{-1/2}.$$

Also $q_1 > \log^2 d$ for sufficiently large d , so

$$y_2^{\text{term}} q_1 > (\log d)^{-1/2} \log^2 d = \log^{3/2} d.$$

Since $t_1 < 128 \log d$, the right-hand side eventually exceeds $\frac{9}{16} t_1$. Hence any $N \in \mathcal{M}_1 [1, y_2^{\text{term}}, 2^{-b_1}]$ contains a further restriction whose row set is $Q \times \text{Rows}(\mathcal{M}_0)$ for some $Q \subseteq [q_1]$ with

$$|Q| = \frac{9}{16} t_1.$$

Since $2^{-b_1} > h_2^\downarrow$, applying Lemma 50 with $r' = 1$ yields

$$D(N) \geq D(\mathcal{M}_0) + \log t_1 = D(\mathcal{M}_1) - \log r_1$$

by Corollary 51. $\square$

C.4 The Stage-2 Certification Package

Proof of Lemma 54. Write

$$\varepsilon_2 := \varepsilon_{q_2, t_2}, \quad y_c := \left(\frac{h_2 2^{-c}}{1 + \varepsilon_2} \right)^{1/t_2}.$$

Clause (i). We must show

$$D\left(\mathcal{M}_1^\top \left[1, 2^{-b_1}, y_{\log q_2 + D(\mathcal{M}_1)}\right]\right) \geq 1.$$

By Lemma 9, this is equivalent to

$$D\left(\mathcal{M}_1 \left[1, y_{\log q_2 + D(\mathcal{M}_1)}, 2^{-b_1}\right]\right) \geq 1.$$

By Lemma 53,

$$D\left(\mathcal{M}_1 \left[1, y_{\log q_2 + D(\mathcal{M}_1)}, 2^{-b_1}\right]\right) \geq D(\mathcal{M}_1) - \log r_1 \geq 1.$$

Clause (ii). Fix integers $0 \leq k \leq D(\mathcal{M}_1)$ and $0 \leq c \leq \log t_2 + k$. By Lemma 9, it is enough to prove that for each $j \in \{0, 1, 2\}$,

$$D\left(\mathcal{M}_1 \left[1, y_c/2^j, 2^{-b_1}\right]\right) \geq D(\mathcal{M}_1) - k - j.$$

Fix $j \in \{0, 1, 2\}$, and write

$$K := k + j.$$

Let

$$y_2^* := \left(\frac{h_2 2^{-\log t_2}}{1 + \varepsilon_2}\right)^{1/t_2}.$$

Since $h_2 = (\log d)^{-3}$, $\log t_2 < \log \log d + 3$, and $t_2 \geq 3 \log d / \log \log d$, we have

$$y_2^* \geq 2^{-\frac{(4 \log \log d + 4) \log \log d}{3 \log d}}.$$

For all sufficiently large d , the exponent $\frac{(4 \log \log d + 4) \log \log d}{3 \log d}$ is smaller than $\log(5/3)$, so $y_2^* > 0.6$. Hence

$$y_c \geq y_2^* 2^{-k/t_2} \geq 0.6 \cdot 2^{-k},$$

and therefore

$$\frac{y_c}{2^j} \geq 0.6 \cdot 2^{-K}.$$

If $K \leq \log r_1$, set $r_{1,K} := r_1/2^K$. Then any matrix in $\mathcal{M}_1 [1, y_c/2^j, 2^{-b_1}]$ retains at least

$$0.6 \cdot 2^{-K} q_1 = 0.6 \cdot \frac{r_{1,K}}{r_1} (r_1 t_1 - 2)$$

rows. Since $0.6 > \frac{9}{16}$, this exceeds $\frac{9}{16} r_{1,K} t_1$ for all sufficiently large d . We may therefore choose $Q \subseteq [q_1]$ of size $\frac{9}{16} r_{1,K} t_1$ inside the surviving row set. Because $2^{-b_1} > h_2^\downarrow$, Lemma 50 with $r' = r_{1,K}$ gives

$$D\left(\mathcal{M}_1 \left[1, y_c/2^j, 2^{-b_1}\right]\right) \geq D(\mathcal{M}_0) + \log t_1 + \log r_{1,K} = D(\mathcal{M}_1) - K = D(\mathcal{M}_1) - k - j.$$

If $K > \log r_1$, then it is enough to reach $r' = 1$. Since $c \leq \log t_2 + D(\mathcal{M}_1)$, the same estimates as in Lemma 53 give

$$y_c > (\log d)^{-1/2}.$$

Since $j \leq 2$, we therefore have

$$\frac{y_c}{2^j} > 2^{-2} (\log d)^{-1/2}.$$

Hence a member of $\mathcal{M}_1 [1, y_c/2^j, 2^{-b_1}]$ retains more than

$$2^{-2} \log^{3/2} d$$

rows. Since $t_1 < 128 \log d$, this eventually exceeds $\frac{9}{16}t_1$. Choosing $Q \subseteq [q_1]$ of that size and applying Lemma 50 with $r' = 1$ yields

$$D\left(\mathcal{M}_1 \left[1, y_c/2^j, 2^{-b_1}\right]\right) \geq D(\mathcal{M}_0) + \log t_1 = D(\mathcal{M}_1) - \log r_1 \geq D(\mathcal{M}_1) - K = D(\mathcal{M}_1) - k - j.$$

Since this holds for $j = 0, 1, 2$, the definition of Λ_M gives

$$\Lambda_{\mathcal{M}_1^\top}(1, 2^{-b_1}, y_c) \geq D(\mathcal{M}_1) - k.$$

This proves clause (ii). $\square$

Proof of Corollary 56. The upper bound is the direct protocol that first sends the active Stage 2 outer index and then solves $\mathcal{M}_1^\top$.

For the lower bound, set

$$p_2^{\text{seed}} := \frac{9}{16}t_2, \quad x_2^{\text{seed}} := 2^{-b_1}t_2, \quad T_2 := \left\lceil q_2 2^{-b_1} |C_1| \right\rceil.$$

Choose any $Q \subseteq [q_2]$ with

$$|Q| = \frac{9}{16}q_2 = r_2 p_2^{\text{seed}},$$

and choose any row set $R' \subseteq Q \times C_1$ that is (Q, T_2) -equipartitioned. Consider

$$N := \mathcal{M}_2 \upharpoonright_{R', C_2}.$$

Since

$$T_2 = \left\lceil r_2 x_2^{\text{seed}} |C_1| \right\rceil,$$

Corollary 39 applies to N with

$$M := \mathcal{M}_1^\top, \quad b := b_1, \quad t := t_2, \quad r := r_2, \quad h := h_2,$$

and with

$$p_{\text{seed}} := p_2^{\text{seed}}, \quad x_{\text{seed}} := x_2^{\text{seed}}, \quad h_{\text{seed}} := h'_2, \quad r' := r_2, \quad a := 0.$$

By Lemmas 54 and 55, it only remains to verify the seed comparison

$$\frac{h_2 2^{-(\log t_2 + D(\mathcal{M}_1))}}{1 + \varepsilon_{q_2, t_2}} \geq h'_2.$$

Since

$$\frac{h'_2}{h_2} = 2^{-5 \log \log d} = (\log d)^{-5},$$

while $q_1 + 2 < 4 \log^2 d$, $t_2 < 6 \log d / \log \log d$, $1 + \varepsilon_{q_2, t_2} \leq 2$, and $D(\mathcal{M}_1) = a + 1 \leq \log(q_1 + 2) + 1$, we obtain

$$\frac{2^{-(\log t_2 + D(\mathcal{M}_1))}}{1 + \varepsilon_{q_2, t_2}} \geq \frac{1}{4(q_1 + 2)t_2} \geq \frac{\log \log d}{96 \log^3 d} \geq (\log d)^{-5}$$

for all sufficiently large d . Hence the seed comparison holds, and Corollary 39 yields

$$D(N) \geq D(\mathcal{M}_1) + \log q_2.$$

Since N is a restriction of $\mathcal{M}_2$, monotonicity gives

$$D(\mathcal{M}_2) \geq D(\mathcal{M}_1) + \log q_2.$$

Together with the upper bound, this proves the claim. $\square$

Proof of Lemma 57. By Lemma 9, it suffices to prove that for

$$\beta_0 := \frac{1}{2} + \delta = 0.6, \quad \beta_1 := \frac{1}{4} + \frac{\delta}{2} = 0.3, \quad \beta_2 := \frac{1}{8} + \frac{\delta}{4} = 0.15,$$

every matrix

$$N = \mathcal{M}_2 \upharpoonright_{\tilde{R}, \tilde{C}} \in \mathcal{M}_2[1, \beta_j, h_2]$$

satisfies

$$D(N) \geq D(\mathcal{M}_2) - j \quad (j \in \{0, 1, 2\}).$$

Set

$$p_2^{\text{seed}} := \frac{9}{16}t_2, \quad x_2^{\text{seed}} := 2^{-b_1}t_2, \quad r'_0 := r_2, \quad r'_1 := \frac{r_2}{2}, \quad r'_2 := \frac{r_2}{4}, \quad q_j'' := \frac{9}{16}r'_j t_2.$$

Let

$$\tilde{\beta} := \frac{|\tilde{R}|}{q_2|C_1|} \geq \beta_j, \quad T_{2,j} := \left\lceil r'_j x_2^{\text{seed}} |C_1| \right\rceil.$$

Since

$$\beta_j \geq \frac{3}{20} \quad \text{for } j = 0, 1, 2,$$

and

$$r'_j x_2^{\text{seed}} \leq r_2 x_2^{\text{seed}} = q_2 2^{-b_1} < \frac{2}{d},$$

we have $\tilde{\beta} \geq r'_j x_2^{\text{seed}}$ for all sufficiently large d . Thus the hypothesis of Lemma 31 is satisfied. Apply Lemma 31 to $M := \mathcal{M}_1^\top$, $q := q_2$, $x := r'_j x_2^{\text{seed}}$, and the row set $\tilde{R} \subseteq [q_2] \times C_1$. This yields a set $J \subseteq [q_2]$ such that $\tilde{R} \cap (J \times C_1)$ is $(J, T_{2,j})$ -equipartitioned and

$$|J| = \left\lceil q_2 \cdot \frac{\tilde{\beta} - r'_j x_2^{\text{seed}}}{1 - r'_j x_2^{\text{seed}}} \right\rceil.$$

Moreover,

$$r'_j x_2^{\text{seed}} \leq r_2 x_2^{\text{seed}} = q_2 2^{-b_1} < \frac{2}{d},$$

so

$$|J| \geq q_2 \left(\beta_j - \frac{2}{d} \right).$$

Since

$$\beta_0 - \frac{9}{16} = \frac{3}{80}, \quad \beta_1 - \frac{9}{32} = \frac{3}{160}, \quad \beta_2 - \frac{9}{64} = \frac{3}{320},$$

it follows that $|J| \geq q_j''$ for $j = 0, 1, 2$ once d is large enough. Choose any $Q \subseteq J$ with $|Q| = q_j''$, and set

$$R' := \tilde{R} \cap (Q \times C_1).$$

Then R' is $(Q, T_{2,j})$ -equipartitioned, and the column density remains $\geq h_2$.

Apply Corollary 39 to

$$N' := \mathcal{M}_2 \upharpoonright_{R', \tilde{C}}$$

with

$$M := \mathcal{M}_1^\top, \quad b := b_1, \quad t := t_2, \quad r := r_2, \quad h := h_2,$$

and with

$$p_{\text{seed}} := p_2^{\text{seed}}, \quad x_{\text{seed}} := x_2^{\text{seed}}, \quad h_{\text{seed}} := h'_2, \quad r' := r'_j, \quad a := 0.$$

By Lemmas 54 and 55, and by the same seed-comparison estimate already verified in the proof of Corollary 56, all hypotheses of Corollary 39 hold. Hence

$$D(N') \geq D(\mathcal{M}_1) + \log(r'_j t_2).$$

Since N' is a restriction of N , monotonicity gives

$$D(N) \geq D(\mathcal{M}_1) + \log t_2 + \log r'_j = D(\mathcal{M}_2) - j.$$

This proves the required robust lower bounds for $\mathcal{M}_2^\top$, and $D(\mathcal{M}_2^\top) = D(\mathcal{M}_2) \geq 1$ again follows from Lemma 8. $\square$

C.5 The Stage-4 Gadget Proofs

Proof of Lemma 62. On the slice $X_{p,\alpha}$, the template rows are determined by the Stage 2 component $j_p = (\alpha, \gamma_p)$, so after deleting duplicate columns coming from the unused choices of γ_m for $m \neq p$, the template part reduces to one copy of each column $\gamma_p \in C_1$. On that representative family, the template rows are exactly the first q_1 outer row blocks of the full $(q_1 + 5)$ -coordinate Stage 1 subgame over C_1 .

By the Stage 4 definition, every vector row $i \in A_\alpha(B)$ agrees on $X_{p,\alpha}$ with the local row $\mathcal{M}_0 \langle 5 \rangle (\pi_\alpha(i), \text{tail}(\gamma_p))$. Since π_α is injective, these rows are pairwise distinct and occupy non-default positions in the 5-row gadget. Every vector row in $B \setminus A_\alpha$ agrees on $X_{p,\alpha}$ with the default row $\mathcal{M}_0 \langle 5 \rangle (5, \text{tail}(\gamma_p))$.

Keeping the template rows, all rows of $A_\alpha(B)$, and at most one row from $B \setminus A_\alpha$ when $\nu_\alpha(B) = 1$ therefore yields a subgame obtained from

$$\mathcal{M}_0 \langle 2^a + 3, S_{2^a+3, t_1}(\text{Cols}(\mathcal{M}_0)) \rangle \upharpoonright_{Q_{p,\alpha}(B) \times \text{Rows}(\mathcal{M}_0), S_{2^a+3, t_1}(\text{Cols}(\mathcal{M}_0))}$$

by adding duplicate columns and, when $\nu_\alpha(B) = 1$, possibly duplicate copies of the neutral row. The same conclusion holds for every shifted slice $X_{p,\alpha}^{(s)}$: replacing k by $k \oplus s$ is a column permutation of the local 5-row gadget, so the extracted subgame is isomorphic to the one on $X_{p,\alpha} = X_{p,\alpha}^{(0)}$.

If $\nu_\alpha(B) = 1$ and $\ell_\alpha(B) \geq 2$, then

$$|Q_{p,\alpha}(B)| = q_1 + \ell_\alpha(B) + \nu_\alpha(B) = 2^a - 2 + \ell_\alpha(B) + \nu_\alpha(B) \geq 2^a + 1.$$

Applying Corollary 48 with

$$S' = S_{2^a+3, t_1}(\text{Cols}(\mathcal{M}_0)) \quad \text{and} \quad \rho = 0$$

gives lower bound $B_{\text{cap}} + 1$ for the displayed chosen-coordinate subgame. Duplicate rows and duplicate columns do not change deterministic communication complexity, so the same lower bound holds for the selected branch. $\square$

Proof of Corollary 64. Let

$$S' := \{ \gamma \in C_1 : (\text{tail}(\gamma), ((\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma), (\alpha, \gamma))) \in Y \}.$$

Since $|\widehat{X}_{p,\alpha}| = |C_1|$, we have

$$|S'| \geq (1 - \rho)|C_1| = (1 - \rho) |S_{2^a+3,t_1}(\text{Cols}(\mathcal{M}_0))|.$$

On the canonical diagonal copy $\widehat{X}_{p,\alpha}$, the distinguished component is exactly γ , and the local 5-row gadget sees $k = \text{tail}(\gamma)$. Therefore the template rows and vector rows behave exactly as in the proof of Lemma 62, with one column per $\gamma \in S'$. Keeping those columns yields a subgame obtained from

$$\mathcal{M}_0 \langle 2^a + 3, S' \rangle \upharpoonright_{Q_{p,\alpha}(B) \times \text{Rows}(\mathcal{M}_0), S'}$$

by adding duplicate rows and duplicate columns.

If $\nu_\alpha(B) = 1$ and $\ell_\alpha(B) \geq 2$, then

$$|Q_{p,\alpha}(B)| \geq 2^a + 1.$$

Applying Corollary 48 to the chosen-coordinate subgame over S' gives lower bound $B_{\text{cap}} + 1$, and duplicate rows and duplicate columns preserve deterministic communication complexity. $\square$

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We used OpenAI Codex and ChatGPT Pro to assist with proof auditing, consistency checks, and substantive revisions to proof packaging and exposition. These tools materially influenced the manuscript throughout, with their strongest effect on the parameter-heavy arguments in Sections 4 and 5, and Appendix C. The authors made the final decisions about all mathematical content and manually checked the manuscript, citations, and bibliography entries for correctness and attribution.